

Cloud Infrastructure and Services



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Module-I

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Notes**Unit-1: Introduction to Cloud Computing****Objective:**

After studying this chapter, you will be able to:

- Explain the meaning of Cloud Computing
- Discuss the Characteristics of Cloud computing as per NIST
- Explain the Private, public and hybrid cloud
- Discuss the Cloud types; IaaS, SaaS, PaaS
- Understand the Benefits and challenges of cloud computing

Introduction

To put it very simply, cloud computing is the delivery of computing resources as a service. Moving to the cloud basically means that the resources are owned and managed by a third-party provider, instead of the end-user.

This means that you don't need to worry about hard drives, main-frames, or where any of this hardware and software is located. As far as you, the user, is concerned, it's floating up there in a metaphorical 'cloud' – which you're able to access via the internet.

This shift from software and hardware that was on-premises to a networked, remote resource has meant that companies no longer have to worry about investing in labour, expertise, or capital for the maintenance of these resources. It has spawned a plethora of cloud computing companies, including key players like AWS and Microsoft Azure.

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, software, analytics, intelligence, and more, over the Cloud (Internet).

Most of data is stored on local networks with servers that may be clustered and sharing storage. This approach has had time to be developed into stable architecture, and provide decent redundancy when deployed right. A newer emerging technology, cloud computing, has shown up demanding attention and quickly is changing the direction of the technology landscape. Whether it is Google's unique and scalable Google File System, or Amazon's robust Amazon S3 cloud storage model, it is clear that cloud computing has arrived with much to be gleaned from.

In dealing with the abstract term, "the cloud", it is easy to misunderstand what makes up the structure and function. The basic function is what comes from "the cloud". This is primarily output, however, not only. Input is what makes the cloud tick.

Do not confuse cloud computing with the term data center, as it typically sits on top of the latter. By viewing the cloud as logical rather than a physical, one can see it object describes it better.

In a world that sees new technological trends bloom and fade on almost a daily basis, one new trend promises more longevity. This trend is called cloud computing, and it will change the way one uses computer and the Internet.

Cloud computing portends a major change in how we store information and run applications. Instead of running programs and data on an individual desktop computer,

everything is hosted in the “cloud” a nebulous assemblage of computers and servers accessed via the Internet. Cloud computing lets one access all applications and documents from anywhere in the world, freeing one from the confines of the desktop and making it easier for group members in different locations to collaborate.

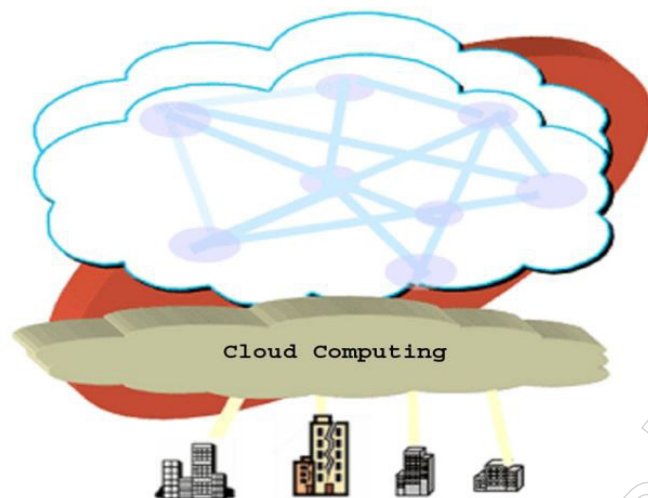


Figure 1.1: Cloud Computing.

The emergence of cloud computing is the computing equivalent of the electricity revolution of a century ago. Before the advent of electrical utilities, every farm and business produced its own electricity from freestanding generators. After the electrical grid was created, farms and businesses shut down their generators and bought electricity from the utilities, at a much lower price (and with much greater reliability) than they could produce on their own.

Look for the same type of revolution to occur as cloud computing takes hold. The desktop-centric notion of computing that we hold today is bound to fall by the wayside as we come to expect the universal access, 24/7 reliability, and ubiquitous collaboration promised by cloud computing.

1.1 Definition of Cloud Computing

Simple Definition of Cloud Computing

“Cloud computing is the use of off-site systems to help computers store, manage, process, and/or communicate information. These off-site systems are hosted on the cloud (or the internet) instead of on your computer or other local storage. They can encompass anything from email servers to software programs, data storage, or even increasing your computer’s processing power.

The “cloud” is a term that simply means “the internet.” Computing involves the infrastructures and systems that allow a computer to run and build, deploy, or interact with information. In cloud computing, this means that instead of hosting infrastructure, systems, or applications on your hard drive or an on-site server, you’re hosting it on virtual/online servers that connect to your computer through secure networks.”

With traditional desktop computing, one runs copies of software programs on each computer. The documents one creates are stored on the computer on which they were

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created. Although documents can be accessed from other computers on the network, they cannot be accessed by computers outside the network.

With cloud computing, the software programs one uses are not run from personal computer, but are rather stored on servers accessed via the Internet. If computer crashes, the software is still available for others to use. Same goes for the documents one creates; they are stored on a collection of servers accessed via the Internet. Anyone with permission cannot only access the documents, but can also edit and collaborate on those documents in real time. Unlike traditional computing, this cloud computing model is not PC centric, it is document-centric. Which PC one uses to access a document simply is not important.

But that is a simplification. Let us look in more detail at what cloud computing is and, just as important, what it is not.

First, cloud computing is not network computing. With network computing, applications/documents are hosted on a single company's server and accessed over the company's network. Cloud computing is a lot bigger than that. It encompasses multiple companies; multiple servers, and multiple networks. Plus, unlike network computing, cloud services and storage are accessible from anywhere in the world over an Internet connection; with network computing, access is over the company's network only.

Cloud computing also is not traditional outsourcing, where a company farms out (subcontracts) its computing services to an outside firm. While an outsourcing firm might host a company's data or applications, those documents and programs are only accessible to the company's employees via the company's network, not to the entire world via the Internet.

So, despite superficial similarities, networking computing and outsourcing are not cloud computing. Key to the definition of cloud computing is the "cloud" itself. For our purposes, the cloud is a large group of interconnected computers. These computers can be personal computers or network servers; they can be public or private.

For example, Google hosts a cloud that consists of both smallish PCs and larger servers. Google's cloud is a private one (that is, Google owns it) that is publicly accessible (by Google's users).

This cloud of computers extends beyond a single company or enterprise. The applications and data served by the cloud are available to broad group of users, cross-enterprise and cross-platform. Access is via the Internet. Any authorized user can access these docs and apps from any computer over any Internet connection. And, to the user, the technology and infrastructure behind the cloud is invisible. It is not apparent (and, in most cases does not matter) whether cloud services are based on HTTP, HTML, XML, JavaScript, or other specific technologies.

It might help to examine how one of the pioneers of cloud computing, Google, perceives the topic. From Google's perspective, there are six key properties of cloud computing:

- ◇ **Cloud computing is user-centric.** Once a user is connected to the cloud, whatever are stored there-documents, messages, images, applications, whatever-becomes? In addition, not only is the data, but one can also share it with others. In effect, any device that accesses data in the cloud.

- ◇ **Cloud computing is task-centric.** Instead of focusing on the application and what it can do, the focus is on what one needs done and how the application can do it for ones. Traditional applications-word processing, spreadsheets, email, and so on-are becoming less important than the documents they create.
- ◇ **Cloud computing is powerful.** Connecting hundreds or thousands of computers together in a cloud creates a wealth of computing power impossible with a single desktop PC.
- ◇ **Cloud computing is accessible.** Because data is stored in the cloud, users can instantly retrieve more information from multiple repositories. One is not limited to a single source of data, as one is with a desktop PC.
- ◇ **Cloud computing is intelligent.** With all the various data stored on the computers in a cloud, data mining and analysis are necessary to access that information in an intelligent manner.
- ◇ **Cloud computing is programmable.** Many of the tasks necessary with cloud computing must be automated. For example, if that one computer goes offline, the cloud's programing automatically redistributes that computer's data to a new computer in the cloud.

All these definitions behind us, what constitutes cloud computing in the real world?

Draft of Web-hosted, Internet-accessible, group collaborative applications are currently available, with many more on the way. Perhaps the best and most popular examples of cloud computing applications today are the Google family of applications-Google Docs Spreadsheets, Google Calendar, Gmail, Picasa, and the like. All of these applications are hosted on Google's servers, are accessible to any user with an Internet connection, and can be used for group collaboration from anywhere in the world.

In short, cloud computing enables a shift from the computer to the user, from application to tasks, and from isolated data to data that can be accessed from anywhere and shared with anyone. The user no longer has to take on the task of data management; it does not even have to remember where the data is. All that matters is that the data is in the cloud, and thus immediately available to that user and to other authorized user.

1.2 Characteristics of Cloud Computing as per NIST

Cloud computing is an evolving paradigm. The National institute of standards and technology (NIST) definition characterizes important aspects of cloud computing and is intended to serve as a means for broad comparisons of cloud services and deployment strategies, and to provide a baseline for discussion from what is cloud computing to how to best use cloud computing. The service and deployment models defined form a simple taxonomy that is not intended to prescribe or constrain any particular method of deployment, service delivery, or business operation. Cloud computing is a model for enabling ubiquitous, convenient, demand network access to a shared on-pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. This cloud model is composed of five essential characteristics.

According to NIST there are five essential characteristics of cloud computing:

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- ◇ On Demand Self Service
- ◇ Broad network access
- ◇ Resource pooling
- ◇ Rapid elasticity
- ◇ Measured service

1. On Demand Self Service

User gets on demand self-services. A consumer can unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider. User can get computer services like email, applications etc. without interacting with each service provider.

Some of the cloud service providers are- Amazon Web Service, Microsoft, IBM, Salesforce.com

2. Broad network access

Cloud services are available over the network and can be accessed through different clients such as mobile, laptops etc.

3. Resource pooling

Same resources can be used by more than one customer at a same time. The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand. There is a sense of location independence in that the customer generally has no control or knowledge over the exact location of the provided resources but may be able to specify location at a higher level of abstraction (e.g., country, state, or datacenter).

For example-storage, network bandwidth can be used by any number of customers and without knowing the exact location of that resource.

4. Rapid elasticity

On users demand cloud services can be available and released. Cloud service capabilities are unlimited and used in any quantity at any time.

5. Measured service

Resources used by the users can be monitored, controlled. This report is available for both cloud providers and consumer. Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

1.3 Private, Public and Hybrid Cloud

The cloud computing term extents a range of classifications, types, and architecture models. This networked computing model has transformed how one

works. But the cloud isn't one thing—cloud computing can be categorized into three general types:

Public cloud is cloud computing that's delivered via the internet and shared across organizations.

Private cloud is cloud computing that is dedicated solely to organization.

Hybrid cloud is any environment that uses both public and private clouds.

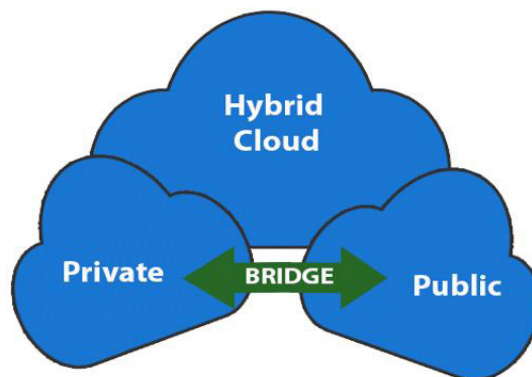


Figure: 1.2

1.3.1. Public Cloud Model

Public cloud is the most popular method to deploy cloud services. A public cloud service is owned and operated by a third party service provider, who will take care of the maintenance of the cloud services and infrastructure. Public cloud services are delivered over the internet and are ideal for small to mid-sized companies. The most popular examples of public cloud are Microsoft Azure, Amazon EC2, and IBM's Blue Cloud.

The infrastructure including the hardware and the software is owned by the service provider and is shared by multiple organizations who are called cloud tenants. Public cloud services follow the pay-as-you-go model, which makes public clouds economical for organizations with varying needs. That factor, in addition to being able to handle smaller amounts of data, makes it ideal for small and mid-sized companies. Since the tenants do not own the services first hand, the pain of maintenance and management of the data centres is offset on the service provider.

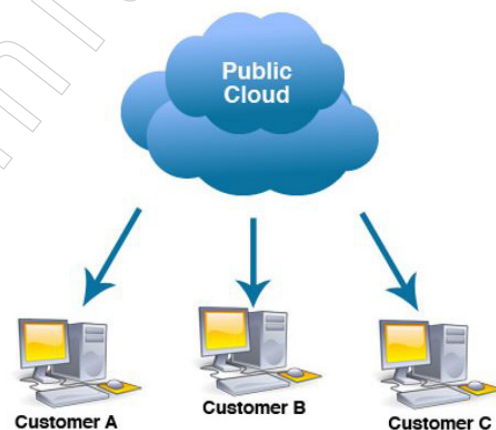


Figure: 1.3

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Public clouds are used when data compliance and control over data is not a major concern for the customer. The major drawback a lot of organizations feel is lack of security and control over the hardware. As the servers are shared and the provider owns the maintenance rights, compliance regulation also becomes a concern.

1.3.1.1. Benefits of Public Cloud

Some advantages of the public cloud are:

It follows the pay-as-you-go model, making it less expensive

The third-party vendor takes care of the maintenance of cloud services

The availability of on-demand resources makes the cloud highly scalable and highly reliable

1. Reduce both Capex & Opex of your business

The payment method in public cloud hosting is metered by the usage -this practice being similar to that of electricity or other utility bill, wherein you pay only for those resources that you procure.

Apart from that, public cloud helps you get the most out of your existing IT system, eliminating the need of up gradation. Thus, it saves your money and resources from getting wasted.

Also, its multi-tenant environment (centralized management) allows multiple users to share computing resources which makes it cost efficient as infrastructure costs are spread across all users.

2. Develop things and scale resources faster

You can develop web, mobile, Internet of Things (IoT) and enterprise apps faster for any device or platform as it is compatible with different operating systems, frameworks, devices, databases, tools and languages.

Public cloud hosting being the vast network of cloud servers, offers individuals or businesses the opportunity to scale the resources –bandwidth, RAM etc. whenever their business necessitates it. You also get the elasticity to accommodate your growing or reducing business needs.

3. Run applications globally

Another major benefit of public cloud service is that it is accessible from any location via an internet connection. This provides multiple opportunities to organizations like remote access to IT infrastructure or online document collaboration from different locations.

4. Get zero risk assurance

Another important advantage of Microsoft public cloud is that it guarantees maximum uptime and no risk of failure.

As it is the system of interconnected servers, so if any one server fails to take your request, then, another one out of the multiple servers will take over the workload of the failed server automatically.

Thus, you experience flawless performance.

5. Maintain backup of all your critical data

It facilitates easy upload and storage of important data, files, documents, videos which you can access from anywhere across the world. Thus, you don't need to deploy and maintain costly data storage appliances. Our Azure public cloud helps you to maintain backup and privacy of your critical data.

1.3.1.2. Drawbacks of Public Cloud

Public clouds are used when data compliance and control over data is not a significant concern for the customer. The major drawback a lot of organizations feel is a lack of security and control over the hardware. As the servers are shared, and the provider owns the maintenance rights, compliance regulation also becomes a concern.

1.3.2. Private Cloud Model

Private clouds are owned and operated by a single organization or entity. In a private cloud environment, the hardware, software, and any related infrastructure is either located at the data center of the organization or is located in a controlled environment of a service provider.

Private clouds differ from public clouds in their flexibility and control over the data. Private clouds, by definition, cannot be provided as a service. The most famous examples of a private cloud are Microsoft, HP Data Centers, Ubuntu, Elasta-private cloud, etc.

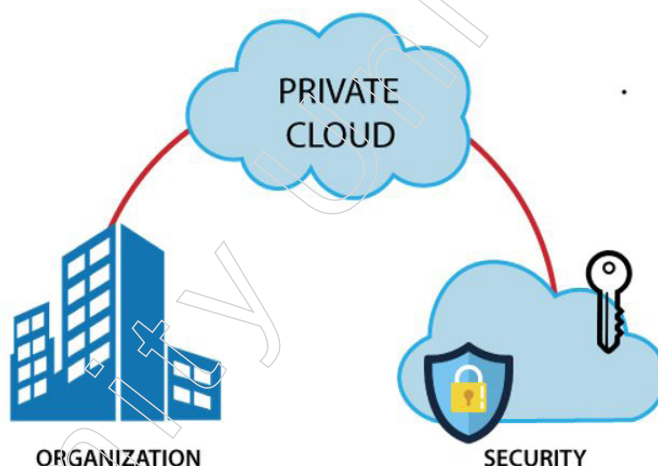


Figure: 1.4

Government institutions, financial institutions like banks, mid to large-sized companies, and any other organization dealing with sensitive information tend to prefer private clouds.

1.3.2.1. Benefits of Private Cloud

The private cloud has a dedicated service provider, so it offers complete control over the data, enhanced flexibility, scalability, automation, security, and it all comes with a price.

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There are the following advantages of Private Cloud –

1) More Control

Private clouds have more control over their resources and hardware than public clouds because it is only accessed by selected users.

2) Security & privacy

Security & privacy are one of the big advantages of cloud computing. Private cloud improved the security level as compared to the public cloud.

3) Improved performance

Private cloud offers better performance with improved speed and space capacity.

1.3.2.2. Drawbacks of Private Cloud

Although private cloud offers scalability and security, it is expensive to set up, and companies will have to maintain the servers constantly and do their own troubleshooting.

In real-world practice, cloud computing services are also offered in another format known as the Hybrid cloud, which tries to deliver the best of both worlds. It incorporates the benefits of both public and private cloud.

1.3.3. Hybrid Cloud Model

Hybrid Cloud is a mixture of public and private cloud. Non-critical activities are performed using public cloud while the critical activities are performed using private cloud. The Hybrid Cloud Model is shown in the diagram below.

The Hybrid Cloud is a cloud computing environment incorporating both private and public cloud services with a coherent synchronization. This cloud allows users to move data and applications between a public cloud and a private cloud.

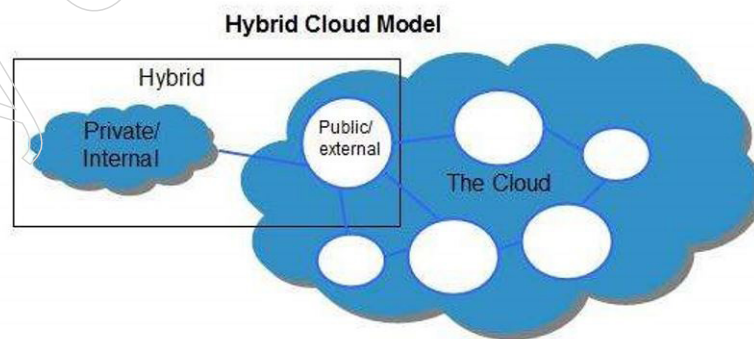


Figure: 1.5

Most companies prefer the hybrid cloud model as it offers numerous business benefits, such as fulfilling regulatory and data requirements, addressing low latency problems, and so on.

1.3.3.1. Benefits of Hybrid Cloud

In a typical Hybrid cloud, the data can be switched between the on-premise and

third-party service provider. This provides enhanced control, flexibility, and cost-saving. Hybrid cloud helps organizations to handle the short-term hike in demands with minimum capital.

Scalability

It offers features of both, the public cloud scalability and the private cloud scalability.

Flexibility

It offers secure resources and scalable public resources.

Cost Efficiency

Public clouds are more cost effective than private ones. Therefore, hybrid clouds can be cost saving.

Security

The private cloud in hybrid cloud ensures higher degree of security.

1.3.3.2. Drawbacks of Hybrid Cloud

Although hybrid cloud offers a gamut of advantages, the significant concern still revolves around the security of the data, which plagues the perception of public clouds as well. As the hybrid cloud is a blend of both, transmitting sensitive information over a network that is subjected to third-party interference is an uncalculated risk for most organizations.

Below are the differences between public, private, and hybrid clouds:

Difference	Public Cloud	Private Cloud	Hybrid Cloud
Data Tenancy	Multi-tenancy: The data of numerous companies is stored in a shared environment	Single Tenancy: The data of only a single organization is stored in the cloud	The data stored in the public cloud is shared, and the data stored in the private cloud is not shared and kept confidential
Cloud Services	Open to public	Only that specific organization can use the cloud services	Services on the public cloud can be accessed by everyone, whereas services in the private cloud can be accessed only by that organization
Connectivity	Over the internet	Over the organization's private network	Over the internet for public cloud services and organization's private network for private cloud services
Management of Cloud Services	Managed by the cloud service provider	Managed by the administrators of that specific organization	The public cloud is managed by the cloud service provider, whereas the administrators of that particular organization manage the private cloud

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Software and Hardware Components	The cloud service provider manages these components	That particular organization operates these components	Public cloud components – Cloud Service provider Private cloud components – Organization
Costs	Less expensive as the cloud service provider offers all the resources	Very expensive as the organization has to purchase all the resources	Less costly for public cloud and more expensive for private cloud resources
Scalability and Flexibility	High	High	High
Security	Low	High	Public Cloud – Low Private Cloud – High

1.4 Cloud Type

Cloud computing is a utility-oriented and Internet-centric way of delivering IT services on demand.

Cloud computing architecture includes:

- ◇ IaaS, Infrastructure as a service
- ◇ PaaS, Platform as a service
- ◇ SaaS, Software as a service

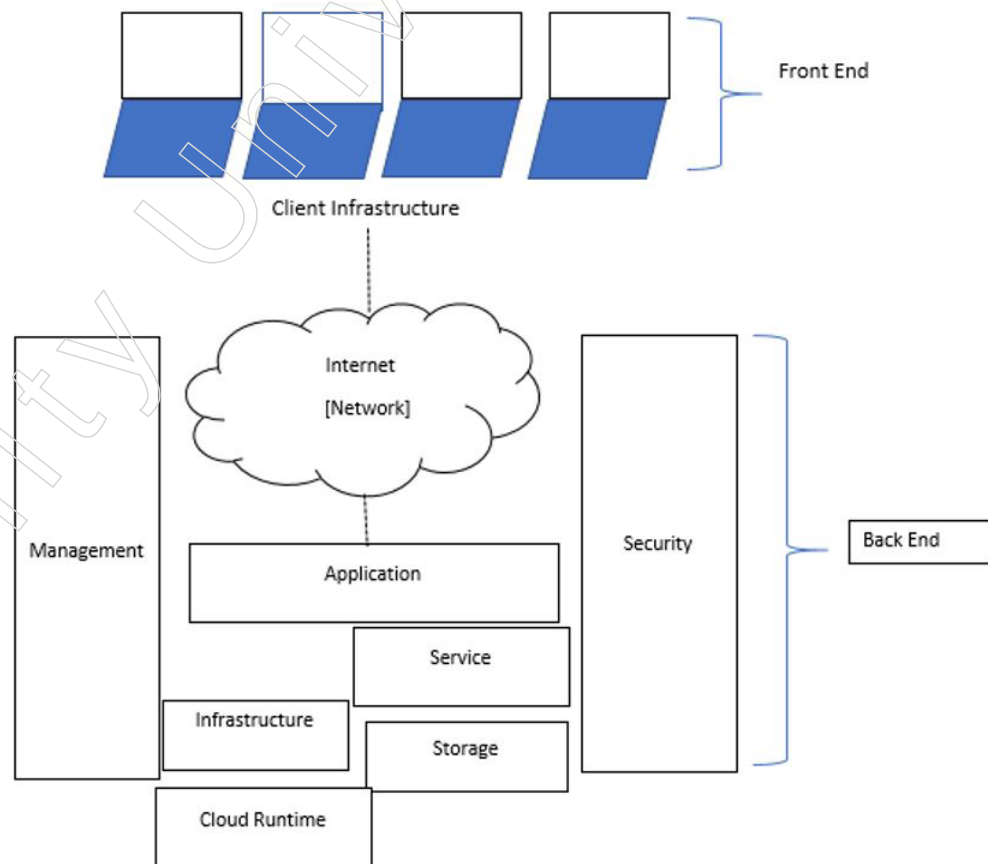


Figure: 1.6 Cloud computing architecture

1.4.1. IaaS Infrastructure as a Service

IaaS stands for infrastructure as a service. It is the basic layer in cloud computing model. The IaaS includes servers, network devices, load balancers, database, Web servers etc. It delivers customizable infrastructure on demand.

IaaS examples can be categorized in two categories

- ◇ IaaS Management layer
- ◇ IaaS Physical infrastructure

Some service providers provide both above categories and some provides only management layer. IaaS management layer also required integration with other IaaS solutions that provide physical infrastructure. On virtual machines applications are installed and deployed. One of the examples of virtual machine is Oracle VM.

Hardware virtualization includes workload partitioning, application isolation, sandboxing, and hardware tuning. Instead of purchasing user can access this virtual hardware on pay per use basis. The users can take advantage of the full customization offered by virtualization to deploy their infrastructure in the cloud. Some virtual machines can be with pre-installed operating systems and other software. On some virtual machines operating systems and others software can be installed as per use.

Some examples are: Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE)

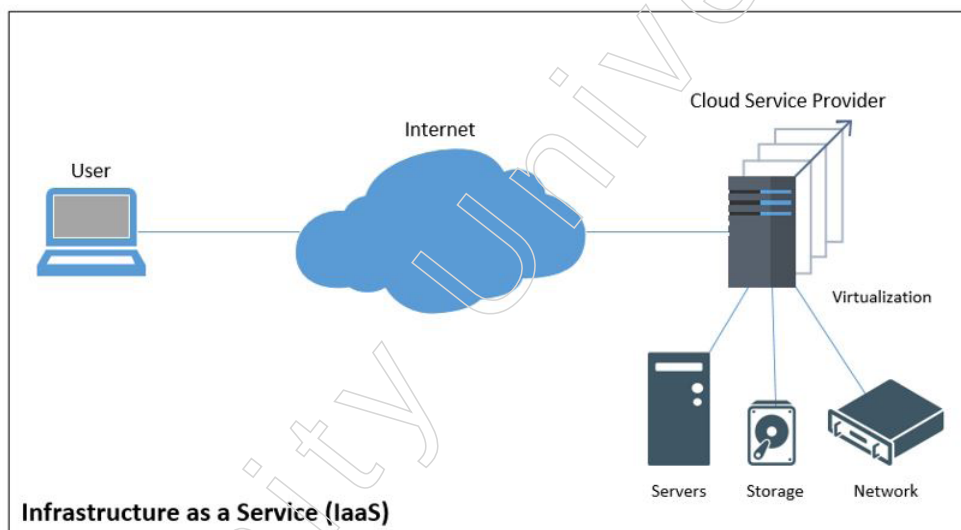


Figure: 1.7 IaaS Infrastructure as a Service

1.4.2. PaaS Platform as a service

PaaS stands for platform as a service. It provides a computing platform with a programming language execution environment. PaaS provide a development and deployment platform for running applications in the cloud. It constitute the middleware on top of which applications are built. Application management is the core functionality of the middleware.

PaaS provides: run time environments for the applications, applications deployment, configuring application components, provisioning and configuring supporting technologies.

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For users PaaS interfaces can be in the form of a Web-based interface or in the form of programming APIs and libraries.

PaaS solutions generally include the infrastructure as well. PurePaaS offered only the user-level middleware.

PaaS classification is as follows:

PaaS-I: Runtime environment with Web-hosted application development platform, rapid application prototyping. For example Force.com which is a combination of middleware and infrastructure product type.

PaaS-II: Runtime environment for scaling Web applications. The runtime could be enhanced by additional components that provide scaling capabilities. For example Google AppEngine which is a combination of middleware and infrastructure product type. App scale is middleware product type.

PaaS-III: Middleware and programming model for developing distributed applications in the cloud. For example Microsoft Azure which is a combination of middleware and infrastructure product type. Manjrasoft Aneka is a middleware product type.

Some examples are:

Google App Engine

Force.com

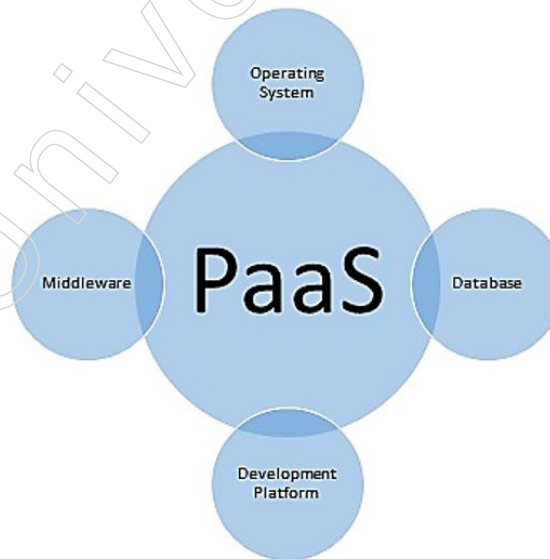


Figure: 1.8 PaaS Platform as a service

1.4.3. SaaS Software as a service

SaaS stands for software as a service. Software as a service (SaaS) allows users to connect to and use cloud-based apps over the Internet. It is the service with which end users interact directly. It provides a means to free users from complex hardware and software management. In SaaS customer do not need to purchase the software and required the license. They simply access the application website, enter their credentials and billing details, and can instantly use the application. Customer can customize their software.

Application is available to the customer on demand. SaaS can be considered as a “one-to-many” software delivery model. In SaaS applications are built as per the user needs.

Some examples are:

Gmail
Google drive
Dropbox
WhatsApp

Characteristics of SaaS:

- ◇ The product sold to customer is application access.
- ◇ The application is centrally managed.
- ◇ The service delivered is one-to-many.
- ◇ The service delivered is an integrated solution delivered on the contract, which means provided as promised.



Figure: 1.9 SaaS Software as a service

1.5 Benefits and Challenges of Cloud Computing

Any serious analysis of cloud computing must address the benefits and challenges offered by this burgeoning technology. What's good-and what's bad-about cloud computing? Let's take a look.

We will start with the benefits offered by cloud computing and there are many.

1.5.1. Benefits of Cloud Computing

1.5.1.1. Lower-cost Computers for Users

There is a quantitative financial advantage one does not need a high-powered (and accordingly high-priced) computer to run cloud computing web-based applications. Because the application runs in the cloud, not on the desktop PC, that desktop PC does not need the processing power or hard disk space demanded by traditional desktop software. Hence the client computers in cloud computing can be lower priced, with smaller hard disks, less memory; more efficient processors, and the like. In fact, a

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client computer in this scenario would not even need a CD or DVD drive, because no software programs have to be loaded and no document files need to be saved.

1.5.1.2. Improved Performance

Let's look further at what results when a desktop PC does not have to store and run a ton of software-based applications. (The apps are run from the cloud, instead.) With fewer bloated programs hogging the computer's memory, users will see better performance from their PCs. Put simply computers in a cloud computing system will boot up faster and run faster, because they will have fewer programs and processes loaded into memory.

1.5.1.3. Lower IT Infrastructure Costs

In a larger organization, the IT department could also see lower costs from the adoption of the cloud computing paradigm. Instead of investing in larger numbers of more powerful servers, the IT staff can use the computing power of the cloud to supplement or replace internal computing resources. Those companies that have peak needs no longer have to purchase equipment to handle the peaks (and then lay fallow the rest of the time); peak computing needs are easily handled by computers and servers in the cloud.

1.5.1.4. Fewer Maintenance Issues

Speaking of maintenance costs, cloud computing greatly reduces both hardware and software maintenance for organizations of all sizes.

First, the hardware with less hardware (fewer servers) necessary in the organization, maintenance costs is immediately lowered. As to software maintenance, remember that all cloud apps are based elsewhere, so there's no software on the organization's computers for the IT staff to maintain. It is that simple.

Did you know?

Cloud services are based on HTTP, HTML, XML, JavaScript, or other specific technologies; to the user, the technology and infrastructure behind the cloud is invisible.

1.5.1.5. Lower Software Costs

There is always the issue of software cost. Instead of purchasing separate software packages for each computer in the organization, only those employees actually using an application need access to that application in the cloud. Even if it costs the same to use web-based applications as it does similar desktop software (which it probably would not), IT staffs are saved the cost of installing and maintaining those programs on every desktop in the organization.

As to the cost of that software, it is possible that some cloud computing companies will charge as much to "rent" their apps as traditional software companies charge for software purchases. However, early indications are that cloud services will be priced substantially lower than similar desktop software. In fact, many companies (such as Google) are offering their web-based applications for free which to both individuals and large organizations is much more attractive than the high costs charged by Microsoft and similar desktop software suppliers.

1.5.1.6. Instant Software Updates

Another software-related advantage to cloud computing is that users are no longer faced with the choice between obsolete software and high upgrade costs. When the app is web-based, updates happen automatically and are available the next time the user logs in to the cloud. Whenever one accesses a web-based application, one is getting the latest version-without needing to pay for or download an upgrade.

1.5.1.7. Increased Computing Power

This is an obvious one. When one is tied into a cloud computing system, one has the power of the entire cloud at disposal. One is no longer limited to what a single desktop PC can do, but can now perform supercomputing-like tasks utilizing the power of thousands of computers and servers. In other words, one can attempt greater tasks in the cloud than one can on desktop.

1.5.1.8. Unlimited Storage Capacity

Similarly, the cloud offers virtually limitless storage capacity. Consider that when desktop or laptop PC is running out of storage space. The computer's 200GB hard drive is peanuts compared to the hundreds of petabytes (a million gigabytes) available in the cloud. Whatever one needs to store, one can.

1.5.1.9. Increased Data Safety

All that data one stores in the cloud, it stays in the cloud-somewhere. Unlike desktop computing, where a hard disk crash can destroy all valuable data, a computer crashing in the cloud does not affect the storage of data. That is because data in the cloud is automatically duplicated, so nothing is ever lost. That also means if personal computer crashes, all data is still out there in the cloud, still accessible. In a world where few individual desktop PC users back up their data on a regular basis, cloud computing can keep data safe.

1.5.1.10. Improved Compatibility between Operating Systems

Ever try to get a Windows-based computer to talk to a Mac? Or a Linux machine to share data with a Windows PC? It can be frustrating.

Not so with cloud computing. In the cloud, operating systems simply do not matter. One can connect Windows computer to the cloud and share documents with computers running Apple's Mac OS, Linux, or UNIX. In the cloud, the data matters, not the operating system.

1.5.1.11. Improved Document Format Compatibility

One also do not have to worry about the documents one creates on machine being compatible with other user's applications or operating systems. In a world where Word 2007 documents cannot be opened on a computer running Word 2003, all documents created by web-based applications can be read by any other user accessing that application. There are no format incompatibilities when everyone is sharing docs and apps in the cloud.

Notes

1.5.1.12. Easier Group Collaboration

Sharing documents leads directly to collaborating on documents. To many users, this is one of the most important advantages of cloud computing—the ability for multiple users to easily collaborate on documents and projects.

Imagine that one, a colleague in West Coast office, and a consultant in Europe all need to work together on an important project. Before cloud computing, one had to email or snail mail the relevant documents from one user to another, and work on them sequentially. Not so with cloud computing. Now each of one can access the project's documents simultaneously; the edits one user makes are automatically reflected in what the other users see onscreen. It is all possible, of course, because the documents are hosted in the cloud, not on any of individual computers. All one need is a computer with an Internet connection, and one is collaborating.

Of course, easier group collaboration means faster completion of most group projects, with full participation from all involved. It also enables group projects across different geographic locations. No longer does the group have to reside in a single office for best effect. With cloud computing, anyone anywhere can collaborate in real time. It is an enabling technology.

1.5.1.13. Universal Access to Documents

Ever get home from work and realize you left an important document at the office? Or forget to take a file with you on the road? Or get to a conference and discover you forgot to bring along your presentation?

Not a problem not anymore, anyway. With cloud computing, one does not take documents with them. Instead, they stay in the cloud, where one can access them from anywhere one have a computer and an Internet connection. All documents are instantly available. There is simply no need to take documents as long as one has an Internet connection.

1.5.1.14. Latest Version Availability

And here is another document-related advantage of cloud computing. When one edit a document at home, that edited version is what one see when can access the document at work. The cloud always hosts the latest version of documents; one is never in danger of having an outdated version on the computer is working on.

1.5.1.15. Removes the Tether to Specific Devices

Finally, here is the ultimate cloud computing advantage is no longer tethered to a single computer or network. Change computers, and existing applications and documents follow one through the cloud. Move to a portable device, and apps and docs are still available. There's no need to buy a special version of a program for a particular device, or save document in a device-specific format. The documents and the programs that created them are the same no matter what computer is using.

Task

Create a growth chart for cloud computing.

1.5.2. Challenges of Cloud Computing

That is not to say, of course, that cloud computing is without its disadvantages. There are a number of reasons why one might not want to adopt cloud computing for particular needs. Let's examine a few of the risks related to cloud computing.

1.5.2.1. Requires a Constant Internet Connection

Cloud computing is, quite simply, impossible if one cannot connect to the Internet. Because one uses the Internet to connect to both applications and documents, if one does not have an Internet connection, one cannot access anything, even documents. A dead Internet connection means no work, period-and in areas where Internet connections are few or inherently unreliable, this could be a deal breaker. When one is offline, cloud computing just does not work.

This might be a more significant disadvantage than one might think. Sure, one is used to a relatively consistent Internet connection both at home and at work, but where else does one like to use computer? If one is used to work on documents on desk, or while one is at a restaurant for lunch, or in car, one would not be able to access cloud-based documents and applications-unless one have a strong Internet connection at the all those locations, of course. A lot of what is nice about portable computing becomes problematic when one is depending on web- based application.

1.5.2.2. Does not Work well with low-Speed Connections

Similarly, a low-speed Internet connection, such as that found with dial-up services, makes cloud computing painful at best and often impossible. Web-based apps often require a lot of bandwidth to download, as do large documents. If one is laboring with a low-speed dial-up connection, it might take seemingly forever just to change from page to page in a document, let alone launch a feature-rich cloud service.

In other words, cloud computing is not for the slow or broadband-impaired.

1.5.2.3. Can Be Slow

Even on a fast connection, web-based applications can sometimes be slower than accessing a similar software program on desktop PC. That is because everything about the program, from the interface to the document one is working on, has to be sent back and forth from computer to the computers in the cloud. If the cloud servers happen to be backed up at that moment, or if the Internet is having a slow day, one would not get the instantaneous access one is used to with desktop apps.

Did you know? Cloud application services or "Software as a Service (SaaS)" deliver software as a service over the Internet, eliminating the need to install and run the application on the customer's own computers and simplifying maintenance and support.

1.5.2.4. Futures Might Be Limited

This particular disadvantage is bound to change, but today many web-based applications simply are not as full-featured as their desktop-based brethren. Compare, for example, the feature set of Google Presentations with that of Microsoft PowerPoint; there is just a lot more one can do with PowerPoint than one can with Google's web-based offering. The basics are similar, but the cloud application lacks many of PowerPoint's advanced features.

Notes

So if one is an advanced user, one might not want to leap into the cloud computing waters just yet. That said, many web-based apps add more advanced features over time. This has certainly been the case with Google Docs and Spreadsheets, both of which started out somewhat crippled but later added many of the more niche functions found on Microsoft Word and Excel. Still, one needs to look at the features before one makes the move. Make sure that the cloud-based application can do everything one needs it to do before one gives up on traditional software.

1.5.2.5. Stored Data Might Not Be Secure

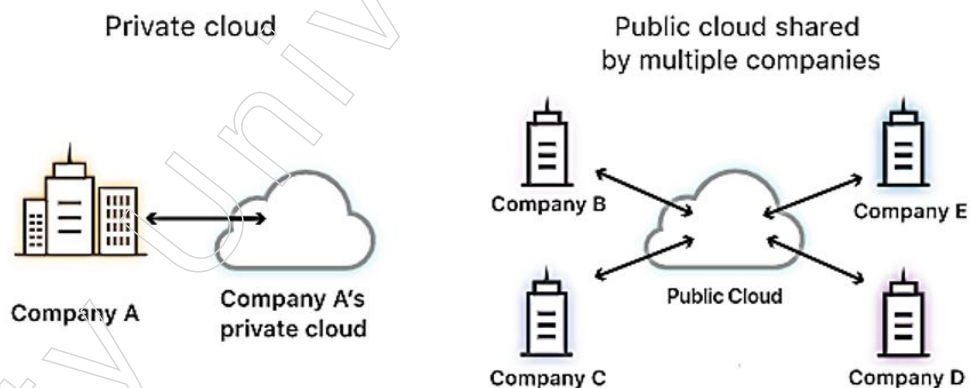
With cloud computing, all data is stored on the cloud. That is all well and good, but how secure is the cloud? Can other, unauthorized users gain access to confidential data? These are all important questions, and well worth further examination.

1.5.2.6. Cloud Loses Data

Theoretically, data stored in the cloud is unusually safe, replicated across multiple machines. But on the off chance that data does go missing, if there is no physical or local backup.

1.6 Private VS Public Cloud

A private cloud is a cloud service that is not shared with any other organization. The private cloud user has the cloud to themselves.



By contrast, a public cloud is a cloud service that shares computing services among different customers, even though each customer's data and applications running in the cloud remain hidden from other cloud customers.

A public cloud is like renting an apartment, while a private cloud is like renting a similarly sized house. The house is more private, but it also typically costs more to rent, and it's not the most efficient use of resources. Maintenance in the apartment is handled by the building supervisor, but it's harder to get a contractor out to fix the house (sometimes, the tenant may have to do it themselves).

There are hosted private clouds, which are offered by a third party cloud provider, and internal private clouds, which are managed and maintained by an organization internally.

Notes

Public Cloud	Private Cloud
Cloud Computing infrastructure shared to public by service provider over the internet. It supports multiple customers i.e, enterprises.	Cloud Computing infrastructure shared to private organisation by service provider over the internet. It supports one enterprise.
Multi-Tenancy i.e, Data of many enterprise are stored in shared environment but are isolated. Data is shared as per rule, permission and security.	Single Tenancy i.e, Data of single enterprise is stored.
Cloud service provider provides all the possible services and hardware as the user-base is world. Different people and organization may need different services and hardware. Services provided must be versatile.	Specific hardware and hardware as per need of enterprise are available in private cloud.
It is hosted at Service Provider site.	It is hosted at Service Provider site or enterprise.
It is connected to the public internet.	It only supports connectivity over the private network.
Scalability is very high, and reliability is moderate.	Scalability is limited, and reliability is very high.
Cloud service provider manages cloud and customers use them.	Managed and used by single enterprise.
It is cheaper than private cloud.	It is costlier than public cloud.
Security matters and dependent on service provider.	It gives high class of security.
Performance is low to medium.	Performance is high.
It has shared servers.	It has dedicated servers.
Example : Amazon web service (AWS) and Google AppEngine etc.	Example : Microsoft KVM, HP, Red Hat & VMWare etc.

Notes

Case Study

Cloud computing and web 2.0 separated at birth?

If you are at all familiar with the concept probably also heard of something called Web 2.0 with, on the face, sounds suspiciously similar to cloud computing. In fact, cloud computing is similar to Web 2.0; in many ways, the one is a subset the other.

What exactly is Web 2.0? To my mind, it is a bit of a buzzword that different people define in different ways. Tim C'Reilly, the so-called God-father father of Web 2.0, defines it as "the network as platform, spanning all connected devices." Others define the concept of Web 2.0 as network as a transition from isolated information silos (Web 1.0) to interlinked computing, platforms that function like locally available, software in the perception of the user. Still others define Web 2.0 in collaborative terms, because all the websites get their value from the actions of users.

These definitions of Web 2.0 sound a lot like cloud computing, but, without the technological underpinnings. That is, cloud computing is defined by its architecture and infrastructure (a grid of interconnected computers/servers functioning as a whole), whereas Web 3.0 is defined by how the user sees/is serviced by the system. In other words, cloud computing is about computers, Web 2.0 is about people

Or, as Tim O'Reilly puts it:

Cloud computing refers specifically to the use of the Internet as a computing platform; Web 2.0, as I have defined it, is an attempt to "explore and explain the business rules of that platform.

Perhaps the terms cloud computing and Web 2.0 are just two different ways of looking at the same phenomenon. Or, equally likely, perhaps cloud computing is a specific Web 2.0 technologies. In any instance, know that both terms sprang up at about the same point on computing evolutionary timeline and both concepts promise similar results to end users.

At the end of the day, it does not matter whether you view something like Google Docs or Microsoft Live Mesh as a cloud service or a Web 2.0 application, or as both. What matters more than a particular label is how that technology impacts the user. In this respect, both cloud computing and Web 2.0 offer very real benefits for all involved.

Questions

1. What is the difference between cloud computing and Web 2.0?
2. What are the benefits of Web 2.0?

Notes**Summary**

- Cloud computing portends a major change in how we store information and run applications.
- Cloud computing is a utility-oriented and Internet-centric way of delivering IT services on demand.
- IaaS stands for infrastructure as a service. It is the basic layer in cloud computing model.
- PaaS stands for platform as a service. It provides a computing platform with a programming language execution environment.
- SaaS stands for software as a service. Software as a service (SaaS) allows users to connect to and use cloud-based apps over the Internet.
- A quantitative financial advantage is one does not need a high-powered (and accordingly high-priced) computer to run cloud computing web-based applications.
- Cloud services include anything from calendar and contact applications to word processing and presentations.
- Cloud computing provides increased amounts of storage and processing power to run the applications they develop.

Keywords

Cloud computing: Cloud computing is the “cloud”-a massive network of servers or even individual PCs interconnected in a grid.

Cloud service: Any web-based application or service offered via cloud computing is called a cloud service.

Cloud storage: One of the primary uses of cloud computing is for data storage. With cloud storage, data is stored on multiple third-party servers, rather than on the dedicated servers used in traditional networked data storage.

Software as a service (SaaS): SaaS stands for software as a service. It allows users to connect to and use cloud-based apps over the Internet.

On Demand Self Service: User gets on demand self-services.

Broad network access: Cloud services are available over the network and can be accessed through different clients such as mobile, laptops etc.

Rapid elasticity: On users demand cloud services can be available and released. Cloud service capabilities are unlimited and used in any quantity at any time.

Self-Assessment Questions

1. Cloud computing has as its antecedents both client/server computing and peer-to-peer distributed computing.
(a) True (b) False
2. Which is not a property of cloud computing?
(a) Powerful (b) Accessible
(c) Intelligent (d) None of these.

Notes

3. A network architecture in which each computer has equivalent capabilities and responsibilities is called:
 - (a) Distributed computing
 - (b) Shared computing
 - (c) Peer-to-Peer computing
 - (d) None of these.
4. Usenet was again created in
 - (a) 1972 (b) 1979
 - (c) 1989 (d) None of these.
5. DEC stands for:
 - (a) Digital Equipment Corporation
 - (b) Dual Equipment Corporation
 - (c) Double Equipment Corporation
 - (d) None of these.
6. The first major Internet-based distributed computing project was launched in:
 - (a) 1975 (b) 1997
 - (c) 1999 (d) None of these.
7. The cloud is a collection of computers and servers that are publicly accessible via the Internet.
 - (a) True (b) False
8. Web-based application or service offered via cloud computing is called:
 - (a) Cloud crash
 - (b) Cloud storage
 - (c) Cloud service
 - (d) None of these.

Review Questions

1. What is cloud computing? What is the need of cloud computing in computation?
2. What are the key properties of cloud computing? Discuss briefly.
3. Discuss the benefits of cloud computing briefly.
4. What are the differences between private, public and hybrid cloud?
5. What are the challenges in cloud computing?
6. How the cloud computing works?
7. Describe the cloud architecture.
8. Cloud computing greatly reduces both hardware and software maintenance for organizations of all sizes. Explain.

9. Write notes on: i) IaaS Infrastructure ii) PaaS Platform iii) SaaS Software
10. Even on a fast connection, web-based applications can sometimes be slower than accessing a similar software program on desktop PC. Explain.

Answers for Self-Assessment Questions

11. 1. (a) 2. (d) 3. (a) 4. (b) 5. (a)
12. 6. (b) 7. (a) 8. (c)

Notes

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Unit-2: Cloud Computing

Objective:

After studying this chapter, you will be able to:

- Explain the meaning of Role of virtualization in enabling the cloud
- Discuss the Cloud computing environment
- Explain the Business Agility
- Discuss the Application, availability, performance, security and disaster recovery
- Understand the Next generation Cloud Application

Introduction

In the era of technology, the cloud environment is important as well as a versatile term. Cloud is a concept that includes cloud servers, cloud-based, cloud storage, cloud infrastructure, cloud computing, etc. The cloud environment architecture enhances the available storage capacity and processing power of the hardware to assist the cloud service provider to create a consolidated and powerful computing network accessible through the internet.

Cloud environment is a very broad term that includes a collection of services offered to enterprises for enhancing their functionality and IT capacity. The major feature of the cloud environment setup is that it centralizes resources and improves business efficiency. This enables mobilization for the workforce and facilitates remote workstations and working possible

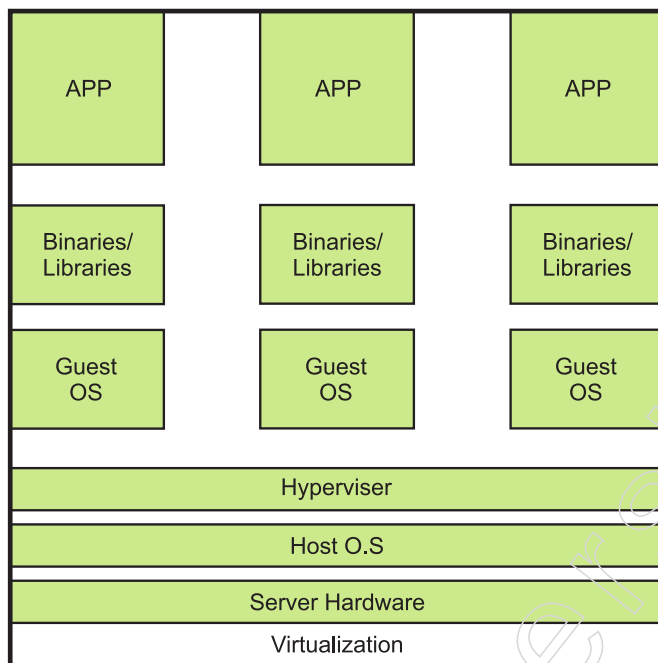
Specific characteristics of the cloud computing environment can be briefed as follows:

- ◇ On-demand self-service
- ◇ Broad network access
- ◇ Resource pooling and multi-tenancy
- ◇ Rapid scalability and elasticity
- ◇ Measured service

Virtualization is a technique of how to separate a service from the underlying physical delivery of that service. It is the process of creating a virtual version of something like computer hardware. It was initially developed during the mainframe era. It involves using specialized software to create a virtual or software-created version of a computing resource rather than the actual version of the same resource. With the help of Virtualization, multiple operating systems and applications can run on same machine and its same hardware at the same time, increasing the utilization and flexibility of hardware.

In other words, one of the main cost effective, hardware reducing, and energy saving techniques used by cloud providers is virtualization. Virtualization allows to share a single physical instance of a resource or an application among multiple customers and organizations at one time. It does this by assigning a logical name to a physical storage and providing a pointer to that physical resource on demand. The

term virtualization is often synonymous with hardware virtualization, which plays a fundamental role in efficiently delivering Infrastructure-as-a-Service (IaaS) solutions for cloud computing. Moreover, virtualization technologies provide a virtual environment for not only executing applications but also for storage, memory, and networking.



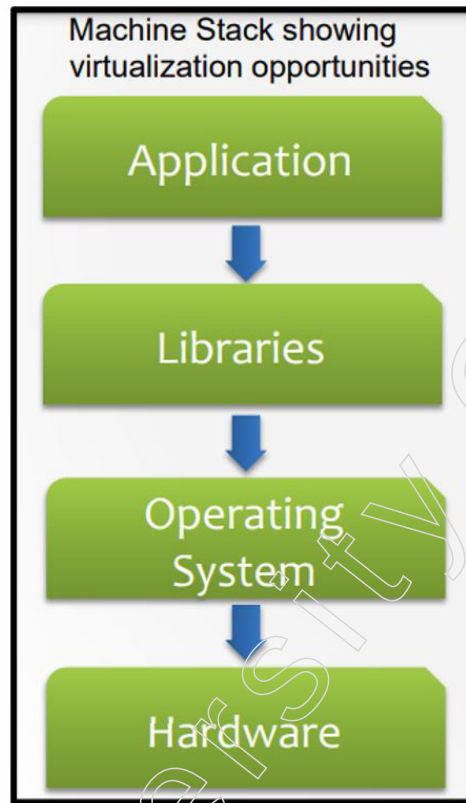
The machine on which the virtual machine is going to be built is known as Host Machine and that virtual machine is referred as a Guest Machine.

On demand provision of computational resources (Infrastructure, Platform, Software).

- ◇ Requires high availability of resources and optimum use.
- ◇ Virtualization is the enabling technology and creates virtual machines that allows a single machine to act as if it were many machines.
- ◇ Benefits of virtualization for cloud computing: Reduces capital expenses and maintenance costs through server consolidation, reduces physical space needed in data centers. Resource Management, Migration, Maintainability, High availability and Fault tolerance are other benefits.
- ◇ Virtualization is implemented using hypervisors.
- ◇ Creation of a virtual version of hardware using software.
- ◇ Runs several applications at the same time on a single physical server by hosting each of them inside their own virtual machine.
- ◇ By running multiple virtual machines simultaneously, a physical server can be utilized efficiently. Primary approaches to virtualization
- ◇ Platform virtualization Ex : Server
- ◇ Resources virtualization Ex : Storage, Network

Notes

Notes



2.1. Role of Virtualization in Enabling the Cloud

Virtual machines. Most IT pros understand this term and the benefits of moving from hardware to virtual machines. But virtualization is far more than just “machines.” It is set to disrupt virtually every computer-related operation for businesses, especially when those businesses make the decision to virtualize in-house or to move to cloud-based IT service providers.

What is Virtualization

The basic concept of virtualization is that a piece of software will function as a physical object, that is, it will “look” and “behave” like hardware. Thus, it will perform all of the functions that a piece of hardware performs without the hardware in place. As such, the software emulates a desktop PC on a server.

And this, in fact, is what cloud-based IT service provides – a place where business functions can occur and be stored without the need for in-house hardware.

How Virtualization is Different from Cloud Computing

Virtualization software allows multiple operating systems and applications to run on the same server at the same time, and, as a result, lowers costs and increases efficiency of a company’s existing hardware. It’s a fundamental technology that powers cloud computing.

Virtualization thus emulates hardware. Cloud computing is a service that results from that manipulation and is an external service. Cloud computing almost always assumes virtualization of certain resources (storage or data) that will be then delivered to the customer on-demand.

The Main Types of Virtualization

There are several types of virtualization, categorized according to the elements they are used on.

1. Server Virtualization

Server space is conserved through by consolidating multiple machines into a single server that then runs multiple virtual environments. It's a method by which businesses can run the same applications on multiple servers, so that there is a "failsafe" position. Because each server is independent, running software on one will not affect the other. Another emerging trend in server virtualization is migration. A server environment can be moved from one place to another, even if the machines have different operating systems. The obvious benefit is the savings on hardware.

2. Storage Virtualization

Disk storage used to be a simple matter. If a business needed more, it simply purchased a larger disk drive. But storage needs continue to grow, and managing them becomes much harder. Virtualization is a great answer. It adds an additional layer of software between systems and servers, and applications no longer need to know where specific data resides. It is managed as if it is a single resource. Servers will see the virtualization layer as one single storage device, and each individual storage device sees the layer as its only server.

3. Network Virtualization

This type of virtualization allows management and monitoring of an entire network as a single entity. Primarily, it is designed to automate administrative tasks, disguising the complexity of the network. Each server (and service) is considered a part of one pool of resources to be used without worry about its physical components.

Understanding the Advantages of Virtualization

The best way to think about the role of virtualization is to understand the difference between private and public clouds. Basically, in a private cloud environment, a business owns/leases both the hardware and the software that provides the service consumption. This is in-house virtualization, and the business maintains full management and control.

The public cloud environment is one in which all of the virtualization is housed somewhere else, and a vendor provides the service to clients on a fee basis. In the public cloud, there are "co-tenants" in the same cloud, and clients pay for the specific services they use, as they use them.

The Private Cloud

A private cloud is thus its own virtualized world. It gives users more control, along with flexibility to manage their own systems, while still having all of the benefits of the cloud. Plus, the owner does not have to worry about coexisting "bad neighbors" or possible slowdowns in performance. Virtualization results in the following benefits:

Maximizing existing resources: Virtualization will allow a user to keep physical systems to a minimum, getting greater value out of existing servers.

Notes

Running multiple applications and their operating systems on the same hardware.

Costs are direct but are fixed. All costs for management, administration, and other requirements are within the in-house IT budget.

For a business to consider whether to use virtualization (a private cloud), it must consider who will be providing the support and how will it be integrated with other in-house systems. Cost (operational expenditures), of course, is a consideration. How much management is a business willing to do? What about scalability and security needs?

In general, businesses that need greater control and security and that have large IT staffs for these purposes will probably find virtualization preferable.

The Public Cloud

Virtualized services through a public cloud environment are usually preferable for businesses that have smaller IT staffs and that tend to have fewer security concerns. A cloud-based solution will provide the following:

IT is basically outsourced. Because there is a service provider, administration and supportive services are taken care of elsewhere. In-house IT staff remain available for other business purposes.

Setup is easy and fast. And servers, hardware, and software licenses are eliminated.

Pay-as-you-use. Cloud-based services are charged based upon scope of use, and, while they can sometimes seem pricey, businesses do not have to put dollars into supportive products (spam/anti-virus resources, data archiving, encryption, off-site storage, etc.)

Scalability. Cloud services allow both permanent and temporary scaling. Thus, a business can off-load high-demand requirements at any time, even on a temporary basis, and pay only for the time of that off-load.

It's important to note that virtualization via private cloud or the move to cloud computing services are not mutually exclusive nor are they competitive.

Many businesses have in-house virtualization for some functions and move to the cloud for others. Still others who begin with virtualization of their own servers may ultimately end up in the cloud, as an evolutionary matter. They simply want more service delivery, scale, and agility.

Private Cloud Virtualization: Advantages and Disadvantages

As discussed above, there are several advantages of private cloud virtualization – in-house control and the flexibility to manage one's own systems, being the most important. And the cost benefits are obvious as well – minimizing the need for physical systems.

Taking a more specific look at the advantages and disadvantages will provide CIO's with the information they need as they make decisions about virtualization.

Advantages

Businesses that "live" in a regulatory environment (e.g. financial services, health) have critical data and protection responsibilities. Building virtualization infrastructures

themselves rather than sharing them with others in a public cloud, can raise issues.

Likewise, companies that have data which they wish to remain confidential (e.g., research), can feel a bit better about in-house virtualization, in which they can protect that data. No other company has access to that infrastructure.

Private Cloud Virtualization has greater reliability. When public clouds are considered, potential users must conduct solid research to determine if the server they select can provide premiere performance for the types of applications and services they need. In building a private cloud, predictable and reliable service for users is generally most assured.

Cost and Flexibility. There are always trade-offs when implementing new hardware and software. In the case of a private cloud, the initial expense of installing servers and storage can be high. On the other hand, great flexibility can be built in so that workloads can easily be shifted during peak usage spikes and when new applications are deployed. There is no need to make a request of a cloud service provider, before changes can be accomplished.

Disadvantages

No software or hardware solution is perfect, and that is certainly the case with private cloud virtualization. Before building and deploying, there are disadvantages to be considered:

Integration with other in-house systems can be an issue.

Managing and supporting virtualization will often require dedicated IT staff, and that may bring costs up, if there is already not a good-sized department. This is the primary reason why smaller businesses opt for external cloud services.

Scaling and security will require specific expertise.

Best practices for server virtualization

Implementing virtualization will require a lot of thoughtful planning, to get the results a business wants – efficiency, cost-savings, etc. Here is a list of some best practices before, during, and after deployment.

Both the host OS and the virtual machines running at the same time must have enough memory. There should be enough for a virtual machine to run its operating system, as well as that of the host OS.

Enough disk space is also critical, including enough for each virtual machine's paging file and for saving each machine's RAM contents.

Identify where large memory support will be needed and enable that.

Consider disabling threading technology. Otherwise, under heavy computing workload events, performance can suffer.

When virtual machines require lots of network activity, there can be a burden on the adapters on the physical computer. Consider configuring two network adapters, one of which is only for use by the operating system.

Be sure to shut down all virtual machines when making any changes.

Notes

Write protect any parent disk.

Anticipate and plan for high disk use, so that there is no “disk time-out.”

Secure all virtual machines just as you would a physical machine. This means enabling a firewall, applying security patches, antivirus software, and lockdown procedures as may be suitable.

Be certain to shut down all virtual machines before shutting down the host server. If not, you could lose data.

Keep an event log.

This is by no means an exhaustive list, but it is a start. Any IT staff that will be managing virtualization should stay current in the latest best practice innovations and trends.

If done right, virtualization can transform business IT tasks – creating an environment of efficiency, of maximizing existing resources, of saving the costs, of continually upgrading hardware systems, and of bringing greater security. It takes planning, the right expertise, and a commitment to ongoing management and administration though.

2.2 Steps Involved in Transitioning from Classic Data Center to Cloud Computing Environment

It's widely understood that cloud-based environments can bring significant benefits to enterprises including improved cost savings and business agility. In fact, 82 percent of US businesses reportedly saved money by moving workloads to the cloud last year. In its 2014 State of the Cloud report, RightScale reveals that enterprises are adopting cloud computing in record numbers: Nearly all organizations (94 percent) surveyed are running applications or experimenting with Infrastructure as a Service (IaaS), while as many as 87 percent of organizations are using public cloud.

However, before businesses make the transition to a cloud-based environment, the challenges associated with the actual onboarding of enterprise application workloads are often overlooked. For businesses migrating to a cloud-based environment, 'onboarding' refers to the deployment of application workloads – or virtual machines and the applications that run on them – into a virtual private cloud or public cloud environment, without the need for major refactoring or rebuilding an enterprise application.

A significant challenge with onboarding is that, for most enterprises, their applications are not designed for a cloud environment which is designed to scale on-demand and has failover capabilities built-in to the infrastructure. Rather, they are designed for the reliability of a dedicated environment.

Not surprisingly, the first applications enterprises typically choose to onboard are those already running in a virtualized environment. But not all virtualized applications are 'cloud friendly.' As such, enterprises must commit to early planning in order to prepare their workloads for cloud environments. This application onboarding process has 7 essential steps, which are outlined below.

Step 1: Define the workload

An application workload comprises all of the components required for the proper performance of an application. As hybrid emerges as the key cloud model, it's

increasingly uncommon for businesses to run single purpose, custom-built applications like Zynga and Netflix do. Rather, they use a combination of off-the-shelf applications like Oracle and SAP, which means their applications usually interact with one another across platforms. For instance, a logistics system which manages deliveries might be integrated with a CRM system. Also, not all of these elements will be migrated to a cloud environment – it's very common for enterprises to maintain their Active Directory in-house, for instance.

For businesses considering migrating to cloud, they must undergo a thorough workload analysis early on. Preparing an application workload for forklifting into the cloud will require virtualization if the application is not already running in a virtualized environment. Ultimately, the nature and scale of the workload will dictate the number and type of virtual machines necessary for migration.

Step 2: Provision cloud resources

Servers, storage and network are required services for cloud environments, and service providers can help enterprises purchase these required amenities. To determine necessary resources, businesses should ask themselves: "What OS, databases and application servers are being used currently and how hard are they to migrate to the cloud?" "What are the CPU, memory, network and storage requirements and what will it cost to provide these in a cloud environment?" "What other software supports the workload?" and "What are the integration touch points with other workloads?"

Step 3: Establish a connectivity bridge

The connectivity bridge is the secure and transparent bi-directional connectivity between the data center and the cloud, which is essential for enabling cloud access. Most enterprises employ Internet VPNs for connectivity to the cloud. However, Internet connectivity is not suitable for all applications. There are alternative, non-Internet connectivity options that enterprises can deploy, such as going through Ethernet (layer 2). Before choosing an alternative, enterprises will need to ensure their cloud providers can support them.

Step 4: Deploy the workload

Once the connectivity bridge is in place, businesses can setup their virtual machines and connect them to applications that remain in-house. Then, enterprises can transfer applications and any associated databases, software and services that they established from Step 2 as entrants for cloud migration.

Step 5: Ensure seamless two-way access

Critical to the success of any cloud migration is smooth integration between the cloud workload and the services not migrated, which remain on premise. With the hybrid model emerging as the dominant cloud environment for enterprises, IT will have additional considerations, including connecting multiple cloud environments as well as managing enterprise-user devices that access applications in the hybrid environment. It's therefore imperative that the two-way connectivity is fluid and secure and enables IT to monitor and manage the applications as well as the cloud infrastructure.

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Step 6: Test and validate

To ensure applications will perform as required it's important to conduct a trial period before launch. Businesses should ensure the application can recover from failure and test all third-party components. They should ask themselves: "Has everything been transferred correctly?" "Do network, storage, compute and database configurations remain intact?" "Can I see and manage the cloud environment properly?" and "Does my cloud backup process work?"

Step 7: Discontinue the old service

Once enterprises have identified and amended any performance glitches, they can then terminate the enterprises services and give all of the users cloud access instead.

2.3. Business Agility: Benefits and Challenges to Cloud Architecture

Agility is the ability of a business to adapt rapidly and cost efficiently in response to changes in the business environment. Business agility can be achieved by quickly adapting goods and services to meet customer demands. Basically, agility is a concept that incorporates the ideas of flexibility, balance, adaptability, and coordination under one umbrella.

In an increasingly challenging business environment, enterprises can no longer expect to thrive on the basis of existing business and IT strategies. Unlike the time-consuming, expensive application modernization projects of a decade ago, transformation initiatives based on social, mobile, analytics and the cloud are cost-effective and efficient. The cloud, in particular, provides a natural platform for optimizing existing IT systems to increase operational efficiencies while driving business agility and growth.

So, let's have a look at how cloud computing strategy drives business agility:

- ◇ **Faster time-to-market** – cloud computing allows companies to significantly decrease the time it takes to provision and deprovision IT infrastructure, speeding delivery of IT projects that are critical to revenue growth or cost reduction. While a physical server could take days or weeks to procure and provision, a cloud server takes minutes.
- ◇ **Automation** – cloud computing simplifies provisioning, de-provisioning and re-deploying resources through automation. The efficiency of cloud computing reduces the amount of time an IT systems administrator has to spend on managing and supporting infrastructure.
- ◇ **Pay for the number of resources you use and need** – There is no denial that cloud computing increases the efficiency and speed of business processes and transactions. The pay-per-use model of cloud computing offers its subscribers the opportunity to either rent or subscribe to the platforms, systems, infrastructure and data services which can be rapidly scaled up or down on need basis. This eliminates the risk of committing a lot of finances on insignificant or less useful resources which are required to store these entities themselves.
- ◇ **Faster innovation** – cloud computing allows companies to support an increased pace of product development and marketing programs that better align IT infrastructure and management costs with the goals and objectives of the business. The flexible infrastructure that cloud offers translates into

businesses enhancing their agility rather than being inhibitors of the change. It allows organizations to step up or tone down their operations to support their business goals such as attracting and retaining new customers or speeding up the time-to-market for latest services.

- ◇ **No need to wait for hardware and software resources** – The cloud makes everything accessible via the Internet. Conventional business processes need weeks or months at times to set up hardware and develop software. On the contrary, such setup can be ready in just a matter of few hours in the cloud.
- ◇ **Flexibility and scalability** – Due to the pay-per-use flexibility of the cloud, end-users are able to scale fast based on the demands of the business. Among the common workloads that require on-demand scalability we can name testing and development, load testing, seasonal spikes in traffic, a new application etc.

Cloud Architecture refers to the various components in terms of databases, software capabilities, applications, etc. engineered to leverage the power of cloud resources to solve business problems. Cloud architecture defines the components as well as the relationships between them.

The various components of Cloud Architecture are:

- ◇ On premise resources
- ◇ Cloud resources
- ◇ Software components and services
- ◇ Middleware

The entire cloud architecture is aimed at providing the users with high bandwidth, allowing users to have uninterrupted access to data and applications, on-demand agile network with possibility to move quickly and efficiently between servers or even between clouds and most importantly network security

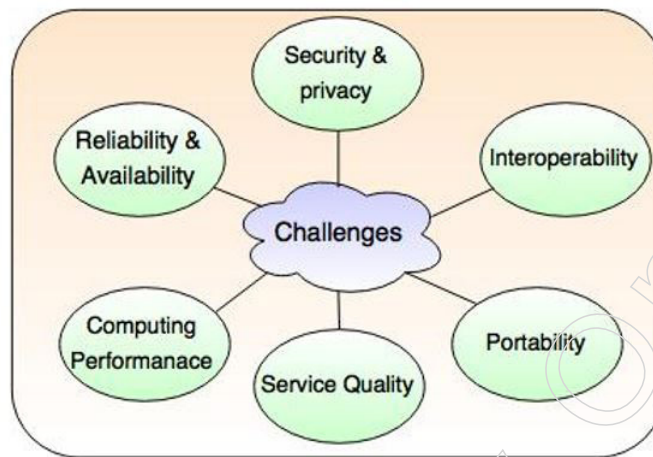
Benefits in Cloud Architecture

- ◇ Cloud architecture uses simple APIs to provide easily accessible services to the user through the internet medium.
- ◇ It provides scale on demand feature to increase the industrial strength. It provides the transparency between the machines so that users don't have to worry about their data.
- ◇ Users can just perform the functionality without even knowing the complex logic's implemented in cloud architecture.

Challenges in Cloud Computing

Following diagram shows the major challenges in cloud computing.

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Challenges in Cloud Computing

Security and Privacy

- ◇ Security and privacy are the main challenge in cloud computing.
- ◇ These challenges can be reduced by using security applications, encrypted file systems, data loss software.

Interoperability

- ◇ The application on one platform should be able to incorporate services from the other platform. This is known as Interoperability.
- ◇ It is becoming possible through web services, but to develop such web services is complex.

Portability

- ◇ The applications running on one cloud platform can be moved to new cloud platform and it should operate correctly without making any changes in design, coding.
- ◇ The portability is not possible, because each of the cloud providers uses different standard languages for their platform.

Service Quality

- ◇ The Service-Level Agreements (SLAs) of the providers are not enough to guarantee the availability and scalability. The businesses disinclined to switch to cloud without a strong service quality guarantee.

Computing Performance

- ◇ High network bandwidth is needed for data intensive applications on cloud, this results in high cost.
- ◇ In cloud computing, low bandwidth does not meet the desired computing performance.

Reliability and Availability

- ◇ Most of the businesses are dependent on services provided by third-party, hence it is mandatory for the cloud systems to be reliable and robust.

Disadvantages of cloud Architecture

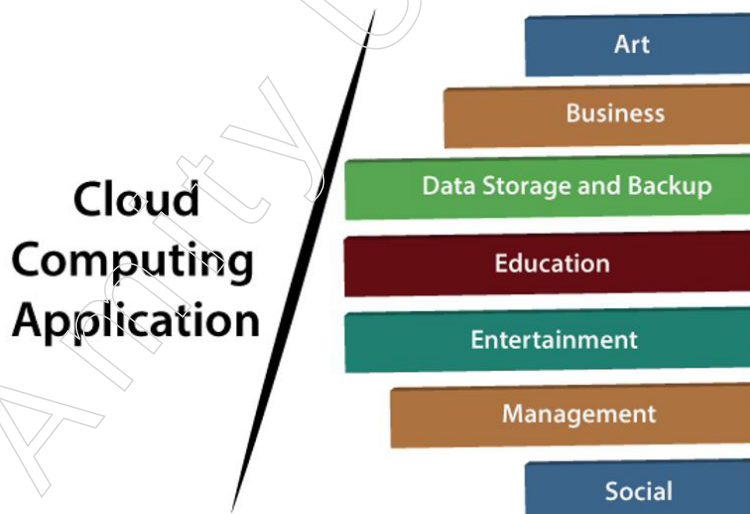
- ◇ Before looking at the disadvantages, let's check the advantages.
- ◇ First and foremost: cost. It's lower than having machines on premises. This is by far the main reason to go to the cloud. There is no justification to keep paying more when you can get the infrastructure you need for less.
- ◇ Plus it is not a Capex cost, no hard investment in infrastructure, and you pay for as much as you consume.
- ◇ Having said that, one possible disadvantage is that you have to manage your use of the cloud and manage your provider(s). But this disadvantage disappears when you implement Devops (Development + operation). Devops make it very clear that you must streamline your IT production, and this can be done by a third party, the cloud, at a lower cost.
- ◇ Second disadvantage: you put your data in the hands of a third party. What about security, privacy, confidentiality, contingency recovery? This has to be checked carefully. What we have today are cloud providers who comply with all the regulations, laws, restrictions, etc. Probably more than your company. Anyway, you have to carefully check, have good contracts, apply all the rules of good management of a third party, and divide your workload and data between more than one provider to be safe. This also applies to the communication lines with your cloud provider.

2.4 Application, Availability, Performance, Security and Disaster Recovery

Application:

Cloud service providers provide various applications in the field of art, business, data storage and backup services, education, entertainment, management, social networking, etc.

The most widely used cloud computing applications are given below -



1. Art Applications

Cloud computing offers various art applications for quickly and easily design attractive cards, booklets, and images. Some most commonly used cloud art applications are given below:

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i. Moo

Moo is one of the best cloud art applications. It is used for designing and printing business cards, postcards, and mini cards.

ii. Vistaprint

Vistaprint allows us to easily design various printed marketing products such as business cards, Postcards, Booklets, and wedding invitations cards.

iii. Adobe Creative Cloud

Adobe creative cloud is made for designers, artists, filmmakers, and other creative professionals. It is a suite of apps which includes PhotoShop image editing programming, Illustrator, InDesign, TypeKit, Dreamweaver, XD, and Audition.

2. Business Applications

Business applications are based on cloud service providers. Today, every organization requires the cloud business application to grow their business. It also ensures that business applications are 24*7 available to users.

There are the following business applications of cloud computing -

i. MailChimp

MailChimp is an email publishing platform which provides various options to design, send, and save templates for emails.

iii. Salesforce

Salesforce platform provides tools for sales, service, marketing, e-commerce, and more. It also provides a cloud development platform.

iv. Chatter

Chatter helps us to share important information about the organization in real time.

v. Bitrix24

Bitrix24 is a collaboration platform which provides communication, management, and social collaboration tools.

vi. Paypal

Paypal offers the simplest and easiest online payment mode using a secure internet account. Paypal accepts the payment through debit cards, credit cards, and also from Paypal account holders.

vii. Slack

Slack stands for Searchable Log of all Conversation and Knowledge. It provides a user-friendly interface that helps us to create public and private channels for communication.

viii. Quickbooks

Quickbooks works on the terminology "Run Enterprise anytime, anywhere, on any device." It provides online accounting solutions for the business. It allows more than 20 users to work simultaneously on the same system.

3. Data Storage and Backup Applications

Cloud computing allows us to store information (data, files, images, audios, and videos) on the cloud and access this information using an internet connection. As the cloud provider is responsible for providing security, so they offer various backup recovery application for retrieving the lost data.

A list of data storage and backup applications in the cloud are given below -

i. Box.com

Box provides an online environment for secure content management, workflow, and collaboration. It allows us to store different files such as Excel, Word, PDF, and images on the cloud. The main advantage of using box is that it provides drag & drop service for files and easily integrates with Office 365, G Suite, Salesforce, and more than 1400 tools.

ii. Mozy

Mozy provides powerful online backup solutions for our personal and business data. It schedules automatically back up for each day at a specific time.

iii. Joukuu

Joukuu provides the simplest way to share and track cloud-based backup files. Many users use joukuu to search files, folders, and collaborate on documents.

iv. Google G Suite

Google G Suite is one of the best cloud storage and backup application. It includes Google Calendar, Docs, Forms, Google+, Hangouts, as well as cloud storage and tools for managing cloud apps. The most popular app in the Google G Suite is Gmail. Gmail offers free email services to users.

4. Education Applications

Cloud computing in the education sector becomes very popular. It offers various online distance learning platforms and student information portals to the students. The advantage of using cloud in the field of education is that it offers strong virtual classroom environments, Ease of accessibility, secure data storage, scalability, greater reach for the students, and minimal hardware requirements for the applications.

There are the following education applications offered by the cloud -

i. Google Apps for Education

Google Apps for Education is the most widely used platform for free web-based email, calendar, documents, and collaborative study.

ii. Chromebooks for Education

Chromebook for Education is one of the most important Google's projects. It is designed for the purpose that it enhances education innovation.

iii. Tablets with Google Play for Education

It allows educators to quickly implement the latest technology solutions into the classroom and make it available to their students.

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iv. AWS in Education

AWS cloud provides an education-friendly environment to universities, community colleges, and schools.

5. Entertainment Applications

Entertainment industries use a multi-cloud strategy to interact with the target audience. Cloud computing offers various entertainment applications such as online games and video conferencing.

i. Online games

Today, cloud gaming becomes one of the most important entertainment media. It offers various online games that run remotely from the cloud. The best cloud gaming services are Shaow, GeForce Now, Vortex, Project xCloud, and PlayStation Now.

ii. Video Conferencing Apps

Video conferencing apps provides a simple and instant connected experience. It allows us to communicate with our business partners, friends, and relatives using a cloud-based video conferencing. The benefits of using video conferencing are that it reduces cost, increases efficiency, and removes interoperability.

6. Management Applications

Cloud computing offers various cloud management tools which help admins to manage all types of cloud activities, such as resource deployment, data integration, and disaster recovery. These management tools also provide administrative control over the platforms, applications, and infrastructure.

Some important management applications are -

i. Toggl

Toggl helps users to track allocated time period for a particular project.

ii. Evernote

Evernote allows you to sync and save your recorded notes, typed notes, and other notes in one convenient place. It is available for both free as well as a paid version.

It uses platforms like Windows, macOS, Android, iOS, Browser, and Unix.

iii. Outright

Outright is used by management users for the purpose of accounts. It helps to track income, expenses, profits, and losses in real-time environment.

iv. GoToMeeting

GoToMeeting provides Video Conferencing and online meeting apps, which allows you to start a meeting with your business partners from anytime, anywhere using mobile phones or tablets. Using GoToMeeting app, you can perform the tasks related to the management such as join meetings in seconds, view presentations on the shared screen, get alerts for upcoming meetings, etc.

7. Social Applications

Social cloud applications allow a large number of users to connect with each other using social networking applications such as Facebook, Twitter, LinkedIn, etc.

There are the following cloud based social applications -

i. Facebook

Facebook is a social networking website which allows active users to share files, photos, videos, status, more to their friends, relatives, and business partners using the cloud storage system. On Facebook, we will always get notifications when our friends like and comment on the posts.

ii. Twitter

Twitter is a social networking site. It is a microblogging system. It allows users to follow high profile celebrities, friends, relatives, and receive news. It sends and receives short posts called tweets.

iii. Yammer

Yammer is the best team collaboration tool that allows a team of employees to chat, share images, documents, and videos.

iv. LinkedIn

LinkedIn is a social network for students, freshers, and professionals.

Availability:

High availability is the ultimate goal of moving to the cloud. The idea is to make your products, services, and tools available to your customers and employees at any, time from anywhere using any device with an internet connection.

Cloud availability is related to cloud reliability.

For example, let's say you have an online store that is available 24/7. But sometimes clicking the "checkout" button kicks customers out of the system before they have completed the purchase. So, your store may be available all the time, but if the underlying software is not reliable, your cloud offerings are basically useless.

Bringing it all together- Cloud availability, cloud reliability, and cloud scalability all need to come together to achieve high availability. This means that your products and services are accessible anytime and anywhere, function reliably and as expected, and that the system can seamlessly scale up or down to accommodate customer demand without suffering a loss in performance.

Cloud service providers offer an Infrastructure as a Service (IaaS) model that gives you access to storage, servers, and other resources.

IaaS provides automation and scalability on demand so that you can spend your time managing and monitoring your applications, data, and other services.

Because IaaS provides scalability based on a pay-as-you-go model, this saves you money and frees you up to track down and address problems that may come up with the software. Having more time to monitor can help you find areas that need improvement so you can do a better job consistently deploying reliable products and services.

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To survive in today's global market, it's inevitable that your company will need to move to the cloud. It won't happen overnight and will require a lot of planning. As you plan what and how you will make solutions available in the cloud, remember that it is important that your products and services and cloud infrastructure are scalable, reliable, and available when and where they are needed.

Performance:

Cloud performance monitoring and testing tools help organizations gain visibility into their cloud environments, using specific metrics and techniques to assess performance.

Efficient cloud performance is critical for maintaining business continuity and ensuring all relevant parties gain access to cloud services. This is true for basic cloud usage of public clouds and complex hybrid cloud and multi-cloud architectures.

Cloud performance metrics enable you to effectively monitor your cloud resources, to ensure all components communicate seamlessly. Typically, cloud performance metrics measure input/output operations per second (IOPS), filesystem performance, caching, and autoscaling.

Security:

Cloud security, also known as cloud computing security, consists of a set of policies, controls, procedures and technologies that work together to protect cloud-based systems, data, and infrastructure. These security measures are configured to protect cloud data, support regulatory compliance and protect customers' privacy as well as setting authentication rules for individual users and devices. From authenticating access to filtering traffic, cloud security can be configured to the exact needs of the business. And because these rules can be configured and managed in one place, administration overheads are reduced and IT teams empowered to focus on other areas of the business.

The way cloud security is delivered will depend on the individual cloud provider or the cloud security solutions in place. However, implementation of cloud security processes should be a joint responsibility between the business owner and solution provider.

Why is cloud security important?

For businesses making the transition to the cloud, robust cloud security is imperative. Security threats are constantly evolving and becoming more sophisticated, and cloud computing is no less at risk than an on-premise environment. For this reason, it is essential to work with a cloud provider that offers best-in-class security that has been customized for your infrastructure.

Cloud security offers many benefits, including:

Centralized security: Just as cloud computing centralizes applications and data, cloud security centralizes protection. Cloud-based business networks consist of numerous devices and endpoints that can be difficult to manage when dealing with shadow IT or BYOD. Managing these entities centrally enhances traffic analysis and web filtering, streamlines the monitoring of network events and results in fewer software and policy updates. Disaster recovery plans can also be implemented and actioned easily when they are managed in one place.

Reduced costs: One of the benefits of utilizing cloud storage and security is that it eliminates the need to invest in dedicated hardware. Not only does this reduce capital expenditure, but it also reduces administrative overheads. Where once IT teams were firefighting security issues reactively, cloud security delivers proactive security features that offer protection 24/7 with little or no human intervention.

Reduced Administration: When you choose a reputable cloud services provider or cloud security platform, you can kiss goodbye to manual security configurations and almost constant security updates. These tasks can have a massive drain on resources, but when you move them to the cloud, all security administration happens in one place and is fully managed on your behalf.

Reliability: Cloud computing services offer the ultimate in dependability. With the right cloud security measures in place, users can safely access data and applications within the cloud no matter where they are or what device they are using.

More and more organizations are realizing the many business benefits of moving their systems to the cloud. Cloud computing allows organizations to operate at scale, reduce technology costs and use agile systems that give them the competitive edge. However, it is essential that organizations have complete confidence in their cloud computing security and that all data, systems and applications are protected from data theft, leakage, corruption and deletion.

All cloud models are susceptible to threats. IT departments are naturally cautious about moving mission-critical systems to the cloud and it is essential the right security provisions are in place, whether you are running a native cloud, hybrid or on-premise environment. Cloud security offers all the functionality of traditional IT security, and allows businesses to harness the many advantages of cloud computing while remaining secure and also ensure that data privacy and compliance requirements are met.

Disaster recovery:

Data is the most valuable asset of modern-day organizations. Its loss can result in irreversible damage to your business, including the loss of productivity, revenue, reputation, and even customers. It is hard to predict when a disaster will occur and how serious its impact will be. However, what you can control is the way you respond to a disaster and how successfully your organization will recover from it. Get to discover post how you can use disaster recovery in cloud computing for your benefit.

This blog post attempts to answer the following questions:

How does disaster recovery in cloud computing differ from traditional disaster recovery? – Traditional disaster recovery involves building a remote disaster recovery (DR) site, which requires constant maintenance and support on your part. In this case, data protection and disaster recovery are performed manually, which can be a time-consuming and resource-intensive process. Disaster recovery in cloud computing entails storing critical data and applications in cloud storage and failing over to a secondary site in case of a disaster. Cloud computing services are provided on a pay-as-you-go basis and can be accessed from anywhere and at any time. Backup and disaster recovery in cloud computing can be automated, requiring minimum input on your part.

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How does disaster recovery planning work in cloud computing? – Creating, testing, and updating a DR plan can prepare your organization for an unexpected disaster and ensure safety and continuity for your business. A comprehensive DR plan should take into account your infrastructure, potential threats and vulnerabilities, most critical assets and the order of their recovery, and workable DR strategies. Integration of cloud computing services in disaster recovery allows you to design a DR plan and automate each step of the recovery process.

Backup and Disaster Recovery in Cloud Computing

Cloud computing is the on-demand delivery of computing services over the internet (more often referred to as 'the cloud') which operates on a pay-as-you-go basis. Cloud computing vendors generally provide access to the following services:

Infrastructure as a service (IaaS) allows you to rent IT infrastructure, including servers, storages and network component, from the cloud vendor.

Platform as a service (PaaS) allows you to rent a computing platform from the cloud provider for developing, testing, and configuring software applications.

Software as a service (SaaS) allows you to access software applications which are hosted on the cloud.

As you can see, each cloud computing service is designed to help you achieve different business needs. More so, cloud computing can considerably improve data the security and high availability of your virtualized workloads. Let's discuss how you can approach disaster recovery in the cloud computing environment.

Cloud disaster recovery vs. traditional disaster recovery

Cloud disaster recovery is a cloud computing service which allows for storing and recovering system data on a remote cloud-based platform. To better understand what disaster recovery in cloud computing entails, let's compare it to traditional disaster recovery.

The essential element of traditional disaster recovery is a secondary data center, which can store all redundant copies of critical data, and to which you can fail over production workloads. A traditional on-premises DR site generally includes the following:

A dedicated facility for housing the IT infrastructure, including maintenance employees and computing equipment.

Sufficient server capacity to ensure a high level of operational performance and allow the data center to scale up or scale out depending on your business needs.

Internet connectivity with sufficient bandwidth to enable remote access to the secondary data center.

Network infrastructure, including firewalls, routers, and switches, to ensure a reliable connection between the primary and secondary data centers, as well as provide data availability.

However, traditional disaster recovery can often be too complex to manage and monitor. Moreover, support and maintenance of a physical DR site can be extremely expensive and time-consuming. When working with an on-premises data center, you

can expand your server capacity only by purchasing additional computing equipment, which can require a lot of money, time, and effort.

Disaster recovery in cloud computing can effectively deal with most issues of traditional disaster recovery. The benefits include the following:

You don't need to build a secondary physical site, and buy additional hardware and software to support critical operations. With disaster recovery in cloud computing, you get access to cloud storage, which can be used as a secondary DR site.

Depending on your current business demands, you can easily scale up or down by adding required cloud computing resources.

With its affordable pay-as-you go pricing model, you are required to pay only for the cloud computing services you actually use.

Disaster recovery in cloud computing can be performed in a matter of minutes from anywhere. The only thing you need is a device that is connected to the internet.

You can store your backed up data across multiple geographical locations, thus eliminating a single point of failure. You can always have a backup copy, even if one of the cloud data centers fails.

State-of-the-art network infrastructure ensures that any issues or errors can be quickly identified and taken care of by a cloud provider. Moreover, the cloud provider ensures 24/7 support and maintenance of your cloud storage, including hardware and software upgrades.

Why Choose Disaster Recovery in Cloud Computing

The primary goal of disaster recovery is to minimize the overall impact of a disaster on business performance. Disaster recovery in cloud computing can do just that. In case of disaster, critical workloads can be failed over to a DR site in order to resume business operations. As soon as your production data center gets restored, you can fail back from the cloud and restore your infrastructure and its components to their original state. As a result, business downtime is reduced and service disruption is minimized.

Due to its cost-efficiency, scalability, and reliability, disaster recovery in cloud computing has become the most lucrative option for small and medium-sized businesses (SMBs). Generally, SMBs don't have a sufficient budget or resources to build and maintain their own DR site. Cloud providers offer you access to cloud storage, which can become a cost-effective and long-lasting solution to data protection as well as disaster recovery.

How to Design a Cloud-Based Disaster Recovery Plan

After considering the benefits of cloud computing in disaster recovery, it is time to design a comprehensive DR plan. In fact, you can read one of our blog posts which walks you through the entire process of creating a DR plan. Below, we are going to discuss how to create a DR plan which works in the cloud environment.

As a rule, an effective cloud-based DR plan should include the following steps:

Perform a risk assessment and business impact analysis.

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Choose prevention, preparedness, response, and recovery measures.

Test and update your cloud-based DR plan.

Let's discuss how disaster recovery planning works in cloud computing.

Perform a risk assessment and business impact analysis

The first step in a disaster recovery planning in cloud computing is to assess your current IT infrastructure, as well as identify potential threats and risk factors that your organization is most exposed to.

A risk assessment helps you discover vulnerabilities of your IT infrastructure and identify which business functions and components are most critical. At the same time, a business impact analysis allows you to estimate how unexpected service disruption might affect your business.

Based on these estimations, you can also calculate the financial and non-financial costs associated with a DR event, particularly Recovery Time Objective (RTO) and Recovery Point Objective (RPO). The RTO is the maximum amount of time that IT infrastructure can be down before any serious damage is done to your business. The RPO is the maximum amount of data which can be lost as a result of service disruption. Understanding the RTO and RPO can help you decide which data and applications to protect, how many resources to invest in achieving DR objectives, and which DR strategies to implement in your cloud-based DR plan.

2.5 Next Generation Cloud Application

Expand your knowledge regarding the board aspects of cloud computing. This will help enterprises to understand that why it is essential to move data online for the development of their firm. The post is going to provide a brief picture of four next-gen cloud computing technologies with their live examples that exist today.

It is well-known that the ecosystem of online computing technology is vast. The technologies multitude contains the cloud architecture on which the company relies to share/use applications and services in an efficient way and at a massive range. Either in an indirect or a direct manner, organizations are making use of open source tools for maximizing operations and development & keeping their goodwill in the changing market scenario.

What is the Future in Hands of Cloud Computing?

In today's date, everything is connected with the cloud through one or another means. Many predictions are made by people regarding the future of online computing because it is having capability of opening doors for newer platforms, applications, etc. Infinite number of possibilities pave the direction to create and implement innovative ideas. In the next-generation, cloud computing technology role is going to be as an integral element in the life of each human being. It is so because this is the only platform where all devices can be connected to one place. In the next section, one is going to find next-gen cloud computing technologies that will shape your mind with its future scope.

Technologies In Next-Gen Cloud Computing

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1. Unikernels – These are the specialized operating systems, which render enhanced security, fine-grained optimization, and a smaller footprint required for micro-services. They are made up of the library OS technology and can be customized on the basis of different programs and hardware. Unikernels are in the form of executable image, which can be natively executed on specific hypervisor. It does not require any extra supporting OS.

Unikernels comprises of library OS that is nothing just a library collection, which represents the core capability of an operating system. For example – MirageOS is a library operating system, which develops unikernels for networking purpose over a variety of online computing and mobile environments. Another example of Unikernel OS can be Rumpun Unikernel. This operating system comprises of thousands of coding lines and works with POSIX application directly on the raw hardware. It also supports working on the cloud hypervisors like Xen and KVM.

2. Blockchain – Blockchain technology is a new face of internet where digital data is distributed without copying. Information held in the blockchain appears like sharing, having numerous advantages of its use. It is impossible for an individual person to hold blockchain because it does not have a single failure point. The network lives in a 'consensus' mode where a self-auditing kind of ecosystem is available. This system reconciles each transaction, which is accomplished in 10 minutes of intervals.

Well, the very first application that came in the form of blockchain technology is 'bitcoin' in the year 2009. Bitcoin is a cryptocurrency and, it is underpinned by the blockchain. This next-gen cloud computing technology eliminates the involvement of a human in processing cross-border trades. These type of systems can be set up as smart payments or contracts that prove themselves helpful at the time when it is about meeting certain set of conditions.

3. Container-as-a-Service – CaaS (or Container-as-a-Service) is a service provided by the cloud providers who render container orchestration and computing resources. The framework can be utilized by IT industry developers through web interface or API for easy container management. This new generation of cloud computing technology can be considered as a new layer in cloud platform to deploy application. It indicates towards the software that is purposed to give relief from the stress between operational and developmental team in a business. CaaS is useful at the time of pushing application data and monitoring program.

The tools under Container-as-a-Service category simplifies management and renders a framework not only to define the initial deployment level of the container but, also to manage several containers as a single thing. The whole and sole aim of these tools are to deal with scaling, networking, and availability. Azure Container Service, Google Container Engine, Cloud Foundry's Diego, etc., are the live example of this next-gen cloud computing technology.

4. Software Defined Networking – Depending upon the providers and users, the meaning of this term is different. In general, it is a key component in data centers for the automation purpose. SDN renders efficient methods for managing

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virtualization that saves the extra costing in hardware implementation. Managers of the data center have the right to manage every aspect associated with the data center to upgrade their hardware as per the requirements. The world of digitization already comprises of several issues to maintain stability in the market therefore, automated software becomes important. These automated tools eliminate the complications that are faced while managing activities. It helps organizations in enhancing their cloud data security by reducing the human errors.

Summary

- Cloud computing is having a bright scope in future because it is holding several technical barriers and knows solution to deal with them. No human involvement is required for dealing with such type of hurdles. The constant change is the need of today's digitization world and no organization can stay in the market with making updation. The concept of next-gen cloud computing technology has already been adopted by huge firms and now it is your time to upgrade yourself.
- Cloud computing is based on the Internet cloud and depends on Internet access.
- Cloud computing today is attracting the best and biggest companies from across the computing industry.
- Peak computing needs are easily handled by computers and servers in the cloud.
- Money is always tight on college campuses, and a few hundred dollars savings per student adds up quickly.

Keywords

Cloud Backup: Backing up data to a remote, cloud-based server. The data is stored in and accessed from a network of interconnected resources, forming the aforementioned cloud.

Cloud Computing: The delivery of IT services on-demand over shared networked computing resources. The top three cloud offerings are IaaS, PaaS, and SaaS.

Infrastructure: The catchall term describing all IT resources, both virtual and hardware that support a given IT environment.

PaaS: An acronym standing for Platform as a Service, this is a product where the cloud services provider offers users the necessary software and hardware for the creation, deployment, and management of applications, all via the Internet.

Virtualization: is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources

Self-Assessment Questions

- 1) What type of computing technology refers to services and applications that typically run on a distributed network through virtualized resources?
 - a. Distributed Computing
 - b. Cloud Computing

- c. Soft Computing
 - d. Parallel Computing
- 2) Which one of the following options can be considered as the Cloud?
- a. Hadoop
 - b. Intranet
 - c. Web Applications
 - d. All of the mentioned
- 3) Cloud computing is a kind of abstraction which is based on the notion of combining physical resources and represents them as _____resources to users.
- a. Real
 - b. Cloud
 - c. Virtual
 - d. none of the mentioned
- 4) Which of the following has many features of that is now known as cloud computing?
- a. Web Service
 - b. Softwares
 - c. All of the mentioned
 - d. Internet
- 5) Which one of the following cloud concepts is related to sharing and pooling the resources?
- a. Polymorphism
 - b. Virtualization
 - c. Abstraction
 - d. None of the mentioned
- 6) Which one of the following statements is not true?
- a. The popularization of the Internet actually enabled most cloud computing systems.
 - b. Cloud computing makes the long-held dream of utility as a payment possible for you, with an infinitely scalable, universally available system, pay what you use.
 - c. Soft computing addresses a real paradigm in the way in which the system is deployed.
 - d. All of the mentioned
- 7) Which one of the following can be considered as a utility is a dream that dates from the beginning of the computing industry itself?
- a. Computing
 - b. Model

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- c. Software
 - d. All of the mentioned
- 8) Which of the following is an essential concept related to Cloud?
- a. Reliability
 - b. Abstraction
 - c. Productivity
 - d. All of the mentioned
- 9) Which one of the following is Cloud Platform by Amazon?
- a. Azure
 - b. AWS
 - c. Cloudera
 - d. All of the mentioned
- 10) Which of the following statement is not true?
- a. Through cloud computing, one can begin with very small and become big in a rapid manner.
 - b. All applications benefit from deployment in the Cloud.
 - c. Cloud computing is revolutionary, even though the technology it is built on is evolutionary.
 - d. None of the mentioned

Review Questions

1. Explain Role of virtualization in enabling the cloud
2. What are the Steps involved in transitioning from Classic data center to Cloud computing environment?
3. Explain Business Agility with Benefits and challenges to cloud architecture.
4. Explain the terms Application, availability, performance, security and disaster recovery in cloud.
5. What do you understand by the Next generation Cloud Application?

Answers

- | | | | |
|-----|------|-----|-----|
| 1)B | 2)A | 3)C | 4)D |
| 5)B | 6)C | 7)A | 8)B |
| 9)B | 10)B | | |

Module-II

Notes

Structure:

- 3. Cloud Application
 - 3.1 Technologies and the processes required when deploying web services
 - 3.2 Deploying a web service from inside and outside a cloud architecture
 - 3.3 Advantages and disadvantages

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Unit -3: Cloud Application

Objective:

After studying this chapter, you will be able to:

- Explain the meaning Cloud Application
- Discuss the Technologies and the processes required when deploying web services
- Explain the Deploying a web service from inside and outside a cloud architecture
- Discuss the Advantages and Disadvantages

Introduction

Overview

Cloud applications are software that users access primarily through the internet, meaning at least some of it is managed by a server and not users' local machines. Cloud-native application development strategies help development teams design apps with consistent experiences (for front-end users and back-end operations teams) across any IT infrastructure: physical, virtual, or cloud-based.

A brief history (and future predictions)

This basic definition doesn't fully describe how cloud applications have reshaped markets and business models, though. If designed well, cloud applications can offer a user experience like a program installed entirely on a local machine, but with reduced resource needs, more convenient updating, and the ability to access functionality across different devices.

New cloud applications can become key sources of revenue, and cause disruptive shifts to markets and business models. Gartner forecasts \$110.5 billion revenue from worldwide cloud services in 2020, and additionally forecasts a rise to \$143.7 billion by 2022.

If you're interested in cloud applications, you may want to integrate one into your own architecture to support your teams, or develop your own cloud app for internal or customer use.

Definition of a Cloud Application

A cloud application is an Internet-based program where some, or all, of the processing logic and data storage is processed in the cloud. The user interacts with the application via a web browser or a mobile application, and the data processing is managed by a combination of the local device and a cloud computing solution. From the user's perspective, the cloud application behaves like a standard website, but the computing and data processing are handled by the cloud via an API (application program interface) or a hybrid of both.

A cloud application, or cloud app, is a software program where cloud-based and local components work together. This model relies on remote servers for processing logic that is accessed through a web browser with a continual internet connection.

Cloud application servers typically are located in a remote data center operated by a third-party cloud services infrastructure provider. Cloud-based application tasks may encompass email, file storage and sharing, order entry, inventory management, word processing, customer relationship management (CRM), data collection, or financial accounting features.

Cloud Application Designs

Cloud applications are usually designed in one of three ways:

- ◇ A local device runs a browser, and the application looks like a classic web solution using HTML sent by the cloud server to the local device. In this case, the logic, the data and the formatting of the output are controlled by the cloud servers. A well-known example of such an application is DropBox, where the user interacts with the application in a web browser to upload and interact with their files, but all of the data processing and storage happens remotely on the cloud.
- ◇ The local device runs a browser, but at the startup of the application, it downloads a browser-based application, such as javascript, that runs local to the device. Then the local app handles some of the processing and makes API calls to the cloud servers for additional processing and data. An example of this is Google Docs. Once you've launched Google Docs, you can configure the system to allow you to continue editing a document even after losing connection to the Internet. It will store the changes locally and sync with the cloud once the connection is restored.
- ◇ The local device runs a native application handling some of the processing, and the native application uses APIs to send and receive data to the application on the cloud server. An example of this is when a smartphone app is loaded onto your phone which runs much of the program logic. These are often designed so that if the connection is lost, they can still function by storing the changes locally. Once the connection is restored the data will be synced with the cloud.

Cloud Application Categories

Generally, cloud application services fall into one of the following three categories:

- ◇ **SaaS** – Software as a Service: The most common category and the one that most users are already familiar with, this is a third party cloud application where the third party not only supplies the hardware for running the application, but also the software application. Additionally, they provide complete support of the systems and the application. An excellent example of this would be Google's G Suite that includes Gmail, Google Drive, Google Docs, etc. Using a SaaS solution eliminates the need to buy expensive equipment, expensive licensing and they provide support for the hardware and software.
- ◇ **IaaS** – Infrastructure as a Service: IaaS is where a third party provides the infrastructure and infrastructure support in the cloud, but the software designer supplies the middleware, the applications, and the application support. Middleware is the software between the operating system and the application, used to give extended features to the operating system functions

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and simplify programming. An example of this is Google Compute Engine, a sub-product of the Google Cloud Platform, which allows developers access to the same infrastructure that Google uses in their products on demand to handle spikes in usage.

- ◇ **PaaS** – Platform as a Service: PaaS is similar to IaaS, but it also includes some of the software, middleware, and operating systems. The PaaS supplier takes care of supporting the hardware and the software they provide. They also keep the software up to date. The developer provides the applications and the application support. Google App Engine, another sub-product of the Google Cloud Platform, is an example of a hosting service that can scale to the needs of the application and is the preferred solution for cloud application development by Cloudbakers.

Benefits of cloud apps

Fast response to business needs. Cloud applications can be updated, tested and deployed quickly, providing enterprises with fast time to market and agility. This speed can lead to culture shifts in business operations.

Simplified operation. Infrastructure management can be outsourced to third-party cloud providers.

Instant scalability. As demand rises or falls, available capacity can be adjusted.

API use. Third-party data sources and storage services can be accessed with an application programming interface (API). Cloud applications can be kept smaller by using APIs to hand data to applications or API-based back-end services for processing or analytics computations, with the results handed back to the cloud application. Vetted APIs impose passive consistency that can speed development and yield predictable results.

Gradual adoption. Refactoring legacy, on-premises applications to a cloud architecture in steps allows components to be implemented on a gradual basis.

Reduced costs. The size and scale of data centers run by major cloud infrastructure and service providers, along with competition among providers, has led to lower prices. Cloud-based applications can be less expensive to operate and maintain than equivalent on-premises installations.

Improved data sharing and security. Data stored on cloud services is instantly available to authorized users. Due to their massive scale, cloud providers can hire world-class security experts and implement infrastructure security measures that typically only large enterprises can obtain. Centralized data managed by IT operations personnel is more easily backed up on a regular schedule and restored should disaster recovery become necessary.

How cloud apps work

Data is stored and compute cycles occur in a remote data center typically operated by a third-party company. A back end ensures uptime, security and integration and supports multiple access methods.

Cloud applications provide quick responsiveness and don't need to permanently reside on the local device. They can function offline, but can be updated online.

While under constant control, cloud applications don't always consume storage space on a computer or communications device. Assuming a reasonably fast internet connection, a well-written cloud application offers all the interactivity of a desktop application, along with the portability of a web application.

Cloud apps vs. web apps

With the advancement of remote computing technology, clear lines between cloud and web applications have blurred. The term cloud application has gained great cachet, sometimes leading application vendors with any online aspect to brand them as cloud applications.

Cloud and web applications access data residing on distant storage. Both use server processing power that may be located on premises or in a distant data center.

A key difference between cloud and web applications is architecture. A web application or web-based application must have a continuous internet connection to function. Conversely, a cloud application or cloud-based application performs processing tasks on a local computer or workstation. An internet connection is required primarily for downloading or uploading data.

A web application is unusable if the remote server is unavailable. If the remote server becomes unavailable in a cloud application, the software installed on the local user device can still operate, although it cannot upload and download data until service at the remote server is restored.

The difference between cloud and web applications can be illustrated with two common productivity tools, email and word processing. Gmail, for example, is a web application that requires only a browser and internet connection. Through the browser, it's possible to open, write and organize messages using search and sort capabilities. All processing logic occurs on the servers of the service provider (Google, in this example) via either the internet's HTTP or HTTPS protocols.

A CRM application accessed through a browser under a fee-based software as a service (SaaS) arrangement is a web application. Online banking and daily crossword puzzles are also considered web applications that don't install software locally.

An example of a word-processing cloud application that is installed on a workstation is Word's Microsoft Office 365. The application performs tasks locally on a machine without an internet connection. The cloud aspect comes into play when users save work to an Office 365 cloud server.

Cloud apps vs. desktop apps

Desktop applications are platform-dependent and require a separate version for each operating system. The need for multiple versions increases development time and cost, and complicates testing, version control and support. Conversely, cloud applications can be accessed through a variety of devices and operating systems and are platform-independent, which typically leads to significant cost savings.

Every device on a desktop application requires its own installation. Because it's not possible to enforce an upgrade whenever a new version is available, it's tricky to have all users running the same one. The need to provide support for multiple versions

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simultaneously can become a burden on tech support. Cloud applications don't face version control issues since users can access and run only the version available on the cloud.

Why Cloud Applications?

Now that you know what types of cloud applications are out there, what benefits can it bring you?

- ◇ **Keep Costs Down:** Since the majority of the processing power and storage is handled remotely, cloud applications could reduce the cost of your infrastructure - no need to maintain your own servers - as well as the cost of supporting your applications onsite. Depending on what software you are using locally, subscribing to a similar SaaS product could drastically reduce your licensing costs as well.
- ◇ **Accessibility:** Cloud applications aren't tied to a single machine - you can access and interact with the application from wide variety of devices safely and securely from any Internet connection.
- ◇ **Reliability:** Cloud applications have access to more computing resources than it would be feasible to have onsite. Your applications can rapidly scale without an increase in your capital costs. Better yet, this scaling can be dynamic so you'll only use the resources when you need them.
- ◇ **Standardization:** When applications are hosted on trusted partners such as Google, they are guaranteed to be reliable and accessible. It also ensures that your data is safely and securely backed up remotely. Furthermore, it's easy to ensure that everyone in your organization is working from the same version of the cloud application.

3.1. Technologies and the Processes Required when Deploying Web Services

Cloud computing is a next-generation technology based on the internet and network which provides services to the user in multiple ways. It is a simple data outsourcing resource and can be used temporarily also and it is cost-effective because clients can pay for what they use. To work on cloud applications, the requirements are fast internet connections and standard web browsers. It offers scalable access on-demand to the client instantly by sharing its pool resources to client web pages or IP.

Top Cloud Computing Technologies

The different innovation of cloud computing are listed below:

1. Virtualization

It is the process of sharing license keys to physical instances of application among different users of the enterprise. The main purpose of this technology is to provide a standard version of the cloud application to all clients. It is popularly used for its flexibility and instant running process. Few types of virtualization are

- ◇ **Hardware Virtualization:** If the virtual machine is directly installed on hardware servers then it is called Hardware virtualization because it is easy to control a Virtual Machine than a physical server.

- ◇ **Operating System Virtualization:** If the virtual machine is installed in a guest server instead of the hardware system, then it is called as Operating system Virtualization. It increased the testing situation of various software on various OS platforms,
- ◇ **Server Virtualization:** If the virtual machine is installed on the system it is called Server Virtualization. It is divided into multiple resources and utilized for load balancing on-demand basis.
- ◇ **Storage Virtualization:** The process of collecting physical storage from different network storage devices is called Storage Virtualization. It is mainly used for back-up and recovery.

2. Service-Oriented Architecture

SOA is an application which divides the services into individual business functions and procedure daily. This unique component of cloud application enables cloud-related arrangements that can be modified and adjusted on request as business needs. Service-oriented system diffuses two major components, one is Quality as service and other as software as service. The function of Quality of service is to identify the function and behavior of a service from a different view. Software as a service provides a new delivery model of software which is inherited from the world of application service providers

3. Grid Computing

This is a process of connecting multiple servers from multiple to achieve a common goal. Grid computing turns large problems into smaller ones and broadcast to servers and place them within the grid. It is mainly applied in e-commerce and intended to share the resources on huge scale cluster computing. A popular grid computing project is Folding@home. The project aims to find the protein folding, misfolding and related diseases. It involves utilizing the power of unused computers and solve complex scientific problems

4. Utility Computing

This process relies on the pay-per-utilize model. It gives computational services on demand for a metered benefit. It mainly helps in cost-cutting by reducing initial investment. As the computing requirements for a business change, the billing also changes accordingly, without acquiring any additional cost. If the client usage has decreased, then billing cost also reduces accordingly.

Benefits of Cloud Computing Technologies

The Advantages of cloud computing are as follows.

- ◇ **High scalability:** It requests the supply of resources on a large scale without any human intervention.
- ◇ **High availability and reliability:** The servers are available at the right time without any delay or disappointment
- ◇ **Agility:** It divides the resources effectively among the users and executes very quickly.

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- ◇ **Multi-sharing:** By distributed computing, different clients from multiple areas share the same resources by fundamental infrastructure.
- ◇ **Maintenance:** It is user-friendly as they are easy to access from their place and doesn't require any installation set up.
- ◇ **Low cost:** It is very cost-effective and pays to its usage.
- ◇ **Services as pay to use mode:** Application Programming Interface is given to customers to use resources and services and pay on the service basis.
- ◇ **On-Demand Self Service:** Cloud computing offers the required services and application to the client. With the login key, they can start to use without any human interaction and cloud service providers. It includes storage and virtual machines.

Practices for deploying your apps in the cloud

As a developer, you probably hear a lot about new technologies that promise to increase the speed at which you can develop software, as well as ones that can increase the resiliency of your applications once you have deployed them. Your challenge is to wade through these emerging technologies and determine which ones actually hold promise for the projects that you are currently working on.

No doubt, you are aware that cloud computing offers great promise for developers. However, you might not know about the areas where this technology can provide value to you and your projects. You also might not know good practices to employ when implementing a project in the cloud. This article explores the types of cloud computing systems available, and provides guidelines that can help you with real-world application deployments on top of a cloud infrastructure.

Choose between IaaS, PaaS, and SaaS

When people begin discussing cloud computing, they are generally speaking about one of three possible deployment choices for application code: infrastructure as a service (IaaS), platform as a service (PaaS), or software as a service (SaaS). Which one is right for your project depends on your specific needs for the code base that you are working on. Let's examine each one of these cloud choices.

Infrastructure as a service (IaaS)

IaaS is a platform where an infrastructure is provided for you. With a click of a button, you can spin up virtual machines hosted by a provider with an operating system of your choice. The vendor providing the machine is responsible for the connectivity and initial provisioning of the system, and you are responsible for everything else. The vendor provides a machine and an operating system, but you need to install all of the software packages, application runtimes/servers, and databases that your application requires. Generally, IaaS requires that you have a team of system administrators to manage the system and apply firewall rules, patches, and security errata on a frequent basis.

Pro: You have complete control over every aspect of the system.

Con: You need system administration knowledge or a team of system administrators to maintain the systems, since you are responsible for their uptime and security.

Platform as a service (PaaS)

PaaS is a fairly new technology stack that runs on top of IaaS and was created with the developer in mind. With the PaaS platform, everything is provided except the application code, users, and data. Typically, when using a PaaS, the vendor maintains the application server, databases, and all of the necessary operating system components, giving you time to focus on the application code. Since the vendor manages that platform for you, it is often hard to open up ports that are not specifically called for the application server, runtime, or database in use. PaaS also provides features that are specifically meant for applications, including the ability to scale the application tier up based upon the user demand of the application. In most platforms, this happens with little-to-no interaction from the developer.

Pro: PaaS provides a complete environment that is actively managed, letting you focus on your application code.

Con: Developers are often restricted to certain major/minor versions of packages available on the system so that the vendor can manage the platform effectively.

Software as a service (SaaS)

With the SaaS platform, everything is provided for you except the users and the application data. The vendor provides the application code and the developer has limited access to modify the software in use. This is typically not a choice for deploying custom applications, as the vendor provides the entire software stack. Hosted web email clients and hosted sales automation software are two good examples of how SaaS is used.

Pro: The entire stack is provided by the vendor except the application users and the associated data.

Con: You have limited control over the hosted application and it's often hard to integrate external workflows into the system.

Which should you choose?

As an application developer, you should choose PaaS, because that the infrastructure is managed for you, so you can focus on your application code.

Scale your application

As mentioned previously, PaaS provides scaling out of the box for most languages and runtimes. However, as a developer you need to be aware of the types of scaling offered and when it makes sense to scale horizontally or vertically.

Vertical scaling

Vertical scaling refers to a type of scaling that has been the default choice for decades. This type of scaling refers to the notion that to handle load, you simply use larger systems. This is one of the reasons why there are servers in place today with a terabyte of RAM and a massive number of CPUs and cores to serve a single Java® application. Typically when using vertical scaling, a single large system is used to handle most or all of the application requests from the users.

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Horizontal scaling

With horizontal scaling, the application load and requests are spread over a group of smaller servers that are typically behind a load balancer. As a request from a user is made, the load balancer sends the request to a server and then manages the session state across the cluster of servers. There are usually two types of horizontal scaling to use to ensure the best possible experience for the users of your application: manual and automatic scaling.

Manual scaling

With manual scaling, you specify that you want the application to scale up to handle increased traffic when you know you have an upcoming event that will increase application demand. For example, if you know that you are going to be running a marketing campaign to attract more users to your application, you might want to proactively add additional servers to your cluster. Most PaaS providers allow you to accomplish this task with a simple command.

Automatic scaling

With automatic scaling, you specify conditions where your application will automatically scale without any human interaction. This condition can be based on such things as the number of concurrent HTTP requests your application is receiving, or the amount of the CPU that your application is using. This enables the developer to automatically add new servers to the load balancer when the demand for the application is high. Automatic scaling provides a truly hands-off approach to scaling while ensuring that demand from the users is met in a timely fashion. Automatic scaling is crucial when you have unplanned use of your application due to certain circumstances. For example, you might get your mobile application featured on an application store for a short period of time when your back-end services reside in the cloud.

Which application scaling should you choose?

As a developer, you should choose a platform that allows for both manual and automatic horizontal scaling of your application.

Consider application state

Most cloud providers that provide a PaaS want you to start with green field development, which means that projects that are not affected by the constraints of prior work. Porting existing or legacy applications to the platform can be a challenge, mainly because the file systems in place are ephemeral in nature and do not allow for saving application state or resources on the file system.

This restriction is why you might hear that you need to think about future applications as being stateless. To receive the benefits of an infrastructure that resides in the cloud, you need to employ stateless application design in your projects. To achieve that, take into account the following practices for new applications:

- ◇ Allow the application server or container to maintain the session state of the user across the cluster instead of relying on the file system.

- ◇ Do not store files or user assets on the physical file system of the server that your code is deployed to. Instead, consider using a cloud-based storage service and delivering assets through the provided REST API for the storage service.
- ◇ Use a database for storing assets related to a user if you do not have access to use a cloud storage API.

Which application state should you choose?

For green field applications, you should design applications that are stateless, which means they do not store user assets or resources on the file system. For legacy or existing applications, choose a PaaS provider that supports both stateful and stateless applications.

Choose a database for cloud-enabled applications

Almost all applications being created today rely on a database of some type on the back end to store and retrieve information to be presented to the user. When developing applications for the cloud, you must also take into consideration what databases you will be using and where those databases should be located. Should the database be hosted on the same servers as the application, or is it better to house the database on a separate server or container?

In many cases, an application relies on information that's stored in a database that resides behind a corporate firewall, while the application front end is deployed on the public cloud. When this is the case, you have a couple of options for effectively accessing the information that you'll need to present to the user on the front end.

Option 1: Choose a provider that allows you to open up a remote VPC connection back to your database.

Option 2: Communicate to the database through a set of authenticated REST services deployed on the infrastructure that have access to the data.

Both of these options have inherent security risks that you need to consider when connecting to a database behind a corporate firewall from an outside cloud application. When this is the case, your best option is to select a cloud PaaS vendor that allows you to deploy your applications on a non multi-tenant environment.

If your application code does not need to connect to an existing corporate database, the number of options that you have are almost endless. I suggest that you deploy your database in the same geography/datacenter/region as your application code but on different containers or servers than your front-end application code. Use this option to scale the database independently of the web tier. Also, be sure to choose a database that scales quickly and easily regardless of whether it's a SQL or NOSQL database.

Consider multiple geographies

One of the great benefits of cloud computing is that you can deploy your application infrastructure throughout the world with little or no up-front cost. For example, deploying an application that has servers in both North America and EMEA has traditionally incurred a huge up-front cost to purchase and provision hardware and data centers. With an infrastructure that resides in the cloud, you can effortlessly deploy your application across as many geographies as your vendor supports. For

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simple applications that only have a limited number of users, this is not required. However, having access to deploy code in multiple geographies is critical to winning customer satisfaction by locating the application code as close to your target audience as possible.

Throw in the ability to manually or automatically scale your application across different geographies, and you'll have a really interesting value proposition on your hands by incurring a lower cost than deploying a traditional IT infrastructure.

Which cloud provider should you choose for multiple geographies?

Choose a cloud provider that enables you to both deploy and scale your application infrastructure across multiple geographies throughout the world to ensure that your audience has a fast and responsive experience while using your application.

Create and use REST-based web services

As you can see, deploying your application code in the cloud provides many benefits — and one crucial benefit for high-demand applications is the ability to scale out the web and database tiers independently. That being said, it is also good practice to separate your business logic into web services that your front-end code can consume. Use this practice to scale out the web services tier independently from both the database and the front-end code. Separating your application logic from the presentation tier opens new doors for technologies that you might not have considered in the past, such as creating a single-page application using a language like Node.

Implement continuous delivery and integration

DevOps seems to be the latest buzzword that is gaining a lot of attraction across enterprises. To get ahead, you should probably start looking at and implementing both continuous integration and delivery on your next software project. When deploying applications to a cloud-based infrastructure, make sure you have workflows in place on your existing build system so that code can be deployed across the different environments. Fortunately, most of the more popular build systems provide plugins for some of the top cloud providers today, making it easy to configure your deployment rules based upon the correct permissions of who has access to deploy code to each environment. If you are not currently using a build system for your development team, start using one now!

Which cloud provider should you choose for continuous integration and delivery?

Choose a cloud provider that meets all of the requirements above with the added feature of integrated continuous integration and continuous delivery (CI/CD) tools on the platform. The provider you choose should allow you to deploy your own build system or have the ability to easily integrate with existing systems that reside outside of the cloud platform.

Avoid vendor lock-in

If you take one thing away from this article, I hope this is it: While many cloud providers provide great-looking proprietary APIs that reduce the amount of code or work that you have to do, you should avoid them at all costs. This is nothing more than a

simple ploy to get you locked into their ecosystem while making it extremely hard to move your application to another provider or to your own data center running in-house. To avoid these custom APIs, stick with tried-and-true technology stacks across your application, including the database tier, storage tier, and any micro service endpoints that you might want to create. While the up-front investment can be a bit higher than using a proprietary solution out of the box, your technical debt is greatly reduced, which can save you money and time in the long run.

Develop locally or in the cloud

As developers, we often code applications on our local system and then, when we reach a work milestone, we move our code to the team's development environment. Most developers wish they could develop on a daily basis with an infrastructure that resembles production as closely as possible. That goal can often be challenging due to the system administration overhead incurred to provide each developer with a cluster of machines.

Now that PaaS is available, all developers should begin to develop and deploy their code in the cloud. Most integrated development environments (IDE) provide plugins to streamline the process and make it feel as close to developing locally as possible.

Which IDE should you choose?

Choose an IDE that provides a plugin for the cloud provider of your choice. Consider choosing a provider that provides the ability to hot deploy application code as well as the ability to enable remote debugging of your source code. Once you have selected a provider that offers these two things, you can continue to set break points inside of your IDE and step through code just as if you were deploying locally. This enables you to more quickly catch bugs that only appear when moving to a clustered environment.

What to look for in the coming years from cloud providers

This article focused on the current state of applications being deployed to the cloud. One thing to look for and consider this year and next is the mass-industry movement to container-based deployments — you have probably already heard about Docker and rocket containers. When selecting a cloud provider, make sure the roadmap for application migration to container-based deployments is called out clearly with a timeline that clearly defines your migration path. Also, be on the lookout for vendors that are sticking with industry-standard solutions around containers and orchestration, versus creating proprietary solutions.

Conclusion

Cloud computing has many benefits that you should take advantage of in your daily software development and deployment to make your software more stable, scalable, and secure. When moving applications to the cloud, consider the following guidance:

1. For application development, choose PaaS. The infrastructure is managed by a vendor, which gives you more time to focus on your application code.
2. For application development, choose a platform enabled for both manual and automatic horizontal scaling of your application.

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3. For green-field applications, design apps that are stateless.
4. For legacy or existing applications, choose a PaaS provider that supports both stateful and stateless applications.
5. Choose a database that is scalable and located on a separate server or container from your application code. Then you can scale the database independently.
6. Choose a cloud provider that enables you to both deploy and scale your application infrastructure across multiple geographies throughout the world.
7. Develop using REST-based web services.
8. Choose a cloud provider that meets all of the previous requirements with the added feature of integrated continuous integrations and continuous delivery tools in the platform.
9. Avoid being locked in by a vendor.

3.2. Deploying a Web Service from Inside and Outside a Cloud Architecture

Java Web Deployment in Cloud Computing is facing several challenges like performance issue, cost issue, security issue and reliability issue. When a java application is deployed on a java-cloud, the performance of the application is expected to be the equal or better to the non-cloud environments but in the matter of performance, java applications are facing this major challenge. Until unless this major issue is resolved, the cloud computing is not best suited platform for java-applications. Each java-application is based on a particular framework. Thus to overcome these kinds of issues, the framework and application should be optimized or customized as per the requirements. These issues make the application lesser reliable in cloud computing. If these issues doesn't solve within a particular time, the java applications on clouds would never be a better than non-cloud.

Cloud Computing can be considered as a Service over a network and a step ahead to the virtualization as service in the internet which is dynamic. It represents a separate way to explain and management computational resources. It includes delivery of the application as a service throughout the internet and the software that provide services in the data-centre and hardware and the paradigm shift. The data centre-software and hardware is known as a cloud. Cloud Computing could be explained as an Example. Assume one wants to perform a computational task. For the task, he/she turns on the computer, the OS is loaded from hard drive, but if we use cloud computing, the OS is loaded from the network. This service provided over the network is known as Cloud Computing. It is an On-Demand Self-Service based technology. According to this, if one wants any resources to be used, requests for that over the network. Cloud Computing includes services like operating systems, software, data etc. The term cloud refers to the pool of virtualized computer resources including hardware and software resources. Cloud Environment and Non-Cloud Environment could be differentiated as a telephone system. Non-Cloud Environment could be considered as Wired Telephones in Telephone systems. An IP address is assigned like 76.152.781.981. This IP address is like telephone number, as one can determine the location of the user, by watching the first few digits of Telephone number in wired telephony system. Apart from it, in Cellular phones, one cannot determine the location by its number as similar as Cloud Environment. Most of the companies have started their trend with Cloud services Like

Google has started Google App Engine (since July 2008), Microsoft started Windows Azure (since October 2008), Amazon started AWS, EC2 (Early 2006). These are having a revolutionary step in IT industry. Web Deployment in Clouds are as similar as deploying a web-application on the server with many benefits. Web-Deployment on cloud facilitates one by shift paradigm.

You can deploy, run, and test web services client applications.

◇ **Deploying web services applications onto application servers**

After assembling the artifacts required to enable the web module for web services into an enterprise archive (EAR) file, you can deploy the EAR file into the application server.

◇ **Using a third-party JAX-WS web services engine**

In certain situations you might need to set up a third-party JAX-WS web services engine. For example, you must set up a third-party JAX-WS web services engine if you need to deploy applications that use a single run time across various application servers such as WebSphere® Application Server, JBoss, and WebLogic, or if you want to build JAX-WS web services applications using third party JAX-WS run time such as CXF, Axis2, and Metro.

◇ **Deploying web services client applications**

After you have created an enterprise archive (EAR) file for the web services client application, you can deploy the web services client application into the Application Server.

◇ **Making deployed web services applications available to clients**

You can publish WSDL files to the file system. If you are a client developer or a system administrator, you can use WSDL files to enable clients to connect to web services.

◇ **Running an unmanaged web services JAX-RPC client**

WebSphere Application Server Version 8.5 and the Application Client for WebSphere Application Server Version 8.5 provides a thin Java Platform, Standard Edition 6 (Java SE 6) web services client runtime implementation that is based on the Java™ API for XML-based RPC (JAX-RPC) 1.1 specification. The Thin Client for JAX-RPC with WebSphere Application Server is a stand-alone Java SE 6 client environment that enables running unmanaged JAX-RPC web services client applications in a non-WebSphere environment to invoke web services that are hosted by the application server.

◇ **Running an unmanaged web services JAX-WS client**

WebSphere Application Server provides a thin Java Platform, Standard Edition 6 (Java SE 6) web services client runtime implementation that is based on the Java API for XML-based Web Services (JAX-WS) 2.2 specification. The Thin Client for JAX-WS with WebSphere Application Server is a stand-alone Java SE 6 client environment that enables running unmanaged JAX-WS web services client applications in a non-WebSphere environment to invoke web services that are hosted by the application server.

Notes

◇ Testing web services-enabled clients

Once you have developed, assembled, deployed and configured your web service, you can test to confirm your web service runs in the application server environment.

3.3. Advantages and Disadvantages

Positive Aspects

A. Attributes The attributes of cloud computing are:

- 1) **Service Based:** User worries are distant from supplier trepidations through service edges that are fine defined. The edges hide the execution specifics and allow a completely automated response by the provider of the service to the consumer of the service.
- 2) **Elastic:** The service can scale capacity up or down as the consumer demands at the speed of full automation (which may be seconds for some services and hours for others). Elasticity is a trait of shared pools of resources.
- 3) **Shared:** Services share a pool of resources to build economies of scale. IT resources are used with maximum efficiency. The underlying infrastructure, software or platforms are shared among the consumers of the service (usually unknown to the consumers). This enables unused resources to serve multiple needs for multiple consumers, all working at the same time.
- 4) **Metered by Use:** Services are tracked with usage metrics to enable multiple payment models. The service provider has a usage accounting model for measuring the use of the services, which could then be used to create different pricing plans and models. These may include pay-as-you-go plans, subscriptions, fixed plans and even free plans. The implied payment plans will be based on usage, not on the cost of the equipment.
- 5) **Uses Internet Technologies:** The service is delivered using Internet identifiers, formats and protocols, such as URLs, HTTP, IP and representational state transfer Weboriented architecture.

B. Benefits

The most frequently cited benefits of cloud computing are:

- ◇ It is agile, with ease and good speed of deployment
- ◇ Its cost is use-based, and will likely be reduced
- ◇ In-house IT costs are reduced
- ◇ Capital investment is reduced
- ◇ The latest technology is always delivered
- ◇ The use of standard technology is encouraged and facilitated. As an application moves to the cloud, the access to it becomes more simple and ubiquitous. Low cost ultra-light devices and inexpensive hand held devices build on latest operating systems such as android provide access to the internet, the number and types of tasks taking advantage of the new

technology will increase by several orders of magnitude, going far beyond the comparatively modest list of things that we use computers and the Internet for today.

Disadvantage of Web Services

- ◇ Although web services are simple to use but there are some flaws of using it. One of the disadvantage is over Matching Requirements. Any time one create a service to handle a variety of customers, need specialized machine requirements.
- ◇ Second disadvantage of web services is availability. Every user or client who uses web services know that it is not available hundred percent all the time.
- ◇ Third Disadvantage of web services is security. Web services are available to public through http-based protocol. So every one can access web services and use it. This flaw can be avoided using authentication mechanisms.
- ◇ Guaranteed Execution is a major problem of web services because HTTP which is hypertext transport protocol is not a reliable protocol that is it doesnot provide any guarantee of delivery of response.

Summary

Cloud computing has many benefits that you should take advantage of in your daily software development and deployment to make your software more stable, scalable, and secure. When moving applications to the cloud, consider the following guidance:

1. For application development, choose PaaS. The infrastructure is managed by a vendor, which gives you more time to focus on your application code.
2. For application development, choose a platform enabled for both manual and automatic horizontal scaling of your application.
3. For green-field applications, design apps that are stateless.
4. For legacy or existing applications, choose a PaaS provider that supports both stateful and stateless applications.
5. Choose a database that is scalable and located on a separate server or container from your application code. Then you can scale the database independently.
6. Choose a cloud provider that enables you to both deploy and scale your application infrastructure across multiple geographies throughout the world.
7. Develop using REST-based web services.
8. Choose a cloud provider that meets all of the previous requirements with the added feature of integrated continuous integrations and continuous delivery tools in the platform.

Keywords

Cloud computing: It offers a lot to software developers, who can now develop web-based applications that take advantage of the power and reach of cloud computing.

Cloud services development: These are particularly notable to smaller businesses, which otherwise would not have the budget or resources to develop large-scale applications.

Notes

Notes

Cloud services: In the form of centralized web-based applications, also appeal to the IT professional.

EC2: It is just part of Amazon's Web Services (AWS) set of offerings, which provides developers with direct access to Amazon's software and machines.

Google App Engine: It provides a fully integrated application environment.

Mash-ups: It created by users of the Google Maps API, with these custom apps, the data that feeds the map is provided by the developer, where the engine that creates the map itself is provided by Google.

SaaS: It is probably the most common type of cloud service development, with SaaS; a single application is delivered to thousands of users from the vendor's servers.

Self-Assessment Questions

1. The first level of maturity defines the traditional model of software delivery.
 - (a) JSP
 - (b) ASP
 - (c) API
 - (d) AWS
2. The second level of maturity occurs when the vendor hosts a separate instance of the application for each.....
 - (a) Customer
 - (b) Vender
 - (c) Both a and b
 - (d) None of these.
3. defined four primary maturity levels.
 - (a) Linux
 - (b) Google
 - (c) Microsoft
 - (d) None of these.
4. The Google App Engine provides a.....application environment.
 - (a) Product
 - (b) Process
 - (c) Partially integrated
 - (d) fully integrated
5. The third level of maturity occurs when the vendor hosts a separate instance of the application for each customer.
 - (a) True
 - (b) False

Notes

6. IBM is offering a cloud computing solution.
 - (a) True
 - (b) False
7. EC2 enables scalable deployment of applications by letting customers request a set number of virtual machines.
 - (a) True
 - (b) False
8. The second level of maturity defines the traditional application service provider (ASP) model of software delivery.
 - (a) True
 - (b) False

Review Questions

1. Explain Technologies and the processes required when deploying web services
2. What is Deploying a web service from inside and outside a cloud architecture?
3. Discuss the Advantages and disadvantages of web server.

Answers for Self Assessment Questions

- | | | | |
|--------|--------|--------|--------|
| 1. (b) | 2. (a) | 3. (c) | 4. (d) |
| 5. (b) | 6. (a) | 7. (a) | 8. (b) |

Structure:

4. Cloud Services Management
 - 4.1 Reliability, availability and security of services deployed from cloud
 - 4.2 Performance and scalability of services, tools and technologies used to manage cloud services deployment
 - 4.3 Cloud Economics: Cloud Computing infrastructures available for implementing cloud based services
 - 4.4 Economics of choosing a Cloud platform for an organization based on application requirements, economic constraints and business needs

Unit-4: Cloud Services Management

Notes

Objective:

After studying this chapter, you will be able to:

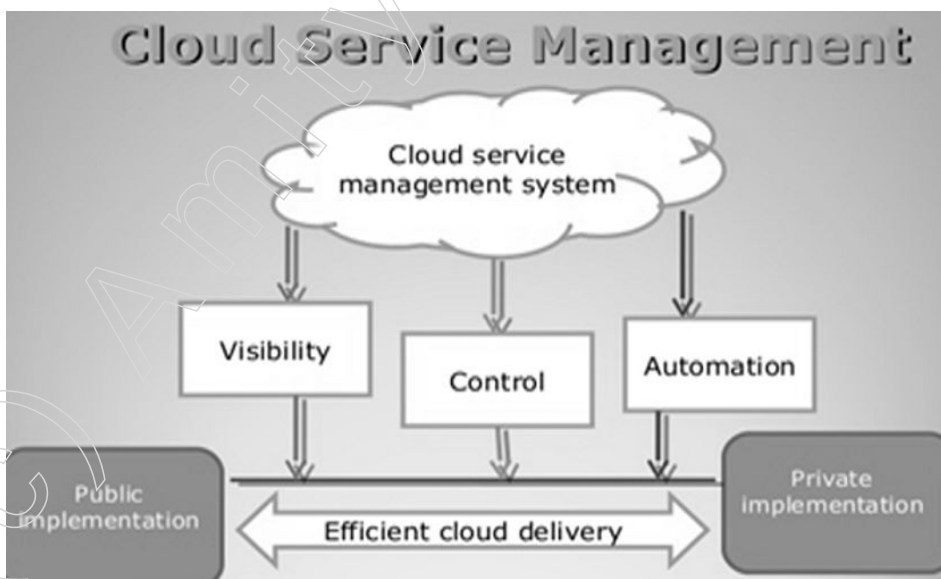
- Explain the meaning of Cloud Services Management
- Discuss the Reliability, availability and security of services deployed from cloud
- Explain the Performance and scalability of services
- Discuss the Cloud Economics
- Understand the economic constraints and business needs

Introduction

The management of cloud infrastructure products and services is cloud management. Public clouds are operated by public cloud service providers, which provide the servers, storage, networking and data centre operations of the public cloud environment. With a third-party cloud management tool, users can also choose to manage their public cloud services.

Public cloud service users can typically choose from three categories of specific cloud provisioning:

- ◇ **User self-provisioning:** Users, usually via a web form or console interface, buy cloud services directly from the provider. On a per-transaction basis, the client pays.
- ◇ **Advanced provisioning:** A pre-determined sum of services scheduled in advance of operation is contracted in advance by customers. A flat fee or a monthly fee is charged by the consumer.
- ◇ **Dynamic provisioning:** When the client requires them, the provider allocates resources, and then decommissions them when they are no longer required. On a pay-per-use basis, the client is paid.



Notes

The purpose and scope of the management of cloud services are listed below:

- ◇ **Purpose:** Establish suitable techniques for managing and running cloud-based services. Insert cloud service management techniques into current frameworks for IT creation and support.
- ◇ **Scope:** Oversight of cloud-based service design, development and change. Cloud-based service management and operation.

Characteristics of Cloud service Management

In a design for handling cloud environments, cloud management incorporates applications and technologies. With a range of cloud management platforms and instruments, software developers have responded to the management challenges of cloud computing. These solutions include native tools provided by public cloud providers, as well as third-party tools designed by various cloud providers to provide consistent functionality. With access to various native features within individual cloud platforms, administrators must balance the conflicting requirements of efficient consistency across various cloud platforms. The need for transparent cross-platform management is motivated by increasing public cloud adoption and increased multi-cloud use. For those technical professionals responsible for maintaining IT systems and facilities, the rapid adoption of cloud services presents a new set of management challenges.

In the following categories, cloud-management systems and instruments should be able to have minimum functionality.

- ◇ **Service request:** receiving and fulfilling user requests to access and deploy cloud services.
- ◇ **Cost management and optimization:** Cloud spending monitors and accurate sizes and aligns resources and efficiency with real demand.
- ◇ **Security and compliance:** handling cloud providers' role-based access and implementing security settings.
- ◇ **Inventory and classification:** discover and maintain pre-existing cloud infrastructure in the brownfield plus track and handle modifications.

4.1. Reliability, Availability and Security of Services Deployed from Cloud

4.1.1 Cloud Service Reliability Modelling and Evaluation

In this section, we develop a holistic model for Cloud Service Reliability, which is defined as the probability that a cloud service under consideration can be successfully completed for a user in a specified period of time.

In particular, this requires that the job request be successfully served by the schedulers in time, the set of subtasks contained by the service be completed, the computing/data resources required by the subtasks be available; and the network be operational during the communications.

From the definition of cloud service reliability will more or less affect this probability to provide a successful service. We classify the above failures in two groups:

1. Request Stage Failures: Overflow and Timeout.

2. Execution Stage Failures: Data resource missing, computing resource missing, Software failure, Database failure, Hardware failure, and Network failure.

The failures in Group 1 may occur before the job request is successfully assigned to computing/data resources; on the other hand, the failures in Group 2 may occur after the job request has been successfully assigned and during the execution of subtasks. Therefore, the two groups of failures could be deemed as independent. Nevertheless, failures within each group are strongly correlated. In summary, the modelling of cloud service reliability can be separated in two parts: modelling of Request Stage Reliability and modelling of execution Stage Reliability.

When you access an app or service in the cloud, you can reasonably expect that:

- ◇ The app or service is up and running.
- ◇ You can access what you need from any device at any time from any location.
- ◇ There will be no interruptions or downtime.
- ◇ Your connection is secure.
- ◇ You will be able to perform the tasks you need to get your job done.

Factors like these measure the reliability of your cloud offerings. In a perfect world, your system would be 100% reliable. But that is probably not an attainable goal. In the real world, things will go wrong. You will see faults from things such as server downtime, software failure, security breaches, user errors, and other unexpected incidents.

Proper planning and cloud visualization can help you address faults quickly so that they don't become huge problems that keep people from accessing your cloud offerings. The cloud makes it easy to build fault-tolerance into your infrastructure. You can easily add extra resources and allocate them for redundancy.

Employing measures that make your cloud system more reliable ensures that:

- ◇ Redundant resources kick in automatically when the system experiences a fault.
- ◇ There is no downtime and products and services remain available.
- ◇ Employees keep doing their jobs without knowing that something went wrong.

Reliability in cloud computing is important for businesses of any size. Buggy software can cause lost productivity, lost revenue, and lost trust in your brand. Before you deploy your applications to the cloud, make sure they are thoroughly tested against a variety of real-world scenarios. This helps to ensure that they are reliable and will meet customer expectations.

4.1.2 Availability

High availability is the ultimate goal of moving to the cloud. The idea is to make your products, services, and tools available to your customers and employees at any time from anywhere using any device with an internet connection.

Cloud availability is related to cloud reliability.

For example, let's say you have an online store that is available 24/7. But sometimes clicking the "checkout" button kicks customers out of the system before they have completed the purchase. So, your store may be available all the time, but if the underlying software is not reliable, your cloud offerings are basically useless.

Notes

Bringing it all together

Cloud availability, cloud reliability, and cloud scalability all need to come together to achieve high availability. This means that your products and services are accessible anytime and anywhere, function reliably and as expected, and that the system can seamlessly scale up or down to accommodate customer demand without suffering a loss in performance.

Cloud service providers offer an Infrastructure as a Service (IaaS) model that gives you access to storage, servers, and other resources. IaaS provides automation and scalability on demand so that you can spend your time managing and monitoring your applications, data, and other services.

Because IaaS provides scalability based on a pay-as-you-go model, this saves you money and frees you up to track down and address problems that may come up with the software. Having more time to monitor can help you find areas that need improvement so you can do a better job consistently deploying reliable products and services.

To survive in today's global market, it's inevitable that your company will need to move to the cloud. It won't happen overnight and will require a lot of planning. As you plan what and how you will make solutions available in the cloud, remember that it is important that your products and services and cloud infrastructure are scalable, reliable, and available when and where they are needed.

4.1.3 Security

Cloud security is the set of control-based security measures and technology protection, designed to protect online stored resources from leakage, theft, and data loss. Protection includes data from cloud infrastructure, applications, and threats. Security applications use a software the same as SaaS (Software as a Service) model.

How to manage security in the cloud?

Cloud service providers have many methods to protect the data.

Firewall is the central part of cloud architecture. The firewall protects the network and the perimeter of end-users. It also protects traffic between various apps stored in the cloud.

Access control protects data by allowing us to set access lists for various assets. For example, you can allow the application of specific employees while restricting others. It's a rule that employees can access the equipment that they required. We can keep essential documents which are stolen from malicious insiders or hackers to maintaining strict access control.

Data protection methods include Virtual Private Networks (VPN), encryption, or masking. It allows remote employees to connect the network. VPN accommodates the tablets and smartphone for remote access. Data masking maintains the data's integrity by keeping identifiable information private. A medical company share data with data masking without violating the HIPAA laws.

For example, we are putting intelligence information at risk in order of the importance of security. It helps to protect mission-critical assets from threats. Disaster recovery is vital for security because it helps to recover lost or stolen data.

Benefits of Cloud Security System

We understand how the cloud computing security operates to find ways to benefit your business.

Cloud-based security systems benefit the business by:

- ◇ Protecting the Business from Dangers
- ◇ Protect against internal threats
- ◇ Preventing data loss
- ◇ Top threats to the system include Malware, Ransomware, and
- ◇ Break the Malware and Ransomware attacks
- ◇ Malware poses a severe threat to the businesses.

More than 90% of malware comes via email. It is often reassuring that employee's download malware without analysing it. Malicious software installs itself on the network to steal files or damage the content once it is downloaded.

Ransomware is a malware that hijacks system's data and asks for a financial ransom. Companies are reluctant to give ransom because they want their data back.

Data redundancy provides the option to pay a ransom for your data. You can get that was stolen with minimal service interruption.

Many cloud data protection solutions identify malware and ransomware. Firewalls keep malicious email out of the inbox.

DDoS Security

Distributed Denial of Service (DDoS) is flooded with requests. Website slows down the downloading until it crashes to handle the number of requests.

DDoS attacks come with many serious side effects. Most of the companies suffering from DDoS attacks lose \$ 10,000 to \$ 100,000. Many businesses damage reputation when customers lose confidence in the brand. If confidential customer data is lost through any DDoS attack, we may face challenges.

The severity of these side effects, some companies shut down after the DDoS attacks. It is to be noted that the last DDoS attack lasted for 12 days.

Cloud security service monitors the cloud to identify and prevent attacks. The cloud service providers protect the cloud service users in real time.

Threat to detect

Cloud computing detects advanced threats by using endpoint scanning for threats at the device level.

Difference between Cloud Security and Traditional IT Security

Notes

Notes

Cloud security	Traditional IT Security
Quick scalable	Slow scaling
Efficient resource utilization	Lower efficiency
Usage-based cost	Higher cost
Third-party data centres	In-house data centres
Reduced time to market	Longer time to market
Low upfront infrastructure	High Upfronts costs

Top 7 Advanced Cloud Security Challenges

It becomes more challenging when adopting modern cloud approaches Like: automated cloud integration, and continuous deployment (CI/CD) methods, distributed serverless architecture, and short-term assets for tasks such as a service and container.

Some of the advanced cloud-native security challenge and many layers of risk faced by today's cloud-oriented organizations are below:

1. Enlarged Surface

Public cloud environments have become a large and highly attractive surface for hackers and disrupt workloads and data in the cloud. Malware, zero-day, account acquisition and many malicious threats have become day-to-day more dangerous.

2. Lack of visibility and tracking

Cloud providers have complete control over the infrastructure layer and cannot expose it to their customers in the IaaS model. The lack of visibility and control is further enhanced in the SaaS cloud models. Cloud customers are often unable to identify their cloud assets or visualize their cloud environments effectively.

3. Ever-changing workload

Cloud assets are dynamically demoted at scale and velocity. Traditional security tools implement protection policies in a flexible and dynamic environment with an ever-changing and short-term workload.

4. DevOps, DevSecOps and Automation

Organizations are adopting an automated DevOps CI/CD culture that ensures the appropriate security controls are identified and embedded in the development cycle in code and templates. Security-related changes implemented after the workload is deployed to production can weaken the organization's security posture and lengthen the time to market.

5. Granular privileges and critical management

At the application level, configured keys and privileges expose the session to security risks. Often cloud user roles are loosely configured, providing broad privileges beyond the requirement. An example is allowing untrained users or users to delete or write databases with no business to delete or add database assets.

6. Complex environment

These days the methods and tools work seamlessly on public cloud providers, private cloud providers, and on-premises manage persistent security in hybrid and multi-cloud environments-it including geographic Branch office edge security for formally distributed organizations.

7. Cloud Compliance and Governance

All the leading cloud providers have known themselves best, such as PCI 3.2, NIST 800-53, HIPAA and GDPR.

It gives the poor visibility and dynamics of cloud environments. The compliance audit process becomes close to mission impossible unless the devices are used to receive compliance checks and issue real-time alerts.

4.2. Performance and Scalability of Services, Tools and Technologies used to Manage Cloud Services Deployment

4.2.1. Performance

Cloud performance monitoring and testing tools help organizations gain visibility into their cloud environments, using specific metrics and techniques to assess performance.

Efficient cloud performance is critical for maintaining business continuity and ensuring all relevant parties gain access to cloud services. This is true for basic cloud usage of public clouds and complex hybrid cloud and multi-cloud architectures.

Cloud performance metrics enable you to effectively monitor your cloud resources, to ensure all components communicate seamlessly. Typically, cloud performance metrics measure input/output operations per second (IOPS), file system performance, caching, and autoscaling.

Cloud Computing Performance Metrics

There are various metrics that can help you monitor and assess the performance of your cloud computing resources, including IOPS, filesystem performance, caching, and autoscaling.

Input/Output Operations per Second (IOPS)

IOPS measures the read and write speed of your storage. It is based on variables like the configuration of the disk array, sequential or random data patterns, data block sizes, and the ratio of write and read processes. IOPS values serve as the performance benchmarks for storage devices, and impact the performance of the servers hosting the devices.

File Storage Performance

There are two primary options for managing storage in the cloud:

- ◇ Running virtual machines and attaching block storage volumes to them - for example, Azure managed disks
- ◇ Using managed storage services, such as Azure Files or Azure NetApp Files

Notes

Because these storage systems interact with applications, they significantly impact cloud performance. You should monitor metrics like latency, IOPS on storage volumes or services, and storage capacity vs. limits on the volume or service.

Caching

The purpose of caching is to improve storage access performance. To achieve this, caching techniques temporarily store data using RAM pools within the compute nodes. This process is implemented before the data is read from or written to a storage device. The data can then be read directly from the disk—this is called cache memory.

Cache memory provides quick access to frequently used files. Because the cache uses RAM, it has faster access rates than disk read operations. When the filesystem needs data, the cache will read it, preventing longer disk read operations. To ensure efficient disk performance, caching solutions orchestrate the process, optimizing performance as needed and freeing up central processing units (CPUs).

Autoscaling

Autoscaling processes are responsible for either decreasing or increasing the provisioning of resources. There are two types of autoscaling:

- ◇ **Vertical scaling**—the process of scaling up, during which you add resources like CPU instances or RAM. You can scale up your network, storage, and compute capabilities. Vertical scaling typically translates into better performance.
- ◇ **Horizontal scaling**—the process of scaling out, during which you add more nodes. This means you increase the amount of servers in your current configuration.

Each cloud vendor provides different scaling options and configurations. Before scaling, check with your vendor to determine costs and specifications for each type of system.

4.2.2. Scalability

If you're wondering whether your company should move to the cloud, the short answer is "yes". And you have a lot of work to do to catch up with other businesses. In fact, 85% of businesses worldwide are using the cloud. And by 2021, 94% of the internet workload will be processed in the cloud.

While researching reasons to migrate to the cloud, you've probably learned that the benefits include "scalability," "reliability," "availability," and more. But what, exactly, do those terms mean?

This article focuses on cloud scalability, cloud reliability, and cloud availability. Spoiler alert: You will learn that:

- ◇ You need cloud scalability to meet customer demand.
- ◇ You need cloud reliability to ensure that your products and services work as expected.
- ◇ You need cloud availability to ensure that customers can access your cloud services whenever they need to and from anywhere in the world.
- ◇ You need to bring all three together to achieve true high availability.

What is scalability in cloud computing?

Cloud scalability refers to how well your system can react and adapt to changing demands. As your company grows, you want to be able to seamlessly add resources without losing quality of service or interruptions. As demand on your resources decreases, you want to be able to quickly and efficiently downscale your system so you don't continue to pay for resources you don't need.

However, there is more to scalability in the cloud than simply adding or removing resources as needed. Let's look at some of the different types of scalability in cloud computing.

Cloud elasticity

This refers to how well your cloud services are able to add and remove resources on demand. Elasticity is important because you want to ensure that your clients and employees have access to the right amount of resources as needed.

Cloud elasticity should be automatic and seamless. People accessing your cloud services should not be able to notice that resources are added or dropped. They should just have the confidence that they can access and use resources without interruptions.

Vertical scaling

Vertical scaling (or "scaling up") refers to upgrading a single resource. For example, installing more memory or storage capacity to a server. In a physical, on-premises setup, you would need to shut down the server to install the updates.

Horizontal scaling

This term is used to describe "building out" a system with additional components. For example, you can add processing power or more memory to a server by linking it with other servers. Horizontal scaling is a good practice for cloud computing because additional hardware resources can be added to the linked servers with minimal impact. These additional resources can be used to provide redundancy and ensure that your services remain reliable and available.

Auto-scaling

This term refers to a cloud computing feature that lets you automatically manage the different types of scalability in the cloud automatically. Cloud providers such as Amazon Web Services offer auto-scaling to enable consistent performance regardless of the current demand on resources.

Implementing and managing a cloud scaling strategy is:

- ◇ **Convenient:** You can easily increase or decrease storage capacity as needed.
- ◇ **Flexible and fast:** You can quickly respond to changing demands to keep customers up and running without delays in service.
- ◇ **Cost-effective:** You don't have to pay for expensive hardware or provide the space to store it.
- ◇ **Fault-tolerant:** Resources can automatically be scaled to accommodate redundancies and to facilitate disaster recovery.

Notes

- ◇ **Time-saving:** Upgrading existing hardware and installing new hardware on-site can be very time-consuming. Cloud computing can take care of the scaling for you. This frees you up to focus on innovation and process improvement rather than troubleshooting errors and other issues.

Cloud computing is so scalable because the cloud service providers have the necessary hardware and software in place. They also use virtual machines (VMs) to scale up or down because:

- ◇ You can easily add resources to VMs at any time with minimal impact.
- ◇ You can easily move VMs to a different server that has more resources.
- ◇ You can host VMs on a server cluster to share resources and balance the load.

4.2.3 Tools and technologies used to manage cloud services deployment

The table below provides a summary of the tools and technologies both formally or informally evaluated by ISC with ratings to reflect our experience. While considering these results, keep in mind that for many technologies we are employing a first-best fit approach, making use of what adds the most value to us quickly as a placeholder until the technology can be formally evaluated. Given that, ratings are subject to change as the application stack evolves and we gain more hands-on experience with each technology.

Here is the list of Cloud Computing tools:

1. Cloudability
2. Cloudyn
3. Informatica
4. CloudHub
5. Chef
6. Puppet
7. AtomSphere
8. RightScale
9. Enstratus
10. Agility Platform

Let us understand them a little deeper....

#1. Cloudability

It is a financial management tool for analyzing all cloud expenses happening across an organization. It helps identify opportunities for lowering costs, prepare reports, and give budget alerts and guidance via SMS and Gmail. It has APIs to link cloud billing and usage data to the business or financial system.

Cloudability Features

- ◇ **Budget Tracking:** Helps in calculating the cost involved and thus keep track of the budget
- ◇ **Multi-Cloud Tracking:** Manages more than one clients at a time on the same platform through a single account

- ◇ **Dashboard:** Provides insights of several reports on a single screen and thus making the analysis more straightforward.
- ◇ **API Integration:** It allows the integration of logic and data with software applications.
- ◇ **Real-time Sync:** Changes made in a single platform are visible in other platforms in real-time.
- ◇ **Compliance Management:** It helps in assessing risk and ensures policy comprehension
- ◇ **Two elastic pricing plans:** Pro and Enterprise.

#2. Cloudyn

This tool is designed to assist corporate IT from over-purchasing Amazon cloud resources. Cloudyn's sets of services give users a control panel viewing detailed data on all of their virtual machine examples, databases, and storage. Cloudyn also provides intuition into unsuccessful suggestions on how to get rid of them.

Cloudyn Features

- ◇ Intuitive dashboards
- ◇ Overall Cost Analysis
- ◇ Usage Breakdown Analysis
- ◇ Resource Cost Analysis
- ◇ Usage Trends Analysis
- ◇ Financial Projections Analysis
- ◇ Unused Resource Elimination Recommendations
- ◇ Downscaling, Relocation and Reallocation Recommendations

#3. Informatica

Informatica is an industry-leading data processing tool used for ETL to extract, transform, and load the data. It is one of the popular cloud computing tools that organizations make use of for ETL purposes. Below we have listed some of the features of Informatica, which are as follows.

Informatica Features:

- ◇ It extracts data from various data sources (structured or unstructured)
- ◇ Gives accurate deliverables and automate deployments
- ◇ Executes multiple processes simultaneously
- ◇ It has a centralized cloud server that allows easy access and tracking of data
- ◇ Easy data maintenance and data monitoring

#4. CloudHub

CloudHub is an integration platform as a service (iPaaS) where you can deploy sophisticated cross-cloud integration applications in the cloud, create new APIs on top of existing data sources, integrate on-premises applications with cloud services, and much more.

Notes

CloudHub Features

- ◇ Scalable Interface
- ◇ Data Mapping
- ◇ Web-Based Dashboard
- ◇ Visual Data Transformation
- ◇ Drag-and-Drop Navigation
- ◇ Certified Integration Apps
- ◇ Connector Development Tool Kit
- ◇ Pre-Built Connector Options
- ◇ One-Click Application Deployment
- ◇ Firewall Options for Sensitive Data

#5. Chef

It is an open-source Ruby-based configuration management product served up by Opscode under the Apache license. With the help of a hosted Chef cloud system, administrators can programmatically configure virtual systems and cut down all repetitive manual operations.

Chef Features

- ◇ Backup and recover
- ◇ Real-time data
- ◇ Scalable automation
- ◇ Test deployment reliability
- ◇ Automation for flexible IT
- ◇ Search-based configuration
- ◇ Software and infrastructure changes
- ◇ Reports on compliance issues and security risks
- ◇ Reports on out of date software
- ◇ Build security and compliance checks
- ◇ Scan entire infrastructures
- ◇ Pre-built profiles for CIS, Linux, and Windows
- ◇ Query, visualize and report on operational events
- ◇ Notifications across multiple platforms
- ◇ High availability solution on-premise or in the cloud
- ◇ Deliver latest tested/approved policies to infrastructure
- ◇ Continuous delivery for applications and infrastructure
- ◇ Code review and automated testing

#6. Puppet

An IT automatic software gives system administrators the power to automate repetitive tasks, fastly deploy critical applications effortlessly, and manage infrastructure

changes, on-premise or in the cloud. It automates tasks at any platform of the IT infrastructure life cycle, including application configuration management, patch management, provisioning, discovery, operating system, and infrastructure audit and compliance.

Puppet Features

- ◇ Scalable
- ◇ Open Source
- ◇ Reporting compliance
- ◇ Cost-Effective
- ◇ Run the same set of configurations multiple times on the same machine

#7. AtomSphere

Boomi AtomSphere is a single instance, a multi-tenant cloud integration platform that has become the best choice for organizations seeking an integration partner to develop software applications by integrating features and benefits of other applications. With the increasing complexity in IT environments, integration has become so demanding and a necessity for us. Dell Boomi AtomSphere is a 100% cloud-native platform that assists you in connecting the applications efficiently.

AtomSphere Features

- ◇ Scalability, high availability, and performance
- ◇ Drag and drop mapping tools
- ◇ A broad set of connectors to integrate into the platform

#8. RightScale

RightScale is the mediator between an application and your cloud infrastructure that allows organizations to deploy and manage applications across public, private, and hybrid clouds. It provides configuration, monitoring, and automation of cloud computing infrastructure and applications.

RightScale Features

- ◇ Migrates workloads in and out of public and private clouds with proper monitoring
- ◇ Governance controls let you manage reporting, budgeting, auditing through a 'single pane of glass' view
- ◇ It is a multi-cloud platform
- ◇ Highly scalable

#9. Enstratus

Enstratus is a cloud computing tool that offers cross-stage cloud infrastructure for private, public, and hybrid clouds related to an enterprise's necessities. It allows self-service provisioning and de-provisioning of cloud resources. With just a single login, users can manage all cloud resources; and support enterprise authentication systems like OpenID and SAML 2.0.

Notes

Enstratus Features

- ◇ Self-service provisioning/de-provisioning of cloud resources
- ◇ Single login to manage all cloud resources
- ◇ Customizable role-based access controls
- ◇ Multi-currency cost/chargeback tracking
- ◇ Support for enterprise authentication systems

#10. Agility Platform

The Agility platform is an integrated control point that provides governance and security across an enterprise's cloud applications. It comprises an affordability policy engine to create and enforce an impressive range of custom governance and security policies. Also, it includes an end-to-end security model, including federated identity management.

Technologies

1. Virtualization

It is the process of sharing license keys to physical instances of application among different users of the enterprise. The main purpose of this technology is to provide a standard version of the cloud application to all clients. It is popularly used for its flexibility and instant running process. Few types of virtualization are

- ◇ **Hardware Virtualization:** If the virtual machine is directly installed on hardware servers then it is called Hardware virtualization because it is easy to control a Virtual Machine than a physical server.
- ◇ **Operating System Virtualization:** If the virtual machine is installed in a guest server instead of the hardware system, then it is called as Operating system Virtualization. It increased the testing situation of various software on various OS platforms,
- ◇ **Server Virtualization:** If the virtual machine is installed on the system it is called Server Virtualization. It is divided into multiple resources and utilized for load balancing on-demand basis.
- ◇ **Storage Virtualization:** The process of collecting physical storage from different network storage devices is called Storage Virtualization. It is mainly used for back-up and recovery.

2. Service-Oriented Architecture

SOA is an application which divides the services into individual business functions and procedure daily. This unique component of cloud application enables cloud-related arrangements that can be modified and adjusted on request as business needs. Service-oriented system diffuses two major components, one is Quality as service and other as software as service. The function of Quality of service is to identify the function and behavior of a service from a different view. Software as a service provides a new delivery model of software which is inherited from the world of application service providers

3. Grid Computing

This is a process of connecting multiple servers from multiple to achieve a common goal. Grid computing turns large problems into smaller ones and broadcast to servers and place them within the grid. It is mainly applied in e-commerce and intended to share the resources on huge scale cluster computing. A popular grid computing project is Folding@home. The project aims to find the protein folding, misfolding and related diseases. It involves utilizing the power of unused computers and solve complex scientific problems

4. Utility Computing

This process relies on the pay-per-utilize model. It gives computational services on demand for a metered benefit. It mainly helps in cost-cutting by reducing initial investment. As the computing requirements for a business change, the billing also changes accordingly, without acquiring any additional cost. If the client usage has decreased, then billing cost also reduces accordingly.

Below is the point that explains the use of cloud computing.

1. Infrastructure as a Service (IaaS)

Cloud computing offers virtual and physical computers. The actual machines are accessed by hypervisors that are grouped into pools and managed by operational supportive networks. Cloud computing introduces operating framework pictures on actual machines and application programming. Infrastructure as a service offers resources like firewalls, IP addresses, monitoring services, storages, bandwidth, virtual machines and so on, all are made available to the clients on cost on a time basis. Examples: Windows Azure, Google Compute Engine, Amazon EC2, Rackspace.

2. Platform as a Service (PaaS)

It is the part of the application development and deployment platform as an available service to programmers or developers. They can utilize the basement to build, deploy, test and handle SaaS applications effectively. The major attributes of PaaS have point-and-snap equipment that empowers the programmer to design web-based applications. Some examples are Google, Force.com, Apache, AWS elastic beanstalk and Windows Azure.

3. Software as a Service (SaaS)

It is the transmission of Applications to end-users by the web browsers. Cloud clients install it and they can enable them to run on the cloud platform. But it is not mandatory for this process because they need software support and reduced maintenance. Instead, they can use SaaS applications which is portable. The best example, an Office Suite. SaaS provides us the Application Programming Interface (API), which allows the developers to build the desired application.

Benefits of Cloud Computing Technologies

The Advantages of cloud computing are as follows.

- ◆ **High scalability:** It requests the supply of resources on a large scale without any human intervention.

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- ◇ **High availability and reliability:** The servers are available at the right time without any delay or disappointment
- ◇ **Agility:** It divides the resources effectively among the users and executes very quickly.
- ◇ **Multi-sharing:** By distributed computing, different clients from multiple areas share the same resources by fundamental infrastructure.
- ◇ **Maintenance:** It is user-friendly as they are easy to access from their place and doesn't require any installation set up.
- ◇ **Low cost:** It is very cost-effective and pays to its usage.
- ◇ **Services as pay to use mode:** Application Programming Interface is given to customers to use resources and services and pay on the service basis.
- ◇ **On-Demand Self Service:** Cloud computing offers the required services and application to the client. With the login key, they can start to use without any human interaction and cloud service providers. It includes storage and virtual machines.

4.3. Cloud Economics: Cloud Computing Infrastructures Available for Implementing Cloud Based Services

Cloud economics is the study of cloud computing's costs and benefits and the economic principles that underpin them. As a discipline, it explores key questions for businesses: What is the return on investment (ROI) of migrating to the cloud or switching current cloud providers? And what is the total cost of ownership (TCO) of a cloud solution versus a traditional on-premises solution?

When individual businesses understand the economics of cloud computing, they can optimize their investments and obtain the greatest value for their organization.

The economic benefits of cloud

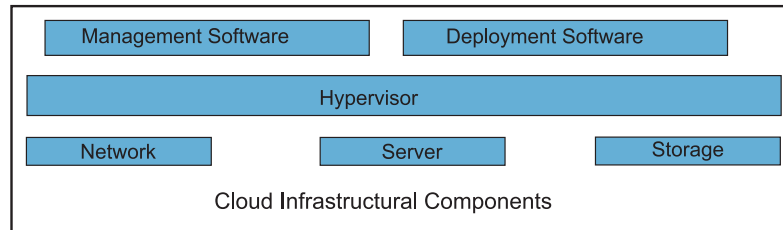
Cloud economics involves two primary principles: economies of scale and global reach. Through economies of scale, cloud providers save organizations money because they purchase computing resources in massive quantities at lower costs. When companies utilize these shared resources, they avoid the substantial up-front CAPEX costs of purchasing their own expensive infrastructure. And with a pay-as-you-go pricing model, companies pay only for the resources they actively use, scaling up or down as needed.

The global reach of cloud computing also brings substantial savings. When servers no longer need to be housed on premises—they can be located and accessed from anywhere in the world—companies can dramatically reduce labor costs. Their IT teams no longer need to devote time to deploying and maintaining complex hardware on site.

By deploying consistent infrastructure and operations across IT environments, IT teams can unlock additional operational savings, as well as reduce complexity and IT silos.

Beyond the tremendous efficiencies and cost savings of cloud computing, there is another economic benefit: business agility. Companies that utilize cloud computing resources can deploy applications faster and ramp up storage and computing power on demand. This IT agility allows businesses to respond to market changes and customer demands more quickly, leading to faster revenue growth.

Cloud infrastructure consists of servers, storage devices, network, cloud management software, deployment software, and platform virtualization.



Hypervisor

Hypervisor is a firmware or low-level program that acts as a Virtual Machine Manager. It allows to share the single physical instance of cloud resources between several tenants.

Management Software

It helps to maintain and configure the infrastructure.

Deployment Software

It helps to deploy and integrate the application on the cloud.

Network

It is the key component of cloud infrastructure. It allows to connect cloud services over the Internet. It is also possible to deliver network as a utility over the Internet, which means, the customer can customize the network route and protocol.

Server

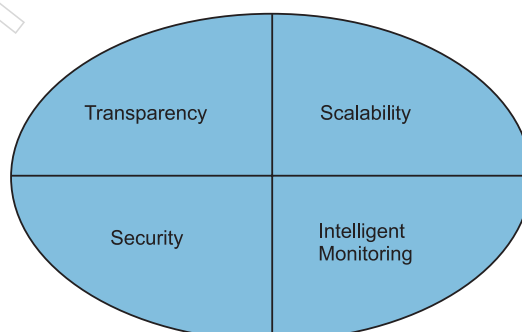
The server helps to compute the resource sharing and offers other services such as resource allocation and de-allocation, monitoring the resources, providing security etc.

Storage

Cloud keeps multiple replicas of storage. If one of the storage resources fails, then it can be extracted from another one, which makes cloud computing more reliable.

Infrastructural Constraints

Fundamental constraints that cloud infrastructure should implement are shown in the following diagram.



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Transparency

Virtualization is the key to share resources in cloud environment. But it is not possible to satisfy the demand with single resource or server. Therefore, there must be transparency in resources, load balancing and application, so that we can scale them on demand.

Scalability

Scaling up an application delivery solution is not that easy as scaling up an application because it involves configuration overhead or even re-architecting the network. So, application delivery solution is need to be scalable which will require the virtual infrastructure such that resource can be provisioned and de-provisioned easily.

Intelligent Monitoring

To achieve transparency and scalability, application solution delivery will need to be capable of intelligent monitoring.

Security

The mega data center in the cloud should be securely architected. Also the control node, an entry point in mega data center, also needs to be secure.

4.4. Economics of Choosing a Cloud Platform for an Organization based on Application Requirements, Economic Constraints and Business Needs

Platform-as-a-Service (PaaS) is the new popular kid on the block of cloud computing. But there is so much confusion around PaaS's identity that even those familiar with this whiz kid don't know how to bring the technology into action.

What is clear, however, is the need to have access to secure, agile development environments and infrastructure. This would free IT departments from repetitive maintenance and allow them to focus on responding to the ever-changing needs and requirements of their wider business stakeholders. Being able to focus IT resource on building new features into existing cloud applications, developing bespoke apps or creating custom dashboards showing real-time data are all compelling possibilities for many businesses.

Cloud platforms offer IT departments the opportunity to make the switch from being a cost centre to a revenue generator; from spending resources on maintenance to focusing on innovation. However, all cloud platforms aren't created equal. And, like many other technology investments, choosing the right cloud platform is an important decision that will have long-term implications. There are many myths circulating around cloud platforms and enterprises need to see through the haze before putting their money on the dock.

Let's look at five most common mistakes IT teams make when developing and executing their company's cloud strategy.

1. All cloud platforms do not offer the same services

The market is growing steadily as specialist cloud start-ups offer vastly different

interpretations of the services that make up a PaaS solution. Most providers offer a very limited number of prebuilt services, and many are dedicated to only one type of service, such as storage. Others cobble together a PaaS offering by purchasing smaller companies. These acquisitions are often separate entities and in no way work together easily. Often, the acquiring company has no intention of investing millions in R&D to bring the disparate acquired services together, which means it is unlikely their customers will benefit from a complete service. If an organisation is spending time and resources on integrating a provider's clouds, then one of the most significant benefits of cloud technology - simplification - is being overlooked from the start. Also, IT staff will find themselves spending too much time trying to understand services from multiple cloud providers and expending time and money trying to connect these different clouds together, something they may not even be able to do at all.

2. PaaS is not a catch-all solution to provide capabilities not found in SaaS

Some SaaS providers want organisations to use their legacy cloud platform so they don't need to improvise the functionality of their SaaS apps. It's like essentially providing organisations with an old horse and asking them to buy it a new saddle. Some cloud providers even bury expensive cloud platform development services within customers' monthly SaaS application fees. While bundling is not necessarily a problem, in some cases, these providers are really asking organisations to spend more for services that should actually be included into their basic SaaS applications as basic functionalities to begin with.

3. Non-standards-based cloud platforms or open source development languages are not necessarily cost-effective

Finding developers for a programming language that is not well-known can be difficult and time-consuming -- not to mention costly. Hard-to-find programmers are more expensive than programmers of industry-standard development languages such as Java.

4. All cloud subscriptions are not the same

Many cloud providers place numerous, complicated restrictions and limitations on how much work customers can demand of the platform for their basic subscription fee. Due to these restrictions, customers may be unable to scale their platform to quickly process large amounts of data that require significant amounts of processing power.

5. All clouds are not built using the same architecture

Many providers have designed their cloud platforms to be as cost-effective as possible for them to "run and manage" - rather than to be more convenient for their customers. This means that their customers may find themselves forced to share the same database and processing power with other clients on the same platform. To deal with this, many cloud providers cap the amount of processing available with their platform services - because this will avoid their entire cloud application going down or underperforming. However, a cloud provider that has all its customers sharing the same application creates a situation that if one customer hogs all the processing power in any given application, all the others' response times will slow down dramatically. Dealing with these caps actually adds hidden costs and development time to the process of extending or creating new applications, unlike a modern personalized cloud platform.

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Selecting the right cloud platform

Enterprises looking to invest in PaaS should select a single, well-known, standards-based cloud platform. This saves organisations significant time and money in the long run, and allows them to innovate faster while avoiding unnecessary headaches.

- ◇ They should use a provider who doesn't lock customers into its platform by using lesser-known proprietary languages or databases.
- ◇ It's also important to ensure the cloud platform connects easily with the business's SaaS applications, and makes it easy for customers to move SaaS applications to and from another platform whenever they need to.
- ◇ Choose a provider that has many prebuilt services integrated at every layer. This approach allows customers to innovate quickly while maintaining governance and standards across their entire organisation.
- ◇ Ultimately, security is the key. It is paramount that the cloud platform is built on the highest standard of enterprise security to safeguard not only the cloud investment but also the business.

Making the business case for cloud economics

Before making the leap to cloud, businesses should analyze the economic pros and cons in depth to get a detailed picture of specific costs and savings. Will it lead to long-term savings and efficiencies? The answers will vary depending on the organizational needs and circumstances and on the cloud solution being considered. The goal is to avoid a cloud adoption strategy that drives up cost, complexity and staffing resources.

When exploring cloud economics for their company, IT and finance managers can follow a basic process to determine cloud computing ROI and TCO, and use those estimates to help make their case to executives. The process should include these three elements:

- ◇ **Benchmarking:** Calculate the cost of operating your current data center, including capital costs over the equipment lifespan, labor costs and any other maintenance and operational costs, from licenses and software to spare parts.
- ◇ **Cloud costs:** Estimate the costs of the cloud infrastructure you're considering (public cloud, private cloud, hybrid cloud, etc.). You'll need a quote from your vendor, but look beyond this basic pricing structure to consider ongoing fees, labor and training costs, ongoing integration and testing of apps, as well as security and compliance.
- ◇ **Migration costs:** Determine the cost to migrate IT operations to the cloud or to switch cloud providers. These costs should include labor and expenses to integrate and test apps.

Summary

Cloud computing has a unique feature as broad network access in which wide resources like storage and virtual machines can be accessed easily with the mobile phone, personal laptops, and computers. Hence it can be accessed at any time. The Resource Pooling allows multiple users to share a common pool like database,

applications and web pages and provides rapid elasticity to resources used by clients or recently assigned to clients is automatically monitored. It is very possible to scale the resource up and down at any time.

Cloud computing is a transformational technology that has helped organizations deliver their products/services and cope with cyber-security issues, managing big data, and quality control. Organizations ranging from startups to MNCs have adopted cloud computing tools to launch their applications and automate business operations. Moving to cloud computing has reduced IT costs, flexibility to scale down or scale up as per business requirements, and last but not the least it allows you to access your data anytime, anywhere.

Keywords

Community Cloud Computing (C3): Community Cloud Computing offers an alternative architecture, created by combining the Cloud with paradigms from Grid Computing, principles from Digital Ecosystems, and sustainability from Green Computing, while remaining true to the original vision of the Internet.

Infrastructure-as-a-Service (IaaS): Infrastructure as a Service is a provision model in which an organization outsources the equipment used to support operations, including storage, hardware, servers and networking components.

Platform-as-a-Service (PaaS): Platform as a service is a category of cloud computing services that provide a computing platform and a solution stack as a service. In the classic layered model of cloud computing, the PaaS layer lies between the SaaS and the IaaS layers.

Quality of Service (QoS): The quality of service refers to several related aspects of telephony and computer networks that allow the transport of traffic with special requirements.

Software-as-a-Service (SaaS): Software as a Service is a software distribution model in which applications are hosted by a vendor or service provider and made available to customers over a network, typically the Internet.

Self-Assessment Questions

- 1) Which one of the following a technology works behind the cloud computing platform?
 - a) Virtualization
 - b) SOA
 - c) Grid Computing
 - d) All of the above
- 2) Which one of the following is a kind of technique that allows sharing the single physical instance of an application or the resources among multiple organizations/customers?
 - a) Virtualization
 - b) Service-Oriented Architecture
 - c) Grid Computing
 - d) Utility Computing

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- 3) Which one of the following statement is true about the Virtualization?
- a) It provides a logical name for a physical resource, and on-demand provides an indicator of that physical resource.
 - b) In Virtualization, we analyze the strategy related problems that customers may face.
 - c) In Virtualization, it is necessary to compile the Multitenant properly.
 - d) All of the above
- 4) In Virtualization, which architecture provides the virtual isolation between the several tenants?
- a) IT Architecture
 - b) Multitenant
 - c) Deployment
 - d) Business Architecture
- 5) On which one of the following utility computing is based?
- a) Grid Computing Model
 - b) SOA Model
 - c) virtual isolation Model
 - d) Pay-per-Use model
- 6) Which one of the following statement is true about the Service-Oriented Architecture?
- a) It is possible to exchange data between applications from different vendors without using additional programming.
 - b) It provides computational resources on-demand as a metered service.
 - c) Service-Oriented Architecture allows using the application as a service for other applications.
 - d) Both A and C
- 7) In Grid Computing, which types of computer resources are there?
- a) heterogeneous dispersed.
 - b) geographically dispersed.
 - c) Both A and B
 - d) None of the above
- 8) Managed IT services are based on the concept of which one of the following?
- a) Virtualization
 - b) Utility Computing
 - c) Grid Computing
 - d) SOA

- 9) Which one of the following refers to the Distributed Computing, in which several sets of computers distributed on multiple geographical locations and are connected with each other to achieve a common goal?
- a) Virtualization
 - b) SOA
 - c) Grid Computing
 - d) Utility Computing
- 10) Which one of the following statement is true about Grid Computing?
- a) Pieces combine small tasks into complex tasks
 - b) The subscription tier plays an important role in grid computing.
 - c) Breaks complex tasks into small operations
 - d) Both A and C

Review Questions

1. Explain Reliability, availability and security of services deployed from cloud
2. What is the Performance and scalability of services? also explain tools and technologies used to manage cloud services deployment?
3. Explain Cloud Economics and Cloud Computing infrastructures available for implementing cloud based services.
4. Discuss the Economics of choosing a Cloud platform for an organization based on application requirements, economic constraints and business needs

Answers 1)D 2)A 3)A 4)B 5)D 6)D 7)C 8)B 9)C 10)C

Notes

Module-IV

Structure:

5. Case Study: Application Development
 - 5.1 Service creation environments to develop cloud based applications
 - 5.2 Development environments for service development
 - 5.3 Amazon, Azure, Google App

Unit-5: Case Study: Application Development

Notes

Objective:

After studying this chapter, you will be able to:

- Explain the meaning of Application Development
- Discuss the Service creation environments to develop cloud based applications
- Explain the Development environments for service development
- Discuss the Amazon, Azure, Google App

Introduction

Cloud-based applications, also known as Cloud apps, seem to be taking over. In theory, a Cloud app is one that uses Cloud-based services. So, whether an app is mobile or web, they probably use some sort of Cloud service. What really differentiates a Cloud app from a native one is the extent to which they use Cloud services.

Increased dependence on the Cloud's processing power is the result of companies building innovative and creative solutions to all sorts of problems that use technology to do things that were previously impossible. Thanks to the ability to process large amounts of data (Big Data) through third party owned IT infrastructure, companies can perform massive calculations and deliver top services.

In particular, Cloud services have opened up the possibility for many web-based Cloud applications, also known as web apps. A web app is one where most of the computation occurs in the Cloud, not on the device itself, and usually built with the use of Cloud application development services. A new form of web app, known as a Progressive Web App (PWA), is also seeing an increase in popularity.

Benefits of a Cloud App

Cloud application development offers various benefits for businesses that wish to use technology to solve a problem. Some of the benefits are:

- ◇ **Improved app performance:** as more computations are performed on the server side of an app, users will experience a faster and more reliable service.
- ◇ **Increased uptime:** thanks to the reliability of Cloud services, a Cloud-based application will remain up easier than through your own IT infrastructure.
- ◇ **Scalability:** businesses can hire on-demand the processing power they need, being this very convenient for moments of high computer processing demand.
- ◇ **Update software easily:** through Cloud technologies, it is possible to update an app easily through a massive deployment.
- ◇ **Security:** Cloud services help reduce the risk of physical IT infrastructure failure.

Cloud Application Development: Developing Applications for the Cloud

Cloud application development is the process through which a Cloud-based app is built. It involves different stages of software development, each of which prepares your app to go live and hit the market. The best Cloud app development teams use DevOps

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practices and tools like Kubernetes. However, an experienced app development company should ideally be technology agnostic, which means being able to build your Cloud app using any technology you prefer. Most apps built using the Cloud are highly dependent on the Cloud to operate.

Application development on Cloud infrastructure allows web and PWA development services to reduce development costs, opens up the possibility to work with remote teams, and reduces project times if used correctly with software development methodologies like Agile. However, not all companies are experienced enough to perform many complex aspects of the app development process using the Cloud. Businesses looking to develop digital products like web-based Cloud applications need to make sure that they work with a trusted Cloud-experienced app development company.

Although some businesses have their own Cloud development teams, most will hire an app development company with experience in Cloud services. A great way to verify an app development company's experience with the Cloud is through certifications like AWS. Koombea, for example, is a certified AWS partner.

5.1. Service Creation Environments to Develop Cloud Based Applications

Cloud app development: Key differences

What about the specifics of developing cloud applications?

- ◇ Developing a cloud application requires deep interaction between programmers, data architects, designers, and quality assurance managers. Developers need to be familiar with various cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, Force.com, and Apache CloudStack. In addition, they should consider connectivity with APIs (application programming interfaces) and CDNs (content delivery networks).
- ◇ Your development team must consider that your final solution should be scalable. One of the important reasons why companies choose to store their data in the cloud is that cloud storage is easy to expand, even within a short period of time.
- ◇ Cloud applications can be technically unlimited in size, but cloud hosting isn't free. Concentrating user requests and optimizing data size should be top development priorities.
- ◇ To convince people to use your application, you need to ensure their data is stored securely, which isn't always easy since you're dealing with cloud technologies that don't have a single data store. This is why an app's codebase should be kept separate from other data.

Cloud-based application development step by step

Developing a cloud application is different from developing a web or mobile application. A mobile app development team builds a cloud solution by relying on your chosen service provider. Amazon Web Services (AWS) is probably the best and most reliable solution on the market right now. It offers a number of great tools and features for developing cloud applications.

You should be willing to invest time and money in creating any digital product. Cloud solutions are no exception. Before you start development, you need to understand the problems your app users face and find a way to solve them using your product.

Step #1. Research your app's target market

When developing a cloud-based mobile app, the first thing you need to consider is the target audience. Understanding your users' needs makes development easier and leads to a better final product. Find out as much as you can about your potential users. You can start by researching the following:

- ◇ **Demographics.** Find out the average age of your users, where they live, what devices they use, etc.
- ◇ **Behavioral trends.** Find out what decreases a user's desire to download an app, your users' security expectations, and so on.

To develop an amazing mobile application, we suggest creating a user persona, or a detailed portrait of your ideal user.

Step #2. Hire a development team

The second step is to find a development team you want to work with. The first phase of development will include business analysis, developing a technical specification, estimating the development cost, and planning the project.

Before diving into the actual development of your mobile app, you and your team should create an app development workflow, choose the main features for the app, and design an app concept. Then your development team should create project milestones and start working on the MVP.

Step #3. Consider the architecture and features

For your application to be successful, you need to carefully consider the architecture and service model. These decisions affect your application's performance, so it's best to consult with specialists who can advise you.

Architecture

It's a good idea to create an advanced data architecture. Classic solutions are always reliable. However, for cloud applications, a microservices architecture is commonly used.

Service model

The service model you select — SaaS, PaaS, IaaS — must match the type of cloud solution you're developing. For example, when developing an application like Slack, you need to take a SaaS approach.

Step #4. Define the tech stack

When choosing tools for developing cloud applications, you should consult with experts. They'll analyze your requirements, features, and designs to select the right set of technologies for your product.

Also, be mindful of your application's scalability to keep your solution up to date.

Notes

Here's a possible tech stack for a cloud-based application:

Application and data

- ◇ Akamai
- ◇ Cloudant
- ◇ ClearDB

Utilities

- ◇ Google Analytics
- ◇ Twilio
- ◇ Optimizely
- ◇ Heap
- ◇ Recurly
- ◇ Zuora
- ◇ Cyfe
- ◇ TransmogrifAI
- ◇ DevOps
- ◇ Jenkins
- ◇ Bitbucket
- ◇ New Relic
- ◇ Datadog
- ◇ Puppet Labs
- ◇ Cloud9 IDE
- ◇ Sauce Labs
- ◇ StillAlive
- ◇ Business tools
- ◇ Jira
- ◇ G Suite
- ◇ InVision
- ◇ Salesforce Sales Cloud
- ◇ Balsamiq
- ◇ DocuSign
- ◇ UXPin

Step #5. Choose a monetization model

The next step is to choose the right monetization model for your mobile application. Now that you know your users' needs, you can predict what your users will pay for. Here are three monetization models to choose from:

- ◇ **Paid.** This monetization model is quite straightforward: users pay once to access your app.

- ◇ **Freemium.** With this model, users can download your app for free. They can then pay to upgrade their accounts or use premium features.
- ◇ **In-app purchases.** With in-app purchases, users can pay for different items, features, or content inside the app.

Step #6. Create an MVP

Creating a cloud-based app is a big and complex project. We recommend launching a minimum viable product (MVP) first and testing its technical and business performance. By using an MVP approach, you'll be able to find out what users like and don't like in your app. Then you'll be able to consider their feedback and improve your app.

Step #7. Test your product carefully

Cloud-based app development should include a testing stage. Before launching your product, your development team has to test it to find any bugs.

At this point, you'll verify that your application is working correctly and provides a satisfying user experience. To do this, it's best to cooperate with a full-cycle development company.

Full-cycle development companies offer development, design, testing, and management services. With one team working on your project from start to finish, communication is vastly simplified. This results in higher product quality.

Mobile and Web App Development

Are you planning to expand your business online? We will translate your ideas into intelligent and powerful solutions.

Step #8. Launch the app and keep it up to date

You can release your app on the App Store (iOS) and Google Play (Android). Google Play uses automated testing to speed up the app store approval process. However, if your application is rejected by Google, it can be difficult to find out why.

The App Store delegates app validation to real people. If validators don't approve your app, they'll ask you to make specific changes.

If you want to distribute your app exclusively within your organization via the App Store, you'll need to pay \$299 a year to join the Apple Developer Enterprise Program. Google Play doesn't charge for its analogous service.

Here's a list of information you need to prepare before submitting your application.

For Google Play:

- ◇ Screenshots
- ◇ App name
- ◇ Description keywords
- ◇ Support URL
- ◇ Marketing URL
- ◇ Privacy policy URL

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- ◇ App icon
- ◇ Categories
- ◇ Rating
- ◇ Copyright
- ◇ Demo account
- ◇ Version information
- ◇ Pricing information

For the App Store:

- ◇ Title (app name)
- ◇ Short description
- ◇ Full description
- ◇ Screenshots
- ◇ High-resolution icon
- ◇ Featured graphic
- ◇ Promo video (optional)
- ◇ Type and category
- ◇ Content rating
- ◇ Languages and translations (if any)
- ◇ Contact details
- ◇ Privacy policy
- ◇ Compatible devices
- ◇ Pricing and distribution

Some of the materials listed will cost you nothing to produce, while others will be quite expensive. Creating a copyright and privacy policy usually takes time and expensive legal services. How much does it cost to list an app on the App Store and Google Play if a development company helps you? Releasing an application can take different amounts of time depending on the amount of work the company has to do.

Also, remember that before your app is published on either app store, it must go through an approval process. This procedure can take some time and require additional development costs. If your app doesn't meet platform rules or requirements, it won't be accepted.

If your app isn't accepted, you may need to make a few changes in order to get it approved. Some mobile app development companies provide their services until your app gets approved, but others don't.

How much does it cost to build a cloud-based app?

The cost of developing a cloud-based mobile app depends on these factors:

- ◇ Product features
- ◇ Product design
- ◇ Hourly rate of your development team
- ◇ Project size and complexity

- ◇ Technology stack
- ◇ Number of team members on the project
- ◇ Time frame

The biggest factor that influences a project's cost is the hourly rate of developers, and that often depends on their location. For example, an application that costs \$40,000 to build in the US will cost around \$9,000 if developed by engineers in Ukraine.

Developing your project with a company based in Eastern Europe is cost-effective and gives you access to top talent.

Quick summary of developing a cloud-based app

Developing a cloud-based app is hard and time-consuming. But if done correctly, your app can become successful and bring a lot of value to users.

Key takeaways:

- ◇ A cloud-based application is a software application that is deployed in a cloud environment.
- ◇ Your development team should consider that the final solution must be scalable. One important reason why companies choose to store their data in the cloud is that cloud storage is easy to expand, even within a short period of time.
- ◇ Cloud applications can be technically unlimited in size, but cloud hosting isn't free. Concentrating user requests and optimizing data size should be top development priorities.
- ◇ When developing a mobile app, the first thing you need to consider is your target audience.
- ◇ Choose the right monetization model for your mobile application: free, paid, freemium, or ad-based.
- ◇ Find a development team to work with. The first phase of development will include business analysis, creating a technical specification, estimating the development cost, and planning the project.
- ◇ We recommend launching a minimum viable product (MVP) first and testing its technical and business performance.

5.2. Development Environments for Service Development

Development environments as a service is an extra layer of abstraction which will automatically configure the source code, runtime, compiler, debugger, editor, relevant editor extensions and more. Dev environments are fast to create and disposable, allowing new team members to quickly on board to a project, or for you to experiment with a new stack, language, or code base, without worrying about it affecting your local configuration.

Why do we need cloud development environments?

While generally moving your development environment to the cloud can keep you from installing and re-installing software locally, there are some more specific benefits, too.

- ◇ Try out a new language, framework, or tool without installing it locally

Notes

- ◇ Ensure your whole team has the same environment setup
- ◇ Use a separate environment for every client to keep projects separate
- ◇ Run unknown code without exposing your machine
- ◇ Help customers get up to speed quickly with your developer tool with ready-to-code environments
- ◇ Switch between computers whenever you want, without taking time to get set up

We have seen the benefits of cloud development environments but what if we could also integrate the ide or the code editors which the developers use for writing code. These ide's could be ported to the browser and provide excellent solution for writing code in reliable and consistent environments which could be accessed from anywhere.

Cloud IDE's have been around for quite some time now, some of the popular cloud ide's are –

- ◇ Cloud9
- ◇ CodeAnywhere
- ◇ CodeEnvy
- ◇ CodeTasty
- ◇ Repl.it & many more

Final thoughts!

With more and more developers, freelancers and teams working remotely and collaborating, there has been a considerable growth of services which provide development environments. Developers are seeking techniques and technologies to help them collaborate more quickly and increase productivity. They do not want to waste time on installation and setting up of the environments anymore.

Cloud Development Environments are surely the future of devops and continuous delivery. The developers could reap the benefits of private run-time containers of multiple languages and stacks with one click solution.

5.3. Amazon, Azure, Google App

5.3.1. Amazon

The full form of AWS is Amazon Web Services. It is a platform that offers flexible, reliable, scalable, easy-to-use and, cost-effective cloud computing solutions.

AWS is a comprehensive, easy to use computing platform offered Amazon. The platform is developed with a combination of infrastructure as a service (IaaS), platform as a service (PaaS) and packaged software as a service (SaaS) offerings.

History of AWS

- ◇ 2002- AWS services launched
- ◇ 2006- Launched its cloud products
- ◇ 2012- Holds first customer event
- ◇ 2015- Reveals revenues achieved of \$4.6 billion

- ◇ 2016- Surpassed \$10 billion revenue target
- ◇ 2016- Release snowball and snowmobile
- ◇ 2019- Offers nearly 100 cloud services
- ◇ 2021- AWS comprises over 200 products and services

Important AWS Services

Amazon Web Services offers a wide range of different business purpose global cloud-based products. The products include storage, databases, analytics, networking, mobile, development tools, enterprise applications, with a pay-as-you-go pricing model.

AWS Compute Services

Here, are Cloud Compute Services offered by Amazon:

1. **EC2(Elastic Compute Cloud)**- EC2 is a virtual machine in the cloud on which you have OS level control. You can run this cloud server whenever you want.
2. **LightSail**- This cloud computing tool automatically deploys and manages the computer, storage, and networking capabilities required to run your applications.
3. **Elastic Beanstalk**- The tool offers automated deployment and provisioning of resources like a highly scalable production website.
4. **EKS (Elastic Container Service for Kubernetes)**- The tool allows you to Kubernetes on Amazon cloud environment without installation.
5. **AWS Lambda**- This AWS service allows you to run functions in the cloud. The tool is a big cost saver for you as you to pay only when your functions execute.

Migration

Migration services used to transfer data physically between your datacenter and AWS.

1. **DMS (Database Migration Service)**– DMS service can be used to migrate on-site databases to AWS. It helps you to migrate from one type of database to another — for example, Oracle to MySQL.
2. **SMS (Server Migration Service)**– SMS migration services allows you to migrate on-site servers to AWS easily and quickly.
3. **Snowball**— Snowball is a small application which allows you to transfer terabytes of data inside and outside of AWS environment.

Storage

1. **Amazon Glacier**- It is an extremely low-cost storage service. It offers secure and fast storage for data archiving and backup.
2. **Amazon Elastic Block Store (EBS)**- It provides block-level storage to use with Amazon EC2 instances. Amazon Elastic Block Store volumes are network-attached and remain independent from the life of an instance.
3. **AWS Storage Gateway**- This AWS service is connecting on-premises software applications with cloud-based storage. It offers secure integration between the company's on-premises and AWS's storage infrastructure.

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Security Services

1. **IAM (Identity and Access Management)**— IAM is a secure cloud security service which helps you to manage users, assign policies, form groups to manage multiple users.
2. **Inspector**— It is an agent that you can install on your virtual machines, which reports any security vulnerabilities.
3. **Certificate Manager**— The service offers free SSL certificates for your domains that are managed by Route53.
4. **WAF (Web Application Firewall)**— WAF security service offers application-level protection and allows you to block SQL injection and helps you to block cross-site scripting attacks.
5. **Cloud Directory**— This service allows you to create flexible, cloud-native directories for managing hierarchies of data along multiple dimensions.
6. **KMS (Key Management Service)**— It is a managed service. This security service helps you to create and control the encryption keys which allows you to encrypt your data.
7. **Organizations**— You can create groups of AWS accounts using this service to manages security and automation settings.
8. **Shield**— Shield is managed DDoS (Distributed Denial of Service protection service). It offers safeguards against web applications running on AWS.
9. **Macie**— It offers a data visibility security service which helps classify and protect your sensitive critical content.
10. **GuardDuty**— It offers threat detection to protect your AWS accounts and workloads.

Database Services

1. **Amazon RDS**- This Database AWS service is easy to set up, operate, and scale a relational database in the cloud.
2. **Amazon DynamoDB**- It is a fast, fully managed NoSQL database service. It is a simple service which allow cost-effective storage and retrieval of data. It also allows you to serve any level of request traffic.
3. **Amazon ElastiCache**- It is a web service which makes it easy to deploy, operate, and scale an in-memory cache in the cloud.
4. **Neptune**- It is a fast, reliable and scalable graph database service.
5. **Amazon RedShift**- It is Amazon's data warehousing solution which you can use to perform complex OLAP queries.

Analytics

1. **Athena**— This analytics service allows perm SQL queries on your S3 bucket to find files.
2. **CloudSearch**— You should use this AWS service to create a fully managed search engine for your website.

3. **ElasticSearch**— It is similar to CloudSearch. However, it offers more features like application monitoring.
4. **Kinesis**— This AWS analytics service helps you to stream and analyzing real-time data at massive scale.
5. **QuickSight**— It is a business analytics tool. It helps you to create visualizations in a dashboard for data in Amazon Web Services. For example, S3, DynamoDB, etc.
6. **EMR (Elastic Map Reduce)**— This AWS analytics service mainly used for big data processing like Spark, Splunk, Hadoop, etc.
7. **Data Pipeline**— Allows you to move data from one place to another. For example from DynamoDB to S3.

Management Services

1. **CloudWatch**— Cloud watch helps you to monitor AWS environments like EC2, RDS instances, and CPU utilization. It also triggers alarms depends on various metrics.
2. **CloudFormation**— It is a way of turning infrastructure into the cloud. You can use templates for providing a whole production environment in minutes.
3. **CloudTrail**— It offers an easy method of auditing AWS resources. It helps you to log all changes.
4. **OpsWorks**— The service allows you to automated Chef/Puppet deployments on AWS environment.
5. **Config**— This AWS service monitors your environment. The tool sends alerts about changes when you break certain defined configurations.
6. **Service Catalog**— This service helps large enterprises to authorize which services user will be used and which won't.
7. **AWS Auto Scaling**— The service allows you to automatically scale your resources up and down based on given CloudWatch metrics.
8. **Systems Manager**— This AWS service allows you to group your resources. It allows you to identify issues and act on them.
9. **Managed Services**— It offers management of your AWS infrastructure which allows you to focus on your applications.

Internet of Things

1. **IoT Core**— It is a managed cloud AWS service. The service allows connected devices like cars, light bulbs, sensor grids, to securely interact with cloud applications and other devices.
2. **IoT Device Management**— It allows you to manage your IoT devices at any scale.
3. **IoT Analytics**— This AWS IOT service is helpful to perform analysis on data collected by your IoT devices.
4. **Amazon FreeRTOS**— This real-time operating system for microcontrollers helps you to connect IoT devices in the local server or into the cloud.

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Application Services

1. **Step Functions**— It is a way of visualizing what's going inside your application and what different microservices it is using.
2. **SWF (Simple Workflow Service)**— The service helps you to coordinate both automated tasks and human-led tasks.
3. **SNS (Simple Notification Service)**— You can use this service to send you notifications in the form of email and SMS based on given AWS services.
4. **SQS (Simple Queue Service)**— Use this AWS service to decouple your applications. It is a pull-based service.
5. **Elastic Transcoder**— This AWS service tool helps you to changes a video's format and resolution to support various devices like tablets, smartphones, and laptops of different resolutions.

Deployment and Management

1. **AWS CloudTrail**: The services records AWS API calls and send backlog files to you.
2. **Amazon CloudWatch**: The tools monitor AWS resources like Amazon EC2 and Amazon RDS DB Instances. It also allows you to monitor custom metrics created by user's applications and services.
3. **AWS CloudHSM**: This AWS service helps you meet corporate, regulatory, and contractual, compliance requirements for maintaining data security by using the Hardware Security Module(HSM) appliances inside the AWS environment.

Developer Tools

1. **CodeStar**— Codestar is a cloud-based service for creating, managing, and working with various software development projects on AWS.
2. **CodeCommit**— It is AWS's version control service which allows you to store your code and other assets privately in the cloud.
3. **CodeBuild**— This Amazon developer service help you to automates the process of building and compiling your code.
4. **CodeDeploy**— It is a way of deploying your code in EC2 instances automatically.
5. **CodePipeline**— It helps you create a deployment pipeline like testing, building, testing, authentication, deployment on development and production environments.
6. **Cloud9**— It is an Integrated Development Environment for writing, running, and debugging code in the cloud.

Mobile Services

1. **Mobile Hub**— Allows you to add, configure and design features for mobile apps.
2. **Cognito**— Allows users to signup using his or her social identity.
3. **Device Farm**— Device farm helps you to improve the quality of apps by quickly testing hundreds of mobile devices.

4. **AWS AppSync**— It is a fully managed GraphQL service that offers real-time data synchronization and offline programming features.

Business Productivity

1. **Alexa for Business**— It empowers your organization with voice, using Alexa. It will help you to Allows you to build custom voice skills for your organization.
2. **Chime**— Can be used for online meeting and video conferencing.
3. **WorkDocs**— Helps to store documents in the cloud
4. **WorkMail**— Allows you to send and receive business emails.

Desktop & App Streaming

1. **WorkSpaces**— Workspace is a VDI (Virtual Desktop Infrastructure). It allows you to use remote desktops in the cloud.
2. **AppStream**— A way of streaming desktop applications to your users in the web browser. For example, using MS Word in Google Chrome.

Artificial Intelligence

1. **Lex**— Lex tool helps you to build chatbots quickly.
2. **Polly**— It is AWS's text-to-speech service allows you to create audio versions of your notes.
3. **Rekognition** — It is AWS's face recognition service. This AWS service helps you to recognize faces and object in images and videos.
4. **SageMaker**— Sagemaker allows you to build, train, and deploy machine learning models at any scale.
5. **Transcribe**— It is AWS's speech-to-text service that offers high-quality and affordable transcriptions.
6. **Translate**— It is a very similar tool to Google Translate which allows you to translate text in one language to another.

AR & VR (Augmented Reality & Virtual Reality)

1. **Sumerian**— Sumerian is a set of tool for offering high-quality virtual reality (VR) experiences on the web. The service allows you to create interactive 3D scenes and publish it as a website for users to access.

Customer Engagement

1. **Amazon Connect**— Amazon Connect allows you to create your customer care center in the cloud.
2. **Pinpoint**— Pinpoint helps you to understand your users and engage with them.
3. **SES (Simple Email Service)**— Helps you to send bulk emails to your customers at a relatively cost-effective price.

Game Development

1. **GameLift**— It is a service which is managed by AWS. You can use this service to

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host dedicated game servers. It allows you to scale seamlessly without taking your game offline.

Applications of AWS services

Amazon Web services are widely used for various computing purposes like:

- ◇ Web site hosting
- ◇ Application hosting/SaaS hosting
- ◇ Media Sharing (Image/ Video)
- ◇ Mobile and Social Applications
- ◇ Content delivery and Media Distribution
- ◇ Storage, backup, and disaster recovery
- ◇ Development and test environments
- ◇ Academic Computing
- ◇ Search Engines
- ◇ Social Networking

Companies using AWS

- ◇ Instagram
- ◇ Netflix
- ◇ Twitch
- ◇ LinkedIn
- ◇ Facebook
- ◇ Turner Broadcasting: \$10 million
- ◇ Zoopla
- ◇ Smugmug
- ◇ Pinterest
- ◇ Dropbox

Advantages of AWS

Following are the pros of using AWS services:

- ◇ AWS allows organizations to use the already familiar programming models, operating systems, databases, and architectures.
- ◇ It is a cost-effective service that allows you to pay only for what you use, without any up-front or long-term commitments.
- ◇ You will not require to spend money on running and maintaining data centers.
- ◇ Offers fast deployments
- ◇ You can easily add or remove capacity.
- ◇ You are allowed cloud access quickly with limitless capacity.
- ◇ Total Cost of Ownership is very low compared to any private/dedicated servers.
- ◇ Offers Centralized Billing and management

- ◇ Offers Hybrid Capabilities
- ◇ Allows you to deploy your application in multiple regions around the world with just a few clicks

Disadvantages of AWS

- ◇ If you need more immediate or intensive assistance, you'll have to opt for paid support packages.
- ◇ Amazon Web Services may have some common cloud computing issues when you move to a cloud. For example, downtime, limited control, and backup protection.
- ◇ AWS sets default limits on resources which differ from region to region. These resources consist of images, volumes, and snapshots.
- ◇ Hardware-level changes happen to your application which may not offer the best performance and usage of your applications.

Best practices of AWS

- ◇ You need to design for failure, but nothing will fail.
- ◇ It's important to decouple all your components before using AWS services.
- ◇ You need to keep dynamic data closer to compute and static data closer to the user.
- ◇ It's important to know security and performance tradeoffs.
- ◇ Pay for computing capacity by the hourly payment method.
- ◇ Make a habit of a one-time payment for each instance you want to reserve and to receive a significant discount on the hourly charge.

5.3.2. Azure

Microsoft Azure is a cloud computing platform that provides a wide variety of services that we can use without purchasing and arranging our hardware. It enables the fast development of solutions and provides the resources to complete tasks that may not be achievable in an on-premises environment. Azure Services like compute, storage, network, and application services allow us to put our effort into building great solutions without worrying about the assembly of physical infrastructure.

This tutorial covers the fundamentals of Azure, which will provide us the idea about all the Azure key services that we are most likely required to know to start developing solutions. After completing this tutorial, we can crack job interviews or able to get different Microsoft Azure certifications.

What is Azure

Microsoft Azure is a growing set of cloud computing services created by Microsoft that hosts your existing applications, streamline the development of a new application, and also enhances our on-premises applications. It helps the organizations in building, testing, deploying, and managing applications and services through Microsoft-managed data centers.

Azure Services

- ◇ **Compute services:** It includes the Microsoft Azure Cloud Services,

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Azure Virtual Machines, Azure Website, and Azure Mobile Services, which processes the data on the cloud with the help of powerful processors.

- ◇ **Data services:** This service is used to store data over the cloud that can be scaled according to the requirements. It includes Microsoft Azure Storage (Blob, Queue Table, and Azure File services), Azure SQL Database, and the Redis Cache.
- ◇ **Application services:** It includes services, which help us to build and operate our application, like the Azure Active Directory, Service Bus for connecting distributed systems, HDInsight for processing big data, the Azure Scheduler, and the Azure Media Services.
- ◇ **Network services:** It helps you to connect with the cloud and on-premises infrastructure, which includes Virtual Networks, Azure Content Delivery Network, and the Azure Traffic Manager.

How Azure works

It is essential to understand the internal workings of Azure so that we can design our applications on Azure effectively with high availability, data residency, resilience, etc.

Microsoft Azure is completely based on the concept of virtualization. So, similar to other virtualized data center, it also contains racks. Each rack has a separate power unit and network switch, and also each rack is integrated with a software called Fabric-Controller. This Fabric-controller is a distributed application, which is responsible for managing and monitoring servers within the rack. In case of any server failure, the Fabric-controller recognizes it and recovers it. And Each of these Fabric-Controller is, in turn, connected to a piece of software called Orchestrator. This Orchestrator includes web-services, Rest API to create, update, and delete resources.

When a request is made by the user either using PowerShell or Azure portal. First, it will go to the Orchestrator, where it will fundamentally do three things:

1. Authenticate the User
2. It will Authorize the user, i.e., it will check whether the user is allowed to do the requested task.
3. It will look into the database for the availability of space based on the resources and pass the request to an appropriate Azure Fabric controller to execute the request.

Combinations of racks form a cluster. We have multiple clusters within a data center, and we can have multiple Data Centers within an Availability zone, multiple Availability zones within a Region, and multiple Regions within a Geography.

- ◇ **Geographies:** It is a discrete market, typically contains two or more regions, that preserves data residency and compliance boundaries.
- ◇ **Azure regions:** A region is a collection of data centers deployed within a defined perimeter and interconnected through a dedicated regional low-latency network.

Azure covers more global regions than any other cloud provider, which offers the scalability needed to bring applications and users closer around the world. It is globally available in 50 regions around the world. Due to its availability over many regions, it

helps in preserving data residency and offers comprehensive compliance and flexible options to the customers.

- ◇ **Availability Zones:** These are the physically separated location within an Azure region. Each one of them is made up of one or more data centers, independent configuration.

Azure Pricing

It is one of the main reasons to learn Microsoft Azure. Because Microsoft is providing free Credits in the Azure account to access Azure services for free for a short duration. This credit is sufficient for people who are new at Microsoft Azure and want to use the services.

Microsoft offers the pay-as-you-go approach that helps organizations to serve their needs. Typically the cloud services will be charged based on the usage. The flexible pricing option helps in up-scaling and down-scaling the architecture as per our requirements.

Azure Certification

Microsoft Azure helps to fill the gap between the industry requirement and the resource available. Microsoft provides Azure Certification into three major categories, which are:

- ◇ **Azure Administrator:** Those who implement, monitor, and maintain Microsoft Azure solutions, including major services.
- ◇ **Azure Developer:** Those who design, build, test, and maintain cloud solutions, such as applications and services, partnering with cloud solution architects, cloud DBAs, cloud administrators, and clients to implement these solutions.
- ◇ **Azure Solution Architect:** Those who have expertise in compute, network, storage, and security so that they can design the solutions that run on Azure.

All these certifications are divided into different levels. If anyone is planning to get certified, then he/she first has to get an associate-level certification and then go for the advanced level.

Prerequisite

Before Learning AWS, one should have basic knowledge of cloud computing and computer fundamentals.

Audience

Our Microsoft Azure tutorial is designed for students and working IT professionals who are new to Cloud Computing and want to pursue or switch their career path as Microsoft Azure Developer or Administrator.

5.3.3. Google App

The Google App can be confusing. You see it on your phone screen with just a big G that suggests "Google". You may not even end up using it as you already use Chrome for all your web searches and site visits. You may be using Google Maps to go from one place to the next. And then there are other apps that help you with other tasks.

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The Google app is the Google Search app.

But it also does a bit more than the standard Google Search Bar on your browser to help you discover the world around you. Think of it as a single window into the outside world. You can then customize it to discover information around your special interests and stay informed with personalized updates.

You can get the same results with any Google Search, but the Google app just makes it easier to stay in control of the glut out there.

The Google Search app can make your searches faster. So, let's jump into the app and see what it can do.

What Does the Google App Do?

Start with a bit of fun. Tap the Google logo on the home page. If there's no special Google Doodle, the colorful dots do a short jig on the screen. Tap it again to check out all its dance moves.

Tap the weather icon (that displays today's temperature) to see a forecast on the Google Search results page. The details pale in comparison to that of a weather app, but it's sufficient to take in with a glance.

The default home screen gives you news stories, sports scores, and anything else that's happening around the world right now. Sign into the app with your Google account to customize the information you want to see.

The main part of the app is Google Search. Tap on it to start your search or select a search from the What's Trending list below it.

The Google App home screen displays a feed of articles which it thinks you'll be interested in. This data comes from your search history, location, and other data that Google has about you. You can turn off web and app activity monitoring or delete it from the app itself by visiting the Activity Controls page.

What Is Collections on the Google App?

Collections is a group of articles you can collate from your search results. It's a bookmarking feature and uses the same icon next to the URL on the app. You can add anything from the search page to a collection like images, trips, and itineraries.

Add multiple collections and access them from the Collections screen. Edit or delete them anytime. You can also share your collections with others via the app.

Tap the Collection you want to share. Configure the share settings. As you can see in the screenshot below, you have two options.

- ◇ Share it with a view-only link.
- ◇ Share it with a Contributor link so others can add to the collection.

You can use Collections for research and then use the Contributor link to seek the help from others, all from your phone. It can work like a pinboard for gathering more information before a trip or for daily stuff like recipe collections.

When you want to view your collections in a browser, sign into that Google account and type: <https://www.google.com/collections>.

Summary

Cloud application development is a hot topic of 2020. The cloud approach gives companies lots of valuable benefits: development cost reduction, (no need for hardware, servers, or even some software), higher accessibility of the final product, a new level of standardization, and scaling opportunities.

At the same time, cloud-based application development is a bit more complicated than web or desktop development, as it involves more aspects to handle, such as various API integrations, data architecture planning, and data decoupling.

Also, you can't be always sure about your cloud service vendor (it's potential to handle traffic surges, if it meets security standard updates, and what kind of bandwidth limitations it has). That's why we advise you to choose a provider carefully and rely on companies with a good reputation (AWS, Google Cloud Platform, or Microsoft Azure).

Keywords

Cloud computing: It offers a lot to software developers, who can now develop web-based applications that take advantage of the power and reach of cloud computing.

Cloud services development: These are particularly notable to smaller businesses, which otherwise would not have the budget or resources to develop large-scale applications.

Cloud services: In the form of centralized web-based applications, also appeal to the IT professional.

EC2: It is just part of Amazon's Web Services (AWS) set of offerings, which provides developers with direct access to Amazon's software and machines.

Google App Engine: It provides a fully integrated application environment.

Mash-ups: It created by users of the Google Maps API, with these custom apps, the data that feeds the map is provided by the developer, where the engine that creates the map itself is provided by Google.

SaaS: It is probably the most common type of cloud service development, with SaaS; a single application is delivered to thousands of users from the vendor's servers.

Self-Assessment Questions

1. The first level of maturity defines the traditional model of software delivery.
 - (a) JSP
 - (b) ASP
 - (c) API
 - (d) AWS
2. The second level of maturity occurs when the vendor hosts a separate instance of the application for each.....
 - (a) Customer
 - (b) Vender

Notes

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- (c) Both a and b
 - (d) None of these.
3. defined four primary maturity levels.
- (a) Linux
 - (b) Google
 - (c) Microsoft
 - (d) None of these.
4. The Google App Engine provides a.....application environment.
- (a) Product
 - (b) Process
 - (c) Partially integrated
 - (d) fully integrated
5. The third level of maturity occurs when the vendor hosts a separate instance of the application for each customer.
- (a) True
 - (b) False
6. IBM is offering a cloud computing solution.
- (a) True
 - (b) False
7. EC2 enables scalable deployment of applications by letting customers request a set number of virtual machines.
- (a) True
 - (b) False
8. The second level of maturity defines the traditional application service provider (ASP) model of software delivery.
- (a) True
 - (b) False

Review Questions

1. What is Service creation environments to develop cloud based applications
2. Explain the Development environments for service development
3. What is Amazon, Azure, Google App?

Module-V

Notes

Structure:

6. Cloud Security and Migration to Cloud
 - 6.1 Security concerns and counter measures in Cloud environment
 - 6.2 Governance, Risk, and Compliance aspects in Cloud
 - 6.3 Cloud security best practices
 - 6.4 Cloud models suitable for different categories of users
 - 6.5 Considerations for choosing applications suitable for Cloud
 - 6.6 Different phases to adopt the Cloud
7. Best Practice Cloud IT Model
 - 7.1 Analyze of Case Studies when deciding to adopt cloud computing architecture
 - 7.2 How to decide if the cloud is right for your requirements
 - 7.3 Cloud based service, applications and development platform deployment so as to improve the total cost of ownership (TCO)

Notes**Unit-6: Cloud Security and Migration to Cloud****Objective:**

After studying this chapter, you will be able to:

- Explain the meaning of Cloud Security and Migration to cloud
- Discuss the Security concerns and counter measures in Cloud environment
- Explain the Governance, Risk, and Compliance aspects in Cloud
- Discuss the Cloud security best practices
- Understand the Cloud models suitable for different categories of users
- Discuss the Different phases to adopt the Cloud

Introduction**Cloud migration**

Cloud migration is the process of moving digital business operations into the cloud. Cloud migration is sort of like a physical move, except it involves moving data, applications, and IT processes from some data centers to other data centers, instead of packing up and moving physical goods. Much like a move from a smaller office to a larger one, cloud migration requires quite a lot of preparation and advance work, but usually it ends up being worth the effort, resulting in cost savings and greater flexibility.

Most often, “cloud migration” describes the move from on-premises or legacy infrastructure to the cloud. However, the term can also apply to a migration from one cloud to another cloud.

What is legacy infrastructure?

In computing, hardware or software is considered “legacy” if it is outdated but still in use. Legacy products and processes are usually not as efficient or secure as more up-to-date solutions. Businesses stuck running legacy systems are in danger of falling behind their competitors; they also face an increased risk of data breaches.

Legacy software or hardware may become unreliable, may run slowly, or may no longer be supported by the original vendor. Windows XP, for instance, is a legacy operating system: released in 2001, its capabilities have been exceeded by later releases of Windows, and Microsoft no longer supports the operating system by releasing patches or updates for it.

Infrastructure includes servers, networking equipment, applications, databases, and any other business-critical software or hardware. Legacy infrastructure, such as aging servers or physical firewall appliances, may slow down a company's business processes. It may also add more security risks as original vendors drop support for their products and stop releasing security patches.

Legacy infrastructure is typically hosted on-premises, meaning it is physically located in buildings or on property where the organization operates. For instance, many businesses host an on-premises data center in the same building where their employees work.

Companies that rely on on-premises legacy infrastructure are unable to experience the benefits of cloud computing. Because of this, most enterprises today have made at least a partial migration to the cloud.

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What are the main benefits of migrating to the cloud?

- ◇ **Scalability:** Cloud computing can scale up to support larger workloads and greater numbers of users far more easily than on-premises infrastructure, which requires companies to purchase and set up additional physical servers, networking equipment, or software licenses.
- ◇ **Cost:** Companies that move to the cloud often vastly reduce the amount they spend on IT operations, since the cloud providers handle maintenance and upgrades. Instead of keeping things up and running, companies can focus more resources on their biggest business needs – developing new products or improving existing ones.
- ◇ **Performance:** For some businesses, moving to the cloud can enable them to improve performance and the overall user experience for their customers. If their application or website is hosted in cloud data centers instead of in various on-premises servers, then data will not have to travel as far to reach the users, reducing latency.
- ◇ **Flexibility:** Users, whether they're employees or customers, can access the cloud services and data they need from anywhere. This makes it easier for a business to expand into new territories, offer their services to international audiences, and let their employees work flexibly.

What are the main challenges of migrating to the cloud?

- ◇ **Migrating large databases:** Often, databases will need to move to a different platform altogether in order to function in the cloud. Moving a database is difficult, especially if there are large amounts of data involved. Some cloud providers actually offer physical data transfer methods, such as loading data onto a hardware appliance and then shipping the appliance to the cloud provider, for massive databases that would take too long to transfer via the Internet. Data can also be transferred over the Internet. Regardless of the method, data migration often takes significant time.
- ◇ **Data integrity:** After data is transferred, the next step is making sure data is intact and secure, and is not leaked during the process.
- ◇ **Continued operation:** A business needs to ensure that its current systems remain operational and available throughout the migration. They will need to have some overlap between on-premises and cloud to ensure continuous service; for instance, it's necessary to make a copy of all data in the cloud before shutting down an existing database. Businesses typically need to move a little bit at a time instead of all at once.

How does an on-premises-to-cloud migration work?

Every business has different needs and will therefore follow a slightly different process for cloud migrations. Cloud providers can help businesses set up their migration process. Most cloud migrations will include these basic steps:

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1. **Establish goals:** What performance gains does a business hope to see? On what date will legacy infrastructure be deprecated? Establishing goals to measure against helps a business determine if the migration was successful or not.
2. **Create a security strategy:** Cloud cybersecurity requires a different approach compared to on-premises security. In the cloud, corporate assets are no longer behind a firewall, and the network perimeter essentially does not exist. Deploying a cloud firewall or a web application firewall may be necessary.
3. **Copy over data:** Select a cloud provider, and replicate existing databases. This should be done continually throughout the migration process so that the cloud database remains up-to-date.
4. **Move business intelligence:** This could involve refactoring or rewriting code (see below). It can be done piecemeal or all at once.
5. **Switch production from on-premises to cloud:** The cloud goes live. The migration is complete.

Some businesses turn off their on-premises infrastructure at the end of these steps, while others may keep legacy systems in place as backup or as part of a hybrid cloud deployment.

What cloud migration strategy should enterprises adopt?

Gartner, a highly influential information technology research company, describes 5 options for organizations migrating to the cloud. These cloud migration strategies are commonly known as the “5 R’s”:

- ◇ **Rehost** - Rehosting can be thought of as “the same thing, but on cloud servers”. Companies that choose this strategy will select an IaaS (Infrastructure-as-a-Service) provider and recreate their application architecture on that infrastructure.
- ◇ **Refactor** - Companies that choose to refactor will reuse already existing code and frameworks, but run their applications on a PaaS (Platform-as-a-Service) provider’s platform – instead of on IaaS, as in rehosting.
- ◇ **Revise** - This strategy involves partially rewriting or expanding the code base, then deploying it by either rehosting or refactoring (see above).
- ◇ **Rebuild** - To “rebuild” means rewriting and re-architecting the application from the ground up on a PaaS provider’s platform. This can be a labor intensive process, but it also enables developers to take advantage of modern features from PaaS vendors.
- ◇ **Replace** - Businesses can also opt to discard their old applications altogether and switch to already-built SaaS (Software-as-a-Service) applications from third-party vendors.

What cloud deployment style should companies choose?

In addition to cloud migration strategy, businesses need to decide how their cloud deployment will look once the migration is complete.

A hybrid cloud mixes two or more types of environments, combining public clouds, private clouds, or on-premises legacy data centers. For a hybrid cloud deployment to

work well, integration must be tight across all deployed clouds and data centers – just as team members need especially tight communication if they're spread out across different offices.

A multicloud deployment combines two or more public clouds. (Public clouds are shared by more than one customer.) Multicloud can serve several purposes: redundancy/backup, cost savings, or leveraging features from different cloud providers, for instance.

Deploying a single cloud from just one cloud vendor is not always feasible for a business, but it is an option. Cloud providers offer both public clouds and private clouds – the difference being that private clouds are not shared with any other business.

How does Cloudflare help with cloud migrations?

Cloudflare offers a single control plane for the performance and security products necessary to make a move to the cloud successful – such as DNS, load balancing, the Cloudflare WAF, and others. These products are all integrated and can all be managed from a single dashboard. Cloudflare integrates easily with any type of cloud deployment – public cloud, private cloud, hybrid cloud, or multicloud.

Cloud security definition

Cloud security is a discipline of cyber security dedicated to securing cloud computing systems. This includes keeping data private and safe across online-based infrastructure, applications, and platforms. Securing these systems involves the efforts of cloud providers and the clients that use them, whether an individual, small to medium business, or enterprise uses.

Cloud providers host services on their servers through always-on internet connections. Since their business relies on customer trust, cloud security methods are used to keep client data private and safely stored. However, cloud security also partially rests in the client's hands as well. Understanding both facets is pivotal to a healthy cloud security solution.

At its core, cloud security is composed of the following categories:

- ◇ Data security
- ◇ Identity and access management (IAM)
- ◇ Governance (policies on threat prevention, detection, and mitigation)
- ◇ Data retention (DR) and business continuity (BC) planning
- ◇ Legal compliance

Cloud security may appear like legacy IT security, but this framework actually demands a different approach. Before diving deeper, let's first look at what cloud security is.

What is cloud security?

Cloud security is the whole bundle of technology, protocols, and best practices that protect cloud computing environments, applications running in the cloud, and data held in the cloud. Securing cloud services begins with understanding what exactly is being secured, as well as, the system aspects that must be managed.

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As an overview, backend development against security vulnerabilities is largely within the hands of cloud service providers. Aside from choosing a security-conscious provider, clients must focus mostly on proper service configuration and safe use habits. Additionally, clients should be sure that any end-user hardware and networks are properly secured.

The full scope of cloud security is designed to protect the following, regardless of your responsibilities:

- ◇ **Physical networks** — routers, electrical power, cabling, climate controls, etc.
- ◇ **Data storage** — hard drives, etc.
- ◇ **Data servers** — core network computing hardware and software
- ◇ **Computer virtualization frameworks** — virtual machine software, host machines, and guest machines
- ◇ **Operating systems (OS)** — software that houses
- ◇ **Middleware** — application programming interface (API) management,
- ◇ **Runtime environments** — execution and upkeep of a running program
- ◇ **Data** — all the information stored, modified, and accessed
- ◇ **Applications** — traditional software services (email, tax software, productivity suites, etc.)
- ◇ **End-user hardware** — computers, mobile devices, Internet of Things (IoT) devices, etc.

With cloud computing, ownership over these components can vary widely. This can make the scope of client security responsibilities unclear. Since securing the cloud can look different based on who has authority over each component, it's important to understand how these are commonly grouped.

To simplify, cloud computing components are secured from two main viewpoints:

1. Cloud service types are offered by third-party providers as modules used to create the cloud environment. Depending on the type of service, you may manage a different degree of the components within the service:
 - ◆ **The core of any third-party cloud service** involves the provider managing the physical network, data storage, data servers, and computer virtualization frameworks. The service is stored on the provider's servers and virtualized via their internally managed network to be delivered to clients to be accessed remotely. This offloads hardware and other infrastructure costs to give clients access to their computing needs from anywhere via internet connectivity.
 - ◆ **Software-as-a-Service (SaaS)** cloud services provide clients access to applications that are purely hosted and run on the provider's servers. Providers manage the applications, data, runtime, middleware, and operating system. Clients are only tasked with getting their applications. SaaS examples include Google Drive, Slack, Salesforce, Microsoft 365, Cisco WebEx, Evernote.
 - ◆ **Platform-as-a-Service** cloud services provide clients a host for developing their own applications, which are run within a client's own "sandboxed"

space on provider servers. Providers manage the runtime, middleware, operating system. Clients are tasked with managing their applications, data, user access, end-user devices, and end-user networks. PaaS examples include Google App Engine, Windows Azure.

- ◆ **Infrastructure-as-a-Service (IaaS)** cloud services offer clients the hardware and remote connectivity frameworks to house the bulk of their computing, down to the operating system. Providers only manage core cloud services. Clients are tasked with securing all that gets stacked atop an operating system, including applications, data, runtimes, middleware, and the OS itself. In addition, clients need to manage user access, end-user devices, and end-user networks. IaaS examples include Microsoft Azure, Google Compute Engine (GCE), Amazon Web Services (AWS).

2. **Cloud environments** are deployment models in which one or more cloud services create a system for the end-users and organizations. These segments the management responsibilities — including security — between clients and providers.

The currently used cloud environments are:

- ◆ Public cloud environments are composed of multi-tenant cloud services where a client shares a provider's servers with other clients, like an office building or coworking space. These are third-party services run by the provider to give clients access via the web.
- ◆ Private third-party cloud environments are based on the use of a cloud service that provides the client with exclusive use of their own cloud. These single-tenant environments are normally owned, managed, and operated offsite by an external provider.
- ◆ Private in-house cloud environments also composed of single-tenant cloud service servers but operated from their own private data center. In this case, this cloud environment is run by the business themselves to allow full configuration and setup of every element.
- ◆ Multi-cloud environments include the use of two or more cloud services from separate providers. These can be any blend of public and/or private cloud services.
- ◆ Hybrid cloud environments consist of using a blend of private third-party cloud and/or onsite private cloud data center with one or more public clouds.

By framing it from this perspective, we can understand that cloud-based security can be a bit different based on the type of cloud space users are working in. But the effects are felt by both individual and organizational clients alike.

How does cloud security work?

Every cloud security measure works to accomplish one or more of the following:

- ◆ Enable data recovery in case of data loss
- ◆ Protect storage and networks against malicious data theft
- ◆ Deter human error or negligence that causes data leaks
- ◆ Reduce the impact of any data or system compromise

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Data security is an aspect of cloud security that involves the technical end of threat prevention. Tools and technologies allow providers and clients to insert barriers between the access and visibility of sensitive data. Among these, encryption is one of the most powerful tools available. Encryption scrambles your data so that it's only readable by someone who has the encryption key. If your data is lost or stolen, it will be effectively unreadable and meaningless. Data transit protections like virtual private networks (VPNs) are also emphasized in cloud networks.

Identity and access management (IAM) pertains to the accessibility privileges offered to user accounts. Managing authentication and authorization of user accounts also apply here. Access controls are pivotal to restrict users — both legitimate and malicious — from entering and compromising sensitive data and systems. Password management, multi-factor authentication, and other methods fall in the scope of IAM.

Governance focuses on policies for threat prevention, detection, and mitigation. With SMB and enterprises, aspects like threat intel can help with tracking and prioritizing threats to keep essential systems guarded carefully. However, even individual cloud clients could benefit from valuing safe user behavior policies and training. These apply mostly in organizational environments, but rules for safe use and response to threats can be helpful to any user.

Data retention (DR) and business continuity (BC) planning involve technical disaster recovery measures in case of data loss. Central to any DR and BC plan are methods for data redundancy such as backups. Additionally, having technical systems for ensuring uninterrupted operations can help. Frameworks for testing the validity of backups and detailed employee recovery instructions are just as valuable for a thorough BC plan.

Legal compliance revolves around protecting user privacy as set by legislative bodies. Governments have taken up the importance of protecting private user information from being exploited for profit. As such, organizations must follow regulations to abide by these policies. One approach is the use of data masking, which obscures identity within data via encryption methods.

What makes cloud security different?

Traditional IT security has felt an immense evolution due to the shift to cloud-based computing. While cloud models allow for more convenience, always-on connectivity requires new considerations to keep them secure. Cloud security, as a modernized cyber security solution, stands out from legacy IT models in a few ways.

Data storage: The biggest distinction is that older models of IT relied heavily upon onsite data storage. Organizations have long found that building all IT frameworks in-house for detailed, custom security controls is costly and rigid. Cloud-based frameworks have helped offload costs of system development and upkeep, but also remove some control from users.

Scaling speed: On a similar note, cloud security demands unique attention when scaling organization IT systems. Cloud-centric infrastructure and apps are very modular and quick to mobilize. While this ability keeps systems uniformly adjusted to organizational changes, it does pose concerns when an organization's need for upgrades and convenience outpaces their ability to keep up with security.

End-user system interfacing: For organizations and individual users alike, cloud systems also interface with many other systems and services that must be secured. Access permissions must be maintained from the end-user device level to the software level and even the network level. Beyond this, providers and users must be attentive to vulnerabilities they might cause through unsafe setup and system access behaviors.

Proximity to other networked data and systems: Since cloud systems are a persistent connection between cloud providers and all their users, this substantial network can compromise even the provider themselves. In networking landscapes, a single weak device or component can be exploited to infect the rest. Cloud providers expose themselves to threats from many end-users that they interact with, whether they are providing data storage or other services. Additional network security responsibilities fall upon the providers who otherwise delivered products live purely on end-user systems instead of their own.

Solving most cloud security issues means that users and cloud providers — both in personal and business environments — must both remain proactive about their own roles in cyber security. This two-pronged approach means users and providers mutually must address:

Secure system configuration and maintenance.

User safety education — both behaviorally and technically.

Ultimately, cloud providers and users must have transparency and accountability to ensure both parties stay safe.

Cloud security risks

What are the security issues in cloud computing? Because if you don't know them, then how are you supposed to put proper measures in place? After all, weak cloud security can expose users and providers to all types of cyber security threats. Some common cloud security threats include:

- ◇ Risks of cloud-based infrastructure including incompatible legacy IT frameworks, and third-party data storage service disruptions.
- ◇ Internal threats due to human error such as misconfiguration of user access controls.
- ◇ External threats caused almost exclusively by malicious actors, such as malware, phishing, and DDoS attacks.

The biggest risk with the cloud is that there is no perimeter. Traditional cyber security focused on protecting the perimeter, but cloud environments are highly connected which means insecure APIs (Application Programming Interfaces) and account hijacks can pose real problems. Faced with cloud computing security risks, cyber security professionals need to shift to a data-centric approach.

Interconnectedness also poses problems for networks. Malicious actors often breach networks through compromised or weak credentials. Once a hacker manages to make a landing, they can easily expand and use poorly protected interfaces in the cloud to locate data on different databases or nodes. They can even use their own cloud servers as a destination where they can export and store any stolen data. Security needs to be in the cloud — not just protecting access to your cloud data.

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Third-party storage of your data and access via the internet each pose their own threats as well. If for some reason those services are interrupted, your access to the data may be lost. For instance, a phone network outage could mean you can't access the cloud at an essential time. Alternatively, a power outage could affect the data center where your data is stored, possibly with permanent data loss.

Such interruptions could have long-term repercussions. A recent power outage at an Amazon cloud data facility resulted in data loss for some customers when servers incurred hardware damage. This is a good example of why you should have local backups of at least some of your data and applications.

Why Cloud security is important

In the 1990s, business and personal data lived locally — and security was local as well. Data would be located on a PC's internal storage at home, and on enterprise servers, if you worked for a company.

Introducing cloud technology has forced everyone to reevaluate cyber security. Your data and applications might be floating between local and remote systems — and always internet-accessible. If you are accessing Google Docs on your smartphone, or using Salesforce software to look after your customers, that data could be held anywhere. Therefore, protecting it becomes more difficult than when it was just a question of stopping unwanted users from gaining access to your network. Cloud security requires adjusting some previous IT practices, but it has become more essential for two key reasons:

1. **Convenience over security.** Cloud computing is exponentially growing as a primary method for both workplace and individual use. Innovation has allowed new technology to be implemented quicker than industry security standards can keep up, putting more responsibility on users and providers to consider the risks of accessibility.
2. **Centralization and multi-tenant storage.** Every component — from core infrastructure to small data like emails and documents — can now be located and accessed remotely on 24/7 web-based connections. All this data gathering in the servers of a few major service providers can be highly dangerous. Threat actors can now target large multi-organizational data centers and cause immense data breaches.

6.1. Security Concerns and Counter Measures in Cloud Environment

According to Cloud Security Alliance (CSA), over 70 percent of the world's businesses now operate — at least in part — on the cloud.

With benefits like lower fixed costs, higher flexibility, automatic software updates, increased collaboration, and the freedom to work from anywhere, 70 percent isn't a big surprise.

Still, the cloud has its share of security issues.

Recently the "Cloud Security Spotlight Report" showed that "90 percent of organizations are very or moderately concerned about public cloud security." These concerns run the gamut from vulnerability to hijacked accounts to malicious insiders to full-scale data breaches.

Although cloud services have ushered in a new age of transmitting and storing data, many companies are still hesitant or make the move without a clear plan for security in place.

We'll show you a big picture view of the top 10 security concerns for cloud-based services you should be aware of.

1. Data Breaches

Cloud computing and services are relatively new, yet data breaches in all forms have existed for years. The question remains: "With sensitive data being stored online rather than on premise, is the cloud inherently less safe?"

A study conducted by the Ponemon Institute entitled "Man In Cloud Attack" reports that over 50 percent of the IT and security professionals surveyed believed their organization's security measures to protect data on cloud services are low. This study used nine scenarios, where a data breach had occurred, to determine if that belief was founded in fact.

After evaluating each scenario, the report concluded that overall data breaching was three times more likely to occur for businesses that utilize the cloud than those that don't. The simple conclusion is that the cloud comes with a unique set of characteristics that make it more vulnerable.

2. Hijacking of Accounts

The growth and implementation of the cloud in many organizations has opened a whole new set of issues in account hijacking.

Attackers now have the ability to use your (or your employees') login information to remotely access sensitive data stored on the cloud; additionally, attackers can falsify and manipulate information through hijacked credentials.

Other methods of hijacking include scripting bugs and reused passwords, which allow attackers to easily and often without detection steal credentials. In April 2010 Amazon faced a cross-site scripting bug that targeted customer credentials as well. Phishing, keylogging, and buffer overflow all present similar threats. However, the most notable new threat – known as the Man In Cloud Attack – involves the theft of user tokens which cloud platforms use to verify individual devices without requiring logins during each update and sync.

3. Insider Threat

An attack from inside your organization may seem unlikely, but the insider threat does exist. Employees can use their authorized access to an organization's cloud-based services to misuse or access information such as customer accounts, financial forms, and other sensitive information.

Additionally, these insiders don't even need to have malicious intentions.

A study by Imperva, "Inside Track on Insider Threats" found that an insider threat was the misuse of information through malicious intent, accidents or malware. The study also examined four best practices companies could follow to implement a secure strategy, such as business partnerships, prioritizing initiatives, controlling access, and implementing technology.

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4. Malware Injection

Malware injections are scripts or code embedded into cloud services that act as “valid instances” and run as SaaS to cloud servers. This means that malicious code can be injected into cloud services and viewed as part of the software or service that is running within the cloud servers themselves.

Once an injection is executed and the cloud begins operating in tandem with it, attackers can eavesdrop, compromise the integrity of sensitive information, and steal data. Security Threats On Cloud Computing Vulnerabilities, a report by the East Carolina University, reviews the threats of malware injections on cloud computing and states that “malware injection attack has become a major security concern in cloud computing systems.”

5. Abuse of Cloud Services

The expansion of cloud-based services has made it possible for both small and enterprise-level organizations to host vast amounts of data easily. However, the cloud’s unprecedented storage capacity has also allowed both hackers and authorized users to easily host and spread malware, illegal software, and other digital properties.

In some cases this practice affects both the cloud service provider and its client. For example, privileged users can directly or indirectly increase the security risks and as a result infringe upon the terms of use provided by the service provider.

These risks include the sharing of pirated software, videos, music, or books, and can result in legal consequences in the forms of fines and settlements with the U.S. Copyright Law reaching up to \$250,000. Depending on the damage, these fines can be even more cost prohibitive. You can reduce your exposure to risk by monitoring usage and setting guidelines for what your employees host in the cloud. Service providers and legal entities, such as CSA have defined what is abusive or inappropriate behavior along with methods of detecting such behaviors.

6. Insecure APIs

Application Programming Interfaces (API) give users the opportunity to customize their cloud experience.

However, APIs can be a threat to cloud security because of their very nature. Not only do they give companies the ability to customize features of their cloud services to fit business needs, but they also authenticate, provide access, and effect encryption.

As the infrastructure of APIs grows to provide better service, so do its security risks. APIs give programmers the tools to build their programs to integrate their applications with other job-critical software. A popular and simple example of an API is YouTube, where developers have the ability to integrate YouTube videos into their sites or applications.

The vulnerability of an API lies in the communication that takes place between applications. While this can help programmers and businesses, they also leave exploitable security risks.

7. Denial of Service Attacks

Unlike other kind of cyberattacks, which are typically launched to establish a long-term foothold and hijack sensitive information, denial of service assaults do not attempt to breach your security perimeter. Rather, they attempt to make your website and servers unavailable to legitimate users. In some cases, however, DoS is also used as a smokescreen for other malicious activities, and to take down security appliances such as web application firewalls.

8. Insufficient Due Diligence

Most of the issues we've looked at here are technical in nature, however this particular security gap occurs when an organization does not have a clear plan for its goals, resources, and policies for the cloud. In other words, it's the people factor.

Additionally, insufficient due diligence can pose a security risk when an organization migrates to the cloud quickly without properly anticipating that the services will not match customer's expectation.

This is especially important to companies whose data falls under regulatory laws like PII, PCI, PHI, and FERPA or those that handle financial data for customers.

9. Shared Vulnerabilities

Cloud security is a shared responsibility between the provider and the client.

This partnership between client and provider requires the client to take preventative actions to protect their data. While major providers like Box, Dropbox, Microsoft, and Google do have standardized procedures to secure their side, fine grain control is up to you, the client.

As Skyfence points out in its article "Office 365 Security & Share Responsibility," this leaves key security protocols – such as the protection of user passwords, access restrictions to both files and devices, and multi-factor authentication – firmly in your hands.

The bottom line is that clients and providers have shared responsibilities, and omitting yours can result in your data being compromised.

10. Data Loss

Data on cloud services can be lost through a malicious attack, natural disaster, or a data wipe by the service provider. Losing vital information can be devastating to businesses that don't have a recovery plan. Amazon is an example of an organization that suffered data loss by permanently destroying many of its own customers' data in 2011.

Google was another organization that lost data when its power grid was struck by lightning four times.

Securing your data means carefully reviewing your provider's back up procedures as they relate to physical storage locations, physical access, and physical disasters.

Security Concerns for Cloud-Based Services

The cloud has opened up a whole new frontier for storage, access, flexibility, and productivity. It's also opened up a new world of security concerns.

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6.2. Governance, Risk, and Compliance Aspects in Cloud

Cloud adoption continues to grow, which is evident from the fact that annual 2016 revenues for cloud vendors were “within touching distance” of \$150 billion. Gartner also predicts that, a corporate ‘no-cloud’ policy will be as rare by 2020 as a ‘no-Internet’ policy is today. However, a “cloud-ready” security and compliance program is the need of the hour, to manage the risks and the complexities due to cloud adoption. This will enable organizations to face cloud challenges which, according to RightScale’s 2016 State of the Cloud Report include compliance with regulations, a lack of resources and expertise, governance and control and security. Although a challenge mainstay, confidence in cloud security is nonetheless rising; SkyHigh Networks points out that 65 percent of IT leaders think the cloud is as secure, or more secure, than on-premises software.

To maximize the benefits of cloud deployments while mitigating the risks, companies need to prioritize a cohesive approach to governance, risk management and compliance (GRC). A cloud governance framework can automate cloud security, risk, and compliance workflows, enable stakeholder reporting and visibility, and ensure best practices and standards for cloud compliance.

With that in mind, here are five recommendations for ensuring a proper governance, risk and compliance framework for cloud assets and operations:

1. Improve Cloud Asset / Service Visibility

An essential first step is to understand the scope of cloud services in use within the organization and gain visibility into the whole cloud environment. IT and infrastructure managers need to have a complete picture of the processes running on cloud deployments, the underlying assets and their ownership within the organization both from an IT and business standpoint. While they may seem intuitive, alarmingly, the 1H 2016 Shadow Data Report states that organizations use 841 cloud apps on average – an astonishing 20 times more than they thought they did. Organizations also need a well-defined policy to deploy, manage and run the cloud applications and categorize the sensitivity of the data held to ensure that requisite controls are in place to manage the data.

2. Assess the Cloud Service Provider (Csp) Continuously

Businesses always have the thought of losing control on application and infrastructure while deploying an application on the cloud. Assessing and creating a working relationship with the cloud provider based on a mutually agreed framework is very important. The organizations must select a cloud provider who can demonstrate validation of controls including network security, physical datacenter security as well as a standard audit framework conforming to applicable regulatory standards.

Gartner recommends that organizations need to address several key issues when selecting a cloud hosting provider, which include access privileges, regulatory compliance, data provenance, data segregation, data recovery and business continuity.

To gain a complete understanding of the CSP environment, organizations should also ensure that there is no ‘insufficient due diligence’, which Cloud Security Alliance (CSA) rates as one of the ‘notorious nine cloud computing top threats, and establish a due-diligence framework to monitor the cloud service provider performance on a continuous basis.

3. Assign Business Ownership And Accountability For Critical Cloud Assets And Services

Organizations should understand the importance for an effective governance function within the cloud environment. The cloud assets, cloud services, business objectives, business processes, policies must be documented, along with their operational relationships. These processes and policies must be accountable, clearly assigned and consistently understood throughout the business.

Also, it is of utmost importance to establish accountability when customer information is intertwined with that of the cloud service provider. This includes logical separation of your data sets from those of the other customers / users, defining SLAs on both sides and categorizing the services consumed.

4. Know The Cloud Threat Landscape And Evaluate Risks

Inevitably, there are risks with cloud environments as there are with all storage and retrieval systems, both electronic and manual. Businesses must understand the cloud threat landscape, effectively evaluate and mitigate risks and protect themselves and their interested parties from exposure.

The likelihood of threats rarely lessens, but threats do change in nature and for this reason companies should be continually alert and abreast of latest developments. SkyHigh Networks revealed, in its Q4 2016 Cloud Adoption and Risk Report that the average company experiences over 23 cloud-related security incidents each month. Yet, despite this, a different study – the 2016 Global Cloud Data Security Study from Gemalto and the Ponemon Institute – discovered that 54 percent of respondents didn't agree that their companies have a proactive approach to managing security and complying with privacy / data protection regulations for the cloud. Therefore, it is imperative that organizations prepare for security threats to the cloud before becoming a victim.

5. Leverage Standard Risk / Control Frameworks to Assess Compliance

Businesses should assess cloud compliance with regard to security, privacy practices and policies. Among the most well-known risk and control frameworks are best practices is the Cloud Security Alliance (CSA)'s GRC stack, which provides a toolkit to assess private and public clouds against industry standard best practices and compliance requirements.

Others include the CSA's 'Treacherous Twelve' Cloud Computing Top Threats, ISACA's cybersecurity Threats and Controls, the National Institute for Standards and Technology (NIST)'s Framework for Improving Critical Infrastructure Cybersecurity, ISO/IEC 27017, ISO/IEC 27018 and the Center for Internet Security (CIS)'s Critical Security Controls.

Leveraging industry standards provides a level of assurance that best practices are followed both by the organization and by cloud Service Providers.

Businesses can achieve enhanced information security, compliance and risk management as well as reliability, operational control and transparency with effective implementation and extension of the GRC framework to their cloud assets and operations. Adhering to best practices and standards will deliver informed decision-making and ongoing management, placing the business in a better position to reduce risk and realize the benefits of the cloud in enhancing business performance.

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Risk: IT governance is tightly woven with business goals and policies to ensure that services are optimized for customer expectations. Because IT and business goals are tightly woven in a governance strategy, it is important to look at cloud computing governance from a holistic business perspective.

Deducing IT risk in cloud computing governance

In the heterogeneous IT environment, IT needs to juggle various tasks: meeting customer expectations, optimizing business goals, recognizing resource constraints, and adhering to rules and requirements. The cloud can further complicate this juggling act because it is yet another resource that IT is responsible for. This means that the governing body is responsible for overseeing the provider relationship.

Of course, the level of involvement and risk around governance might vary with how your organization is using the cloud. For example, the cloud can be used in the following ways, each of which you must evaluate separately determine the level of governance that your company feels comfortable with:

- ◇ For temporary computing power
- ◇ As a SaaS model
- ◇ As a platform to build a service

Cloud computing risk list

Consider these risks as you move into the cloud:

- ◇ Audit and compliance risks including issues around data jurisdiction, data access control, and maintaining an audit trail.
- ◇ Security risks including data integrity, data confidentiality, and privacy.
- ◇ Information risks (outside of security), including protection of intellectual property.
- ◇ Performance and availability risks, including availability and performance levels that your business requires to successfully operate.
- ◇ Interoperability risks, which are associated with developing a service that might be composed of multiple services.
- ◇ Contract risks associated with not reading between the lines of your contract.
- ◇ Billing risks associated with ensuring that you're billed correctly and only for the resources you consume.

If you move into the cloud, you need to trust the cloud provider and every other provider that the cloud provider is working with. Currently, there are no professional standards or laws related to cloud computing.

6.3. Cloud Security Best Practices

From the very beginning of the cloud computing era, security has been the biggest concern among enterprises considering the public cloud. For many organizations, the idea of storing data or running applications on infrastructure that they do not manage directly seems inherently insecure.

CloudPassage's 2021 AWS Cloud Security Report found that misconfiguration of cloud platforms (71 percent), exfiltration of sensitive data (59 percent), and insecure

APIs (54 percent) are the top cloud security threats facing cybersecurity professionals. In addition, 95 percent of survey respondents confirmed that they are extremely to moderately concerned about public cloud security.

Those concerns are certainly justified. According to IDC's 2021 State of Cloud Security Report, 79 percent of surveyed companies reported a cloud data breach in the last 18 months. Public cloud infrastructure as a service (IaaS) may be less vulnerable than traditional data centers, but that doesn't mean it's without its own set of risks. Enterprises that don't want to be part of that statistic should understand and implement cybersecurity best practices when it comes to their cloud infrastructure.

"Cloud security consists of all the technologies and processes that ensure an organization's cloud infrastructure is protected against internal and external cybersecurity threats. As more enterprises look to the cloud as the future of business, cloud security is an absolute necessity to maintain continuity. Cloud security makes sure the lights stay on so businesses can focus on driving progress."

Top 12 best practices for cloud security

Cloud security is constantly evolving, but a handful of best practices have remained constant for ensuring the security of cloud environments. Organizations that have existing cloud solutions in place or are looking to implement them should consider these tips and tools to ensure that sensitive applications and data don't fall into the wrong hands.

1. Understand Your Shared Responsibility Model

In a private data center, the enterprise is solely responsible for all security issues. But in the public cloud, things are much more complicated. While the buck ultimately stops with the cloud customer, the cloud provider assumes the responsibility for some aspects of IT security. Cloud and security professionals call this a shared responsibility model.

Leading IaaS and platform as a service (PaaS) vendors like Amazon Web Services (AWS) and Microsoft Azure provide documentation to their customers so all parties understand where specific responsibilities lie according to different types of deployment. The diagram below, for example, shows that application-level controls are Microsoft's responsibility with software as a service (SaaS) models, but it is the customer's responsibility in IaaS deployments. For PaaS models, Microsoft and its customers share the responsibility.

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Responsibility	On-Prem	IaaS	PaaS	SaaS
Data classification & accountability	Cloud Customer	Cloud Customer	Cloud Customer	Cloud Customer
Client & end-point protection	Cloud Customer	Cloud Customer	Cloud Customer	Cloud Customer
Identity & access management	Cloud Customer	Cloud Customer	Cloud Customer	Cloud Customer
Application level controls	Cloud Customer	Cloud Customer	Cloud Customer	Cloud Customer
Network controls	Cloud Customer	Cloud Customer	Cloud Customer	Cloud Customer
Host infrastructure	Cloud Customer	Cloud Customer	Cloud Customer	Cloud Customer
Physical security	Cloud Customer	Cloud Customer	Cloud Customer	Cloud Customer

Legend: Cloud Customer (Blue), Cloud Provider (Grey)

Enterprises that are considering a particular cloud vendor should review its policies about shared security responsibilities and understand who is handling the various aspects of cloud security. That can help prevent miscommunication and misunderstanding. More importantly, though, clarity about responsibilities can prevent security incidents that occur as a result of a particular security need falling through the cracks.

2. Ask Your Cloud Provider Detailed Security Questions

In addition to clarifying shared responsibilities, organizations should ask their public cloud vendors detailed questions about the security measures and processes they have in place. It's easy to assume that the leading vendors have security handled, but security methods and procedures can vary significantly from one vendor to the next.

To understand how a particular cloud provider compares, organizations should ask a wide range of questions, including:

- ◇ Where do the provider's servers reside geographically?
- ◇ What is the provider's protocol for suspected security incidents?
- ◇ What is the provider's disaster recovery plan?
- ◇ What measures does the provider have in place to protect various access components?
- ◇ What level of technical support is the provider willing to provide?
- ◇ What are the results of the provider's most recent penetration tests?
- ◇ Does the provider encrypt data while in transit and at rest?

- ◇ Which roles or individuals from the provider have access to the data stored in the cloud?
- ◇ What authentication methods does the provider support?
- ◇ What compliance requirements does the provider support?

3. Deploy An Identity And Access Management Solution

The fourth biggest threat to public cloud security identified in CloudPassage's report is unauthorized access (and growing – 53 percent, up from 42 percent in 2020). While hackers' methods of gaining access to sensitive data are becoming more sophisticated with each new attack, a high-quality identity and access management (IAM) solution can help mitigate these threats.

Experts recommend that organizations look for an IAM solution that allows them to define and enforce access policies based on least privilege. These policies should also be based on role-based permission capabilities. Additionally, multi-factor authentication (MFA) can further reduce the risk of malicious actors gaining access to sensitive information, even if they manage to steal usernames and passwords.

Organizations may also want to look for an IAM solution that works in hybrid environments that include private data centers as well as cloud deployments. This can simplify authentication for end users and make it easier for security staff to ensure that they are enforcing consistent policies across all IT environments.

4. Train Your Staff

To prevent hackers from getting their hands on access credentials for cloud computing tools, organizations should train all workers on how to spot cybersecurity threats and how to respond to them. Comprehensive training should include basic security knowledge like how to create a strong password and identify possible social engineering attacks as well as more advanced topics like risk management.

Perhaps most importantly, cloud security training should help employees understand the inherent risk of shadow IT. At most organizations, it's all too easy for staff to implement their own tools and systems without the knowledge or support of the IT department. Without top-to-bottom visibility of all systems that interact with the company's data, there's no way to take stock of all vulnerabilities. Enterprises need to explain this risk and hammer home the potential consequences for the organization.

Organizations also need to invest in specialized training for their security staff. The threat landscape shifts on a daily basis, and IT security professionals can only keep up if they are constantly learning about the newest threats and potential countermeasures.

5. Establish And Enforce Cloud Security Policies

All organizations should have written guidelines that specify who can use cloud services, how they can use them, and which data can be stored in the cloud. They also need to lay out the specific security technologies that employees must use to protect data and applications in the cloud.

Ideally, security staff should have automated solutions in place to ensure that everyone is following these policies. In some cases, the cloud vendor may have a policy enforcement feature that is sufficient to meet the organization's needs. In others, the

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organization may need to purchase a third party solution like CASB that offers policy enforcement capabilities.

Zero trust is one such technology that offers a refined control over policy enforcement. Tools in this category work with other systems to determine how much access each user needs, what they can do with that access, and what it means for the broader organization.

6. Secure Your Endpoints

Using a cloud service doesn't eliminate the need for strong endpoint security—it intensifies it. New cloud computing projects offer an opportunity to revisit existing strategies and ensure the protections in place are adequate to address evolving threats.

A defense-in-depth strategy that includes firewalls, anti-malware, intrusion detection, and access control has long been the standard for endpoint security. However, the array of endpoint security concerns has become so complex that automation tools are required to keep up. Endpoint detection and response (EDR) tools and/or endpoint protection platforms (EPP) can help in this area.

EDR and EPP solutions combine traditional endpoint security capabilities with continuous monitoring and automated response. Specifically, these tools address a number of security requirements, including patch management, endpoint encryption, VPNs, and insider threat prevention among others.

7. Encrypt Data In Motion And At Rest

Encryption is a key part of any cloud security strategy. Not only should organizations encrypt any data in a public cloud storage service, but they should also ensure that data is encrypted during transit—when it may be most vulnerable to attacks.

Some cloud computing providers offer encryption and key management services. Some third-party cloud and traditional software companies offer encryption options as well. Experts recommend finding an encryption product that works seamlessly with existing work processes, eliminating the need for end users to take any extra actions to comply with company encryption policies.

8. Use Intrusion Detection And Prevention Technology

Intrusion prevention and detection systems (IDPS) are among some of the most effective cloud security tools on the market. They monitor, analyze, and respond to network traffic across both on-premises and public cloud environments. When they encounter signature-based, protocol-based, or anomaly-based threats, IDPS solutions add them to a log, alert administrators to unusual activity, and block the threats so admins have enough time to take action.

These tools are important for round-the-clock monitoring and real-time alerts. Without IDPS, it's nearly impossible to analyze network traffic for the telltale signs of a sophisticated attack.

9. Double-Check Your Compliance Requirements

Organizations that collect personally identifiable information (PII) like those in retail, healthcare, and financial services face strict regulations when it comes to customer privacy and data security. Some businesses in certain geographic locations—

or businesses that store data in particular regions—may have special compliance requirements from local or state governments as well.

Before establishing a new cloud computing service, organizations should review their particular compliance requirements and make sure that their service provider will meet their data security needs.

10. Consider A CASB Or Cloud Security Solution

Dozens of companies offer solutions or services specifically designed to enhance cloud security. If an organization's internal security staff doesn't have cloud expertise or if the existing security solutions don't support cloud environments, it may be time to bring in outside help.

Cloud access security brokers (CASBs) are tools purpose-built to enforce cloud security policies. They have become increasingly popular as more organizations have started using cloud services. Experts say that a CASB solution may make the most sense for organizations that use multiple cloud computing services from several different vendors. These solutions can also monitor for unauthorized apps and access too.

11. Conduct Audits and Penetration Testing

Whether an organization chooses to partner with an outside security firm or keep security teams in-house, experts say all enterprises should run penetration testing to determine whether existing cloud security efforts are sufficient to protect data and applications.

Additionally, organizations should conduct regular security audits that include an analysis of all security vendors' capabilities. This should confirm that they are meeting the agreed upon security terms. Access logs should also be audited to ensure only appropriate and authorized personnel are accessing sensitive data and applications in the cloud.

12. Enable Security Logs

In addition to conducting audits, organizations should enable logging features for their cloud solutions. Logging helps system administrators keep track of which users are making changes to the environment—something that would be nearly impossible to do manually. If an attacker gains access and makes changes, the logs will illuminate all their activities so they can be remediated.

Misconfigurations are one of the most significant challenges of cloud security, and effective logging capabilities will help connect the changes that led to a particular vulnerability so they can be corrected and avoided in the future. Logging also helps identify individual users who may have more access than they actually need to do their jobs, so administrators can adjust those permissions to the bare minimum.

6.4. Cloud Models Suitable for Different Categories of Users

Today, organizations have plenty of exciting opportunities to reimagine, repurpose and reinvent their businesses with the cloud. The last decade has seen even more businesses rely on it for quicker time to market, better efficiency and scalability. It helps them achieve long-term digital goals as part of their digital strategy.

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Though the answer to which cloud model is an ideal fit for a business depends on the computing and business needs of your organization. Choosing the right one from the various types of cloud service deployment models is essential. This would ensure your business is equipped with the performance, scalability, privacy, security, compliance & cost-effectiveness it requires. It is important to learn and explore what different deployment types can offer – around what particular set of problems it can solve.

Read on as we cover the various cloud computing deployment and service models to help discover the best choice for your business.

A Cloud Deployment Model?

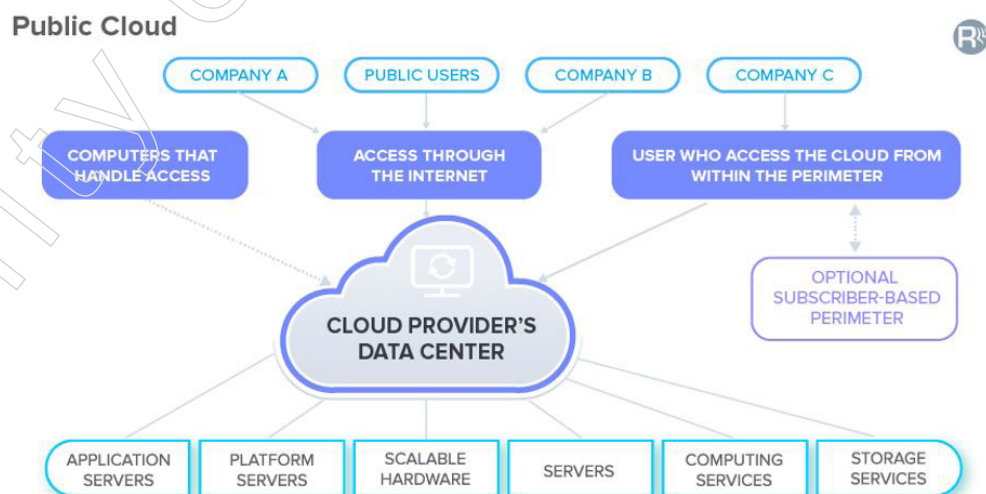
It works as your virtual computing environment with a choice of deployment model depending on how much data you want to store and who has access to the infrastructure.

Different Types Of Cloud Computing Deployment Models

Most cloud hubs have tens of thousands of servers and storage devices to enable fast loading. It is often possible to choose a geographic area to put the data “closer” to users. Thus, deployment models for cloud computing are categorized based on their location. To know which model would best fit the requirements of your organization, let us first learn about the various types.

Public Cloud

The name says it all. It is accessible by the public. Public deployment models in the cloud are perfect for organizations with growing and fluctuating demands. It also makes a great choice for companies with low-security concerns. Thus, you pay a cloud service provider for networking services, compute virtualization & storage available on the public internet. This is also a great delivery model for the teams with development and testing. Its configuration and deployment are quick and easy, making it an ideal choice for test environments.



Benefits of Public Cloud

- ◇ **Minimal Investment** – As a pay-per-use service, there is no large upfront cost and is ideal for businesses who need quick access to resources

- ◇ **No Hardware Set-up** – The entire infrastructure is fully funded by the cloud service providers
- ◇ **No Infrastructure Management** – Does not require an in-house team to utilize the public cloud

Limitations of Public Cloud

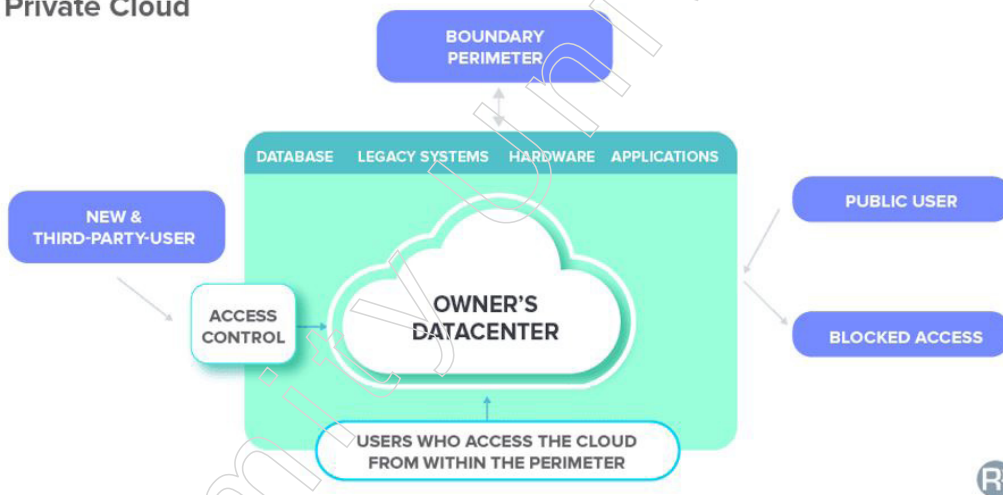
- ◇ **Data Security and Privacy Concerns** – Since it is accessible to all, it does not fully protect against cyber-attacks and could lead to vulnerabilities
- ◇ **Reliability Issues** – Since the same server network is open to a wide range of users, it can lead to malfunction and outages
- ◇ **Service/License Limitation** – While there are many resources that you can exchange with tenants, there is a cap on usage

Private Cloud

Now that you understand what the public cloud could offer you, of course, you are keen to know what a private cloud can do. Companies that look for cost efficiency and greater control over data & resources will find the private cloud a more suitable choice.

What it means is that it will be integrated with your data center and managed by your IT team. Alternatively, you can also choose to host it externally. When it comes to customization, the private cloud offers bigger opportunities that help meet specific organization's requirements. It's also a wise choice for mission-critical processes that may have frequently changing requirements.

Private Cloud



Benefits of Private Cloud

- ◇ **Data Privacy** – It is ideal for storing corporate data where only authorized personnel gets access
- ◇ **Security** – Segmentation of resources within the same infrastructure can help with better access and higher levels of security
- ◇ **Supports Legacy Systems** – This model supports legacy systems that cannot access the public cloud

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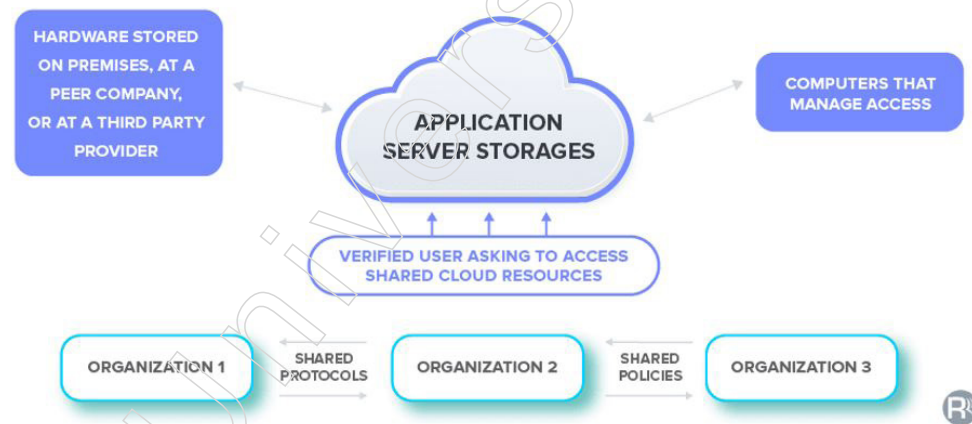
Limitations of Private Cloud

- ◇ **Higher Cost** – With the benefits that you get, the investment will also be larger than the public cloud. Here, you will pay for software, hardware and resources for staff and training.
- ◇ **Fixed Scalability** – The hardware you choose will accordingly help you scale in a certain direction
- ◇ **High Maintenance** – Since it is managed in-house, the maintenance costs also increase

Community Cloud

The community cloud operates in a way that is similar to the public cloud. There's just one difference – it allows access to only a specific set of users who share common objectives and use cases. This type of deployment model of cloud computing is managed and hosted internally or by a third-party vendor. However, you can also choose a combination of all three.

Community Cloud



Benefits of Community Cloud

- ◇ **Smaller Investment** – A community cloud is much cheaper than the private & public cloud and provides great performance
- ◇ **Setup Benefits** – The protocols and configuration of a community cloud must align with industry standards. This allows customers to work much more efficiently

Limitations of Community Cloud

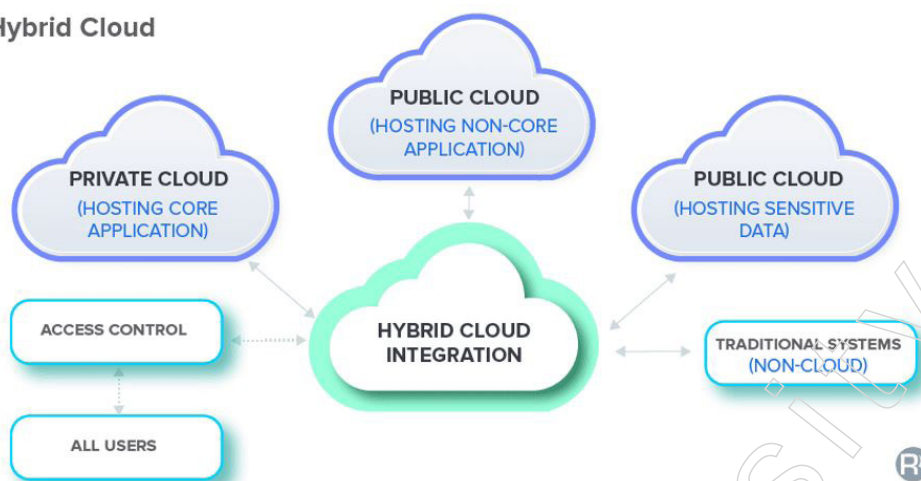
- ◇ **Shared Resources** – Due to restricted bandwidth and storage capacity, community resources often pose challenges
- ◇ **Not as Popular** – Since this is a recently introduced model, it is not that popular or available across industries

Hybrid Cloud

As the name suggests, a hybrid cloud is a combination of two or more cloud architectures. While each model in the hybrid cloud functions differently, it is all part of the same architecture. Further, as part of this deployment of the cloud computing model, the internal, or external providers can offer resources.

Let's understand the hybrid model better. A company that has critical data will prefer storing on a private cloud, while less sensitive data can be stored on a public cloud. The hybrid cloud is also frequently used for 'cloud bursting'. It means, suppose an organization runs an application on-premises, but due to heavy load, they can burst into the public cloud.

Hybrid Cloud



Benefits of Hybrid Cloud

- ◇ **Cost-Effectiveness** – The overall cost of a hybrid solution decreases since it majorly uses the public cloud to store data
- ◇ **Security** – Since data is properly segmented, the chances of data theft from attackers is significantly reduced
- ◇ **Flexibility** – With higher levels of flexibility, businesses can create custom solutions that fit their exact requirements

Limitations of Hybrid Cloud

- ◇ **Complexity** – It is complex setting up a hybrid cloud since it needs to integrate two or more cloud architectures
- ◇ **Specific Use Case** – This model makes more sense for organizations that have multiple use cases or need to separate critical and sensitive data

A Comparative Analysis of Cloud Deployment Models

With the below table we've attempted to analyze the key models with an overview of what each one can do for you;

Important Factors to Consider	Public	Private	Community	Hybrid
Setup and ease of use	Easy	Requires professional IT Team	Requires professional IT Team	Requires professional IT Team
Data Security and Privacy	Low	High	Very High	High
Scalability and flexibility	High	High	Fixed requirements	High

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Cost Effectiveness	Most affordable	Most expensive	Cost is distributed among members	Cheaper than private but more expensive than public
Reliability	Low	High	Higher	High

3 Service Models of Cloud Computing

Cloud computing makes it possible to render several services, which can be defined according to the roles, service providers and user companies. Cloud computing models and services are broadly classified as below:

IAAS: Changing Its Hardware Infrastructure on Demand

The Infrastructure As A Service (IAAS) means the hiring & utilizing of the physical infrastructure of IT (network, storage, and servers) from a third-party provider. The IT resources are hosted on external servers and users can access them via an internet connection.

The Benefits

- ◇ Time and cost savings: No installation and maintenance of IT hardware in-house,
- ◇ Better flexibility: On-demand hardware resources that can be tailored to your needs,
- ◇ Remote access and resource management.

6.5. Considerations for Choosing Applications Suitable for Cloud

Hybrid cloud computing is an attractive option for businesses that want to combine the advantages of public and private clouds.

An organization's hybrid cloud provider will be an important partner as it integrates on-premises systems with cloud-based ones. Before awarding this important contract, there are seven important aspects to consider.

1. Get to know your workloads.

Before even talking to a hybrid cloud provider, understand the workloads that you want to pull into the hybrid environment and where you will locate them. For example, data backup and disaster recovery require a different kind of hybrid cloud service than complex analytics applications.

At the same time, ensure your provider can grow with you as your cloud strategy matures. Look to providers that can offer the services you will need as your cloud environment evolves. Seek out solutions that can integrate well with other providers' platforms if you need to allocate different hybrid cloud contracts in a multicloud environment.

2. Evaluate performance.

Your choice of workload informs the next question to ask a potential provider:

for what kind of workload is its infrastructure optimized? As cloud services evolve, providers are beginning to specialize in the kinds of workloads that they support. For example, some might focus on supporting developers, while others might serve a particular kind of application such as systems, applications and products (SAP).

Another aspect of performance is latency. Latency requirements are strict, especially in hybrid cloud environments where on-premises workloads communicate with cloud infrastructure. In these instances, your organization might require a provider with a local edge data center or at least one that can support the appropriate direct connectivity options.

3. Match public and private infrastructure.

Your hybrid cloud provider must also be able to support the technology options that you already use in on-premises infrastructure. Look for easy mappings between the virtual machine choices you've made in house and the formats that the service provider supports, for example.

Aligning the two infrastructures will make it easier to migrate workloads between one environment and the other.

4. Look for easy onboarding.

Ask your potential cloud provider what assistance it offers with migrating data and workloads to its infrastructure. Migration can be a challenging task, especially when working with large data sets. How can the provider help make it simpler and cheaper?

Some may offer hardware appliances to help you ship large data sets manually. At the very least, it should provide migration tools to help you map data between your on-premises infrastructure and its own or provide a consulting service to walk you through the process.

5. Assess security.

The provider should also be able to help you as you secure your data in a hybrid environment.

Hybrid workloads often involve security controls such as tokens. These tokens protect sensitive information in cloud data centers by pointing to records kept on customer premises. Ensure that hybrid cloud providers can help you implement these security measures.

A cloud provider should also be able to answer questions about their compliance processes and risk management. For a list of questions to ask, look through this cloud security assessment list from the Object Management Group.

6. Ensure availability and redundancy.

Security is only one aspect of computing risk. Another is availability. Check your provider's approach to making your data available.

Service-level agreements (SLAs) will be a key factor here. They should not only include availability guarantees, but also escalation and compensation procedures in case the service provider cannot meet them. Consider the provider's ability to help you

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support multiple cloud service providers so that you can failover between each in the event of a problem.

7. Weigh out pricing.

Cost was one of the main initial drivers for cloud computing. While other considerations such as scalability have become increasingly prevalent as cloud computing strategies mature, budget is still a key factor.

“Cloud shock” is an issue in cloud computing contracts. It often happens when customers don’t keep track of the online resources they are using. Check operating fees with the hybrid cloud provider, including the cost of unplanned service expansions to cover spikes in demand.

Be mindful that ending a contract may come with a fee. Plan for any extraction costs to ensure you can migrate your data successfully at the close of the relationship.

Like any business partnership, a hybrid computing contract is something that customers should approach carefully and with an understanding of what they hope to achieve. This will help you choose the right hybrid cloud provider and craft a solid platform on which to build a long-term hybrid cloud strategy.

6.6. Different Phases to Adopt the Cloud

Four Stages of Cloud Adoption

Alibaba Cloud has been working hard for the slogan “more than just cloud” by breaking down barriers of emerging technologies and redefining the digital world. Throughout Alibaba services, the whole development process of Alibaba service can be roughly divided into four stages.

Stage One: Migrating IT system on cloud. The initial products of Alibaba Cloud are to provide flexible and scalable cloud servers for enterprises through hybrid cloud with existing IT architecture. Thus, realizing the migration of IT systems in the cloud, the results of this phase meet the enterprise pursuit of cloud services. Many enterprises purchase servers and deploy non-core enterprise applications on Alibaba Cloud.

Stage Two: Deploying core business on global cloud. After the first step of service cloud, Alibaba Cloud strives to provide customers with more global cloud services. Nowadays, many regions have been deployed in China and even in the world, and then these IT resources have been utilized to realize the global public cloud and the global service strategic deployment of itself and its customers. At this stage, more companies are willing to deploy enterprise productivity or office applications on the cloud.

Stage Three: Microservices on cloud. With more and more applications coming to the cloud, deploying cloud servers is cumbersome. Alibaba Cloud provides customers with better cloud Microservices so that enterprises do not have to worry about complicated server operations and maintenance work, but focus on the purchase of Microservices they need and improve the enterprise in the cloud office efficiency.

Stage Four: Intelligent Cloud. With the advent of the era of big data, Alibaba Cloud is also embracing the challenge. Give full play to the advantages of many years of experience and provide customers with intelligent learning and large data service based on IoT so that they can promote visits through Big Data or AI.

Three Stages of Cloud Network Adoption

The continuous progress of cloud service is inseparable from the continuous development of network solutions. The development of Alibaba Cloud network solutions can also be roughly divided into three stages.

Stage One: Internet access to cloud. The early solutions combined the virtual machine's network with the physical network to form a flat network architecture, such as the large layer two network. With the increase of virtual network scalability, problems are getting more serious for the early solutions. These problems include ARP's spoofing, broadcast storms, host scanning and more. Various network isolation technologies emerged to solve these problems by completely isolating the physical networks from the virtual networks.

One technology isolates user with VLAN, but VLAN only supports up to 4,096 nodes. It cannot support the huge number of users in the cloud.

Stage Two: Private Hybrid Cloud. With the continuous development of cloud computing, virtual network requirements are getting higher and higher, such as scalability, security, reliability, privacy and higher requirements of connection performance.

This gives a rise to a variety of network virtualization technologies. With the tunneling technology, Alibaba Cloud has developed V-Switch, Software-Defined Network (SDN) and Hardware Gateway, and thus created VPC.

V-Switches are distributed the nodes. The gateway and the controller are deployed in clusters, and all links have redundant disaster recovery. This improves the overall availability of the VPC.

Stage Three: Global Cloud Network. In order to meet the needs of a global deployment, Alibaba Cloud provides Cloud Enterprise Network (CEN) and Global Accelerator (GA) and other services to provide users with a high-quality, efficient and stable network transmission environment. Cloud Enterprise Network (CEN) will provide a global network that can quickly build hybrid cloud and distribute business systems to help users build a cloud and network with enterprise skills and communication capabilities. Alibaba Cloud networking services provide you a highly stable, low latency and high-speed network with flexible hybrid cloud connections. Alibaba Cloud networking services deliver secure and reliable communication to and from RDCs in 19 regions globally.

Summary

- ◇ To sum up, selecting Alibaba Cloud service is to select the best quality and reliable service.
- ◇ Let's have a look at Alibaba Cloud network services portfolios. Alibaba Cloud network technology system is divided into three parts -- Cloud Network, Interconnected Cloud Network and Connect-to-Cloud Network.
- ◇ Cloud Network provides rich network functions on Alibaba Cloud and the management of access to public network traffic. These functions include Elastic IP (EIP), NAT Gateway, Virtual Private Cloud, VPC and Server Load Balancer (SLB).

Notes

- ◇ Interconnected Cloud Network is responsible for the network connection between VPCs, mini-level construction of a cloud across 19 regions in Alibaba Cloud. The Interconnected Cloud Network includes Cloud Enterprise Network (CEN) and Global Accelerator (GA).
- ◇ Connect-to-Cloud Network is a safe way for users to go to the cloud. It connects the POP points, distributed all over the world to the VPC channels on the cloud, and realizes the communication between the POP points. The Connect-to-Cloud Network includes the following services -- VPN Gateway, Express Connect and Smart Access Gateway.
- ◇ The Salesforce.com is probably best known for its sales management SaaS, but it is also a leader in cloud computing development.
- ◇ Developers can use App Exchange applications uploaded by others, share their own applications in the directory, or publish private applications accessible only by authorized companies or clients.
- ◇ The Google App Engine provides a fully integrated application environment. Using Google's development tools and computing cloud, App Engine applications are easy to build, easy to maintain, and easy to scale.
- ◇ Blue Cloud is a series of cloud computing offerings that enables enterprises to distribute their computing needs across a globally accessible resource grid.
- ◇ The first level of maturity defines the traditional application service provider (ASP) model of software delivery, and dates back to the 1990.
- ◇ Amazon's Web Services (AWS) set of offerings, which provides developers with direct access to Amazon's software and machines

Keywords

Infrastructure-as-a-Service (IaaS): Infrastructure as a Service is a provision model in which an organization outsources the equipment used to support operations, including storage, hardware, servers and networking components.

Platform-as-a-Service (PaaS): Platform as a service is a category of cloud computing services that provide a computing platform and a solution stack as a service. In the classic layered model of cloud computing, the PaaS layer lies between the SaaS and the IaaS layers.

Quality of Service (QoS): The quality of service refers to several related aspects of telephony and computer networks that allow the transport of traffic with special requirements.

Software-as-a-Service (SaaS): Software as a Service is a software distribution model in which applications are hosted by a vendor or service provider and made available to customers over a network, typically the Internet.

Self-Assessment Questions

- 1) Which of the following can be referred to the location and management of the cloud's infrastructure?
 - a) Service
 - b) Deployment

- c) Application
 - d) None of the mentioned
- 2) _____ model consists of the particular types of services that you can access on a cloud computing platform.
- a) Service
 - b) Deployment
 - c) Application
 - d) None of the mentioned
- 3) Which one of the following statements is wrong?
- a) The use of the word “cloud” makes reference to the two essential concepts.
 - b) Cloud computing abstracts systems by pooling and sharing resources
 - c) Cloud computing is nothing more than the Internet.
 - d) All of the mentioned
- 4) Which of the following is the deployment model?
- a) Public
 - b) Private
 - c) Hybrid
 - d) All of the mentioned
- 5) Which one of the following statements is wrong?
- a) Cloud Computing has two distinct sets of models.
 - b) Amazon has built a worldwide network of data centers to service its search engine.
 - c) Azure enables .NET Framework applications to run over the Internet.
 - d) None of the mentioned
- 6) Which one of the following is considered the best-known service model?
- a) SaaS
 - b) IaaS
 - c) PaaS
 - d) All of the mentioned
- 7) The _____ model originally did not require a cloud to use virtualization to pool resources.
- a) NEFT
 - b) NIST
 - c) NIT
 - d) All of the mentioned

Notes

- 8) Which one of the following dimensions is related to the organization's boundaries?
- a) The physical location of data
 - b) Ownership
 - c) Security boundary
 - d) All of the mentioned
- 9) How many types of dimensions exist in the Cloud Cube Model?
- a) One
 - b) Two
 - c) Three
 - d) Four
- 10) _____ model attempts to categorize a cloud network based on four-dimensional factors.
- a) Cloud Square
 - b) Cloud Service
 - c) Cloud Cube
 - d) All of the mentioned

Review Questions

1. Discuss the Security concerns and counter measures in Cloud environment
2. Explain Governance, Risk, and Compliance aspects in Cloud
3. Explain the Cloud security best practices
4. What is Cloud models suitable for different categories of users?
5. What do understand by Considerations for choosing applications suitable for Cloud?
6. Explain the Different phases to adopt the Cloud

Answers. 1)B 2)A 3)B 4)D 5)B 6)D 7)B 8)A 9)D 10)C

Unit-7: Cloud IT Model

Notes

Objective:

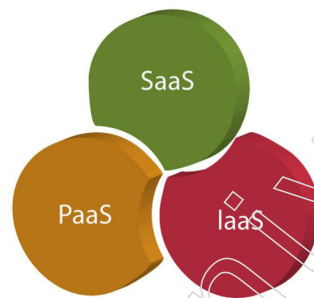
After studying this chapter, you will be able to:

- Explain the Best Practice Cloud IT Model
- Discuss the Analyze of Case Studies
- Explain the How to decide if the cloud is right for your requirements
- Discuss the Cloud based service

7.1. Best Practice Cloud IT Model

There are the following three types of cloud service models -

1. Infrastructure as a Service (IaaS)
2. Platform as a Service (PaaS)
3. Software as a Service (SaaS)



Infrastructure as a Service (IaaS)

IaaS is also known as Hardware as a Service (HaaS). It is a computing infrastructure managed over the internet. The main advantage of using IaaS is that it helps users to avoid the cost and complexity of purchasing and managing the physical servers.

Characteristics of IaaS

There are the following characteristics of IaaS -

- ◇ Resources are available as a service
- ◇ Services are highly scalable
- ◇ Dynamic and flexible
- ◇ GUI and API-based access
- ◇ Automated administrative tasks

Example: DigitalOcean, Linode, Amazon Web Services (AWS), Microsoft Azure, Google Compute Engine (GCE), Rackspace, and Cisco Metacloud.

Platform as a Service (PaaS)

PaaS cloud computing platform is created for the programmer to develop, test, run, and manage the applications.

Notes

Characteristics of PaaS

There are the following characteristics of PaaS -

- ◇ Accessible to various users via the same development application.
- ◇ Integrates with web services and databases.
- ◇ Builds on virtualization technology, so resources can easily be scaled up or down as per the organization's need.
- ◇ Support multiple languages and frameworks.
- ◇ Provides an ability to "Auto-scale".

Example: AWS Elastic Beanstalk, Windows Azure, Heroku, Force.com, Google App Engine, Apache Stratos, Magento Commerce Cloud, and OpenShift.

Software as a Service (SaaS)

SaaS is also known as "on-demand software". It is a software in which the applications are hosted by a cloud service provider. Users can access these applications with the help of internet connection and web browser.

Characteristics of SaaS

There are the following characteristics of SaaS -

- ◇ Managed from a central location
- ◇ Hosted on a remote server
- ◇ Accessible over the internet
- ◇ Users are not responsible for hardware and software updates. Updates are applied automatically.
- ◇ The services are purchased on the pay-as-per-use basis

Example: BigCommerce, Google Apps, Salesforce, Dropbox, ZenDesk, Cisco WebEx, ZenDesk, Slack, and GoToMeeting.

Difference between IaaS, PaaS, and SaaS

The below table shows the difference between IaaS, PaaS, and SaaS -

IaaS	Paas	SaaS
It provides a virtual data center to store information and create platforms for app development, testing, and deployment.	It provides virtual platforms and tools to create, test, and deploy apps.	It provides web software and apps to complete business tasks.
It provides access to resources such as virtual machines, virtual storage, etc.	It provides runtime environments and deployment tools for applications.	It provides software as a service to the end-users.
It is used by network architects.	It is used by developers.	It is used by end users.
IaaS provides only Infrastructure.	PaaS provides Infrastructure+Platform.	SaaS provides Infrastructure+Platform +Software.

7 Best Practice

Notes

Best Practice #1: Create a strategy and a set of (realistic) goals upfront

Some businesses are jumping in without articulating a long-term cloud strategy and how it relates to their overall business. As with any project, establish realistic goals and priorities, a clear budget and deadline, as well as a shared understanding of what resources are available for implementation and maintenance. Although cloud computing promises significant ROI -- productivity gains of 50 percent or more -- keeping complexity and cost to a minimum requires planning and strategy.

Best Practice #2: Learn from the mistakes of those who came before you

Early adopters took a standalone approach to cloud computing. The services were readily available, easy to consume, and economical. Implementation challenges were few. However, for enterprises where traditional IT infrastructure often serves core business operations, the “detached” cloud might deliver only short-term value and potentially require future re-implementation or migration. Although a standalone approach risks creating silo-ed applications, an integrated cloud strategy will deliver long-term results.

Best Practice #3: Avoid upfront costs

One reason businesses turn to cloud computing and software-as-a-service (SaaS) is to mitigate risk. With a pay-as-you-go model, you can simply turn the system off if it is not working for you, and you only pay for what you use. Be sure not to jeopardize return on investment with expensive software licenses -- your integration solution, like every other part of your cloud computing investment, should use a flexible subscription model, too.

Best Practice #4: Get serious about autonomy

Integration of on-premise applications has traditionally required a team of IT specialists who have a deep understanding of underlying application frameworks and processes. SaaS applications are designed to be managed by business users -- non-domain experts who will need to quickly and easily connect data with other enterprise systems. Cloud integration should complement the model by minimizing development, implementation, and maintenance resources, allowing users to focus on their core business.

Best Practice #5: Address security concerns

According to analysts, nearly 75 percent of CIO and IT executives cite security as their number one concern when it comes to cloud computing. Because integrated cloud computing involves moving sensitive data between the cloud and on-premise networks, guaranteeing security is vital. When vetting an integration solution, determine which standards are supported for securing the data in transit. Keep in mind that as enterprises move more processes to the cloud, the volume of sensitive data flowing to and from the cloud increases.

Best Practice #6: Ensure performance and availability of the data

Today, popular cloud computing services provide levels of availability and performance that outperform internal infrastructure, in many cases 99.9 percent uptime

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or better. When designing a cloud strategy, identify integration requirements for each system (real-time, near real-time, batch), and determine the number of simultaneous requests to be handled, and specify all special architecture requirements. Success depends on ensuring information will not be lost if the cloud or on-premise source goes down. (Yes, even the cloud needs a backup plan.)

Best Practice #7: Maximize connectivity options

Cloud computing has become a loose definition for services on the Web: everything from SaaS and platform-as-a-service (PaaS), to Web-based utility and storage solutions, and emerging Web 2.0 properties such as Google Docs, LinkedIn, and Twitter. According to research¹ from Saugatuck Technology, by the end of 2010, one-quarter of business process improvement initiatives will include integration of information from enterprise social computing solutions. Connectivity requirements will continue to evolve beyond standard enterprise applications, legacy systems, and databases, to modern Web service and Web 2.0 APIs.

Solving the Integration Puzzle

Current economic conditions are driving businesses toward cloud computing, and with good reason. No one can argue that the cloud doesn't deliver value. It is already proven to be a disruptive force. However, the cloud's "Achilles heel" is a lack of integration with the rest of the enterprise. Realizing its full potential relies, for the foreseeable future, on integrating data in the cloud with on-premise applications and databases.

7.2. Analyze of Case Studies when Deciding to Adopt Cloud Computing Architecture

Cloud architecture has grown to be so normal that we forget what its definition includes. Many of us can remember a list of features like on-demand resources, optimal performance, and delivery of services. Others may know the "-ilities" like availability or scalability. Developers often start with the inherent promise to the customer: You will have access to the services you need whenever you need them, and with the availability and reliability you expect.

The definition of cloud includes all these details and more, and we will break down those components over a series of articles. The team co-authoring this article regularly supports the creation and maintenance of private and hybrid cloud operations for companies across the globe. We will outline the important aspects of a successful cloud operation based on our own experiences and backed by industry standards.

In order to get into depth on the topic, we must first start with who and what we're defining as cloud.

Defining the key components of cloud computing

The definition of cloud computing in NIST 800-145 covers the required practices ranging from Cloud architecture

Delivering different cloud types, and from private to public cloud. It also covers the various consumption models, from IaaS to PaaS. When we refer to cloud, we are starting from this definition of cloud computing:

Cloud computing is a model for enabling ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources (e.g., networks, servers, storage, applications, and services) that can be rapidly provisioned and released with minimal management effort or service provider interaction. This cloud model is composed of five essential characteristics, three service models, and four deployment models.

NIST definition

Specifically, we propose the usage of the Capability Maturity Model Integration for Services (CMMI-SVC) model. CMMI-SVC provides a framework for the maturity of the processes that combine the people, procedures, and tools to deliver capabilities.

The five practices below structure the work packages that drive continuous improvements around service consumability, security, compliance, availability, and scalability.

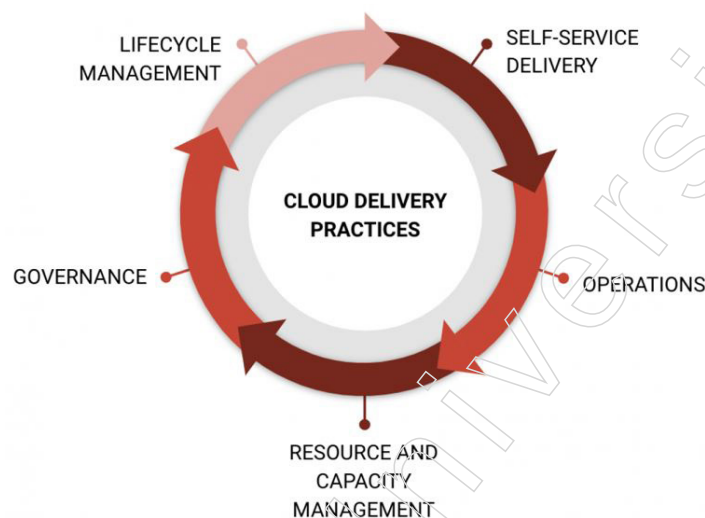


Figure: Cloud Delivery Practices

Here is a brief introduction to these five practices:

- ◇ **Self-Service Delivery:** Enabling developers, operators, and other users to access the cloud resources they need when they need them across the enterprise with minimal friction.
- ◇ **Operations:** The process of delivering the cloud platform in such a way that it meets the agreed Service Level Objectives (SLO) over time, including procedures to ensure scalability, resilience, repairability, security, and continuous compliance.
- ◇ **Resource and Capacity Management:** Establishing and maintaining optimized resource capacity and availability at a justifiable cost. Resource and capacity management covers all layers, from infrastructure (compute, storage, networking) to software and people.
- ◇ **Governance:** The decision-making model. Organizations should be designed to ensure that investments support business objectives and capture the system that drives the cloud team's behavior, decision making, and long-term measures of success. It includes ethics, risk management, compliance, and administration.

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- ◇ **Lifecycle Management:** Cloud technologies often have short release cycles that demand robust processes to ensure timely upgrades and consistent configuration and stability across releases. For example, Kubernetes has a quarterly release cycle for new features while continuously rolling patches and security updates.

7.3. How to Decide if the Cloud is Right for Your Requirements

Selecting the Cloud Platform in 2018: Criteria and Best Vendors

Using cloud technologies has become a must for large businesses and startups. The process of selecting a proper cloud platform is complicated by the fact that there's no common scheme for assessing providers and offerings, as all of them have specific features, and every organization has specific needs. In our post, we want to discuss the basic criteria you can use to help in choosing the right cloud platform and provide a list of platforms that are worth paying attention to.

Criteria for Selecting a Cloud Platform

The cloud market is crowded today. There are several important features that should be considered in order to understand which cloud computing platform you should choose. Figuratively, all the criteria can be divided into three groups.

Your Business Needs

- ◇ The first step is to determine your business-specific needs. To do this, you should consider the following points:
- ◇ The type of data to store. Sensitive or non-sensitive information will directly define the type of platform to use.
- ◇ The number of devices to connect to your platform. Examine the size of your business and the scalability of the service. Remember also that the more people involved, the less secure your data will be.
- ◇ The budget. There are services that are free and those that are not, and their security levels differ. Small vendors are usually less expensive than global leaders.

Cloud Types

The choice of the cloud type depends on the above-mentioned needs.

Public clouds are more suitable for non-sensitive data storage, as they use a shared infrastructure and serve multiple customers. There are both free and pay-per-use options. This option is the most vulnerable to cyber-attacks.

Pros: flexibility; scalability; reasonable price; you don't have to manage it yourself.

Cons: security issues; issues of compliance for certain industries (e.g. finance).

Private clouds are suitable for sensitive data storage (e.g. for the healthcare industry) as they use a proprietary infrastructure and serve just one customer. They work well for changing business models and unpredictable needs.

Pros: high security; scalability.

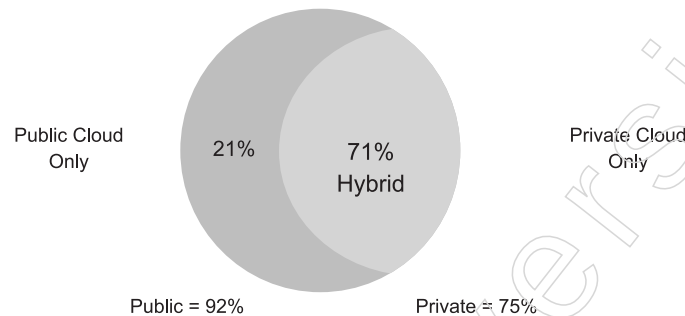
Cons: management (your own team should manage it); price (additional expenses for staffing, maintenance, virtualization, cloud software and tools).

Hybrid clouds combine the best features of the two above-mentioned options. They consist of several private and public clouds, and a company may shift its workloads between them. An on-premise private repository can be used for sensitive data and a public one — for less sensitive information. Hybrids are ideal for highly changeable workloads.

Pros: maximum flexibility; easier access for mobile and remote users; enhanced agility.

Cons: complex management due to complex infrastructure; higher cost; security (have some equal vulnerabilities in public clouds).

96% of Respondents Are Using Cloud



Source: RightScale 2018 State of the Cloud Report

The hybrid cloud market is estimated to grow from \$33 billion in 2016 to \$97.6 billion by 2023.

Multi-clouds have become a new trend in 2018. This approach implies that a client chooses several providers and uses their platforms and environments for different purposes as a coherent system. This is a sound approach as soon as one service maybe not enough to comply with all of the needs of a large organization.

Cloud Features

When you've decided on the cloud type, check that the main cloud features satisfy your business requirements:

- ◇ Security
- ◇ Compliance
- ◇ Architecture
- ◇ Manageability
- ◇ Service levels
- ◇ Support
- ◇ Cost

As you see, all of these criteria are interdependent. If you want to understand how to choose the right cloud solution, you should study them as a system.

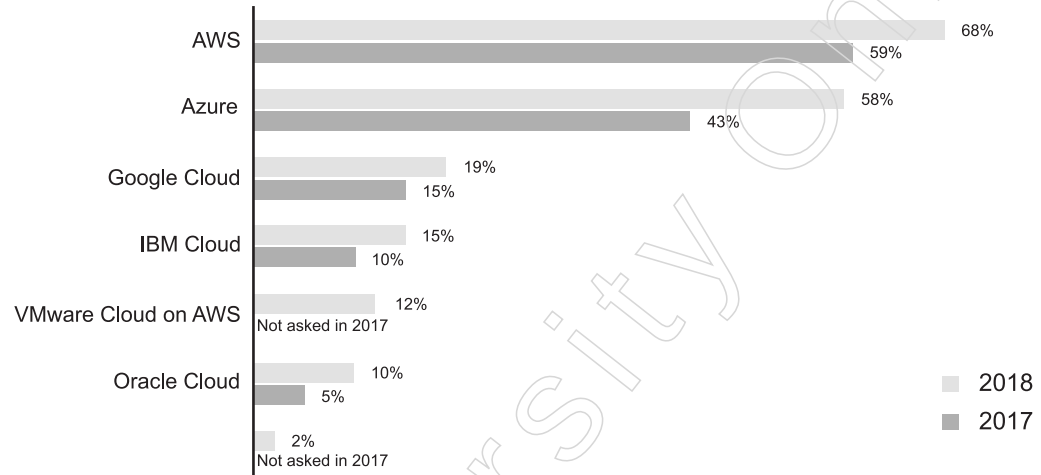
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Best Cloud Platforms for 2018

Now, when you know what to consider when choosing a cloud provider, let's take a look at the list of the most interesting offerings. Among providers, there are undisputed top leaders globally: AWS, Microsoft Azure and Google.

Enterprise Public Cloud Adoption 2018 vs. 2017

% of Respondents Running Applications



Source: RightScale 2018 State of the Cloud Report

These giants are most suitable for large enterprises. For smaller businesses and personal usage, there's a wide choice of vendors.

7.4. Cloud based Service, Applications and Development Platform Deployment so as to Improve the Total Cost of Ownership (TCO)

Cloud based services provide information technology (IT) as a service over the Internet or dedicated network, with delivery on demand, and payment based on usage. Cloud based services range from full applications and development platforms, to servers, storage, and virtual desktops.

Who uses cloud based services and why?

Corporate and government entities utilize cloud based services to address a variety of application and infrastructure needs such as CRM, database, compute, and data storage. Unlike a traditional IT environment, where software and hardware are funded up front by department and implemented over a period of months, cloud based services deliver IT resources in minutes to hours and align costs to actual usage. As a result, organizations have greater agility and can manage expenses more efficiently. Similarly, consumers utilize cloud based services to simplify application utilization, store, share, and protect content, and enable access from any web-connected device.

How cloud based services work

Cloud based services have several common attributes:

Virtualization - cloud computing utilizes server and storage virtualization extensively to allocate/reallocate resources rapidly

Multi-tenancy - resources are pooled and shared among multiple users to gain economies of scale

Network-access - resources are accessed via web-browser or thin client using a variety of networked devices (computer, tablet, smartphone)

On demand - resources are self-provisioned from an online catalogue of pre-defined configurations

Elastic - resources can scale up or down automatically

Metering/chargeback - resource usage is tracked and billed based on service arrangement. Among the many types of cloud based services delivered internally or by third party service providers, the most common are:

Software-as-a-Service (SaaS) – software runs on computers owned and managed by the SaaS provider, versus installed and managed on user computers. The software is accessed over the public Internet and generally offered on a monthly or yearly subscription.

Infrastructure-as-a-Service (IaaS) – compute, storage, networking, and other elements (cloud security, tools) are provided by the IaaS provider via public Internet, VPN, or dedicated network connection. Users own and manage operating systems, applications, and information running on the infrastructure and pay by usage.

Platform-as-a-Service (PaaS) – All software and hardware required to build and operate cloud based applications are provided by the PaaS provider via public Internet, VPN, or dedicated network connection. Users pay by use of the platform and control how applications are utilized throughout their lifecycle.

Benefits of cloud based services

Cloud based services offer numerous benefits to include:

- ◇ Faster implementation and time to value
- ◇ Anywhere access to applications and content
- ◇ Rapid scalability to meet demand
- ◇ Higher utilization of infrastructure investments
- ◇ Lower infrastructure, energy, and facility costs
- ◇ Greater IT staff productivity and across organization
- ◇ Enhanced security and protection of information assets

Dell Technologies Cloud Storage Services

Dell Technologies Cloud Storage Services provide Storage-as-a-Service that's directly connected to multiple clouds for disaster recovery, analytics, test/dev and other use cases. Cloud Storage Services free IT teams from system management while delivering enterprise-grade data security, capacity, replication, performance and availability in the public cloud. With this Dell Technologies offering, enterprises can easily leverage their preferred clouds and switch easily between them at any time based on the needs of applications. Cloud storage services help to avoid vendor lock-in as data is independent of the cloud.

Notes

TCO In Cloud Computing

The total cost of ownership in cloud computing refers to the total cost of adopting, operating, and provisioning cloud infrastructure. Organizations often find it necessary to perform a cloud TCO analysis when they are considering moving to the cloud because it allows them to weigh the cost of cloud adoption against the cost of running their current on-premise systems.

Since TCO is typically used to understand the lifetime cost and value of static or contained resources, however, estimating the TCO of cloud infrastructure — an inherently dynamic ecosystem — can be challenging.

When businesses calculate cloud TCO, they often make an apples-to-apples comparison of the costs of running an on-premise system versus the cost of running the same system in the cloud. That is, comparing the initial purchase price of hardware and software in an on-premise environment to the monthly subscription cost of cloud computing.

For example, a common starting point might be: “How much will it cost to run my on-premises system in the cloud?” or “If I have 3,000 traditional servers in my datacenter, what is the rack-rate for 3,000 compute instances of similar CPU, memory, and networking capacity in the cloud over x period?”

While this is a good place to start, you may not get the full picture. This is because a head-to-head comparison does not capture hidden costs or intangible costs of not switching to the cloud (i.e. the benefits of a cloud solution), such as faster time to market, increased productivity, and elasticity of demand.

To accurately calculate cloud TCO, you must capture not only the purchase price of on-premises vs. cloud solutions but also the intangible costs associated with either solution.

In this article, we'll discuss the best approach and practices when evaluating the total cost of ownership for cloud computing.

Steps for Calculating Cloud TCO

Below are some of the steps you should follow when estimating cloud total cost of ownership.

Step 1: Calculate your current IT infrastructure costs.

Understanding the actual cost of your current IT solution is the first step. This means calculating the direct and indirect costs of running and maintaining your current system as well as estimating your current workloads, including servers, databases, storage, and network bandwidth.

Consider the following cost areas:

- ◇ **Hardware and infrastructure**—Identify the cost of the hardware that powers your on-premise application. These include physical servers, supplies, spare parts, etc.
- ◇ **Datacenter**—How much does it cost to power your data center? How much does it cost to meet your current cooling, power, and space requirements?

- ◇ **Software**—Calculate your current software usage, including the number of licenses and cost of these licenses.
- ◇ **Personnel**—Identify all the personnel involved in system, network, and database administration and how much it costs to payroll them.
- ◇ **Disaster recovery**—If you have a disaster recovery system in place, how much does it cost to maintain and manage that site?
- ◇ **Maintenance**—Calculate the cost of servicing, operating, and maintaining the system, including the cost of both in-house and outsourced maintenance.
- ◇ **Upgrades**—How much will it cost to upgrade the system if the need arises? Would you need to overhaul the system completely?
- ◇ **Security**—Estimate the total cost of securing your current system, from the cost of physical security to firewalls and security experts.
- ◇ **Hidden costs**—How much does downtime cost you? Review log files to determine server downtime frequency, hours lost, and the cost implication of those hours.

Step 2: Estimate the cost of a cloud solution.

Next, calculate the cost of operating your applications in the cloud. Keep in mind that many of the cost areas considered for on-premises infrastructure will not apply in the cloud because they will be offloaded to the cloud service provider. However, it's important to note that a cloud solution is not inherently cheaper than on-premises infrastructure.

When businesses switch to the cloud, they often assume that their cloud bill will be automatically cheaper. But the on-demand nature of cloud services means your cloud bill could quickly spiral out of control as developers deploy instances and move at the speed of the cloud. Understanding the major cost areas in the cloud is key to optimizing your cloud cost and ensuring a lower TCO.

Two of the major cost areas to consider for the cloud are migration costs and the monthly cost of your selected cloud services.

1. Migration costs

Moving your applications and data to the cloud is a key step when switching to the cloud. Your current applications may require modification to function properly in the cloud. Gartner identifies the five ways to move applications into the cloud, namely:

- ◇ Rehosting applications without making any changes to their architecture
- ◇ Refactoring or running applications on a cloud provider's infrastructure
- ◇ Revising the application, i.e. modifying or extending the existing code base
- ◇ Rebuilding or rearchitecting the entire application for the cloud
- ◇ Replacing the application with commercial software delivered as a service

Each application migration method has its cost implications and you need to determine the costs associated with the method you choose. In addition to application migration costs, estimate data transfer charges that will accrue when moving your application.

Notes

2. Monthly cloud cost

Your monthly cloud cost will depend on your workloads, and the specific cloud services consumed and method of purchase. The goal here is to estimate your potential monthly cloud bill based on your current workloads. Since this calculation differs considerably for each organization, major cloud platforms provide pricing calculators that make it easier to estimate your monthly cloud bill. The AWS pricing calculator, for example, allows you to estimate your infrastructure cost based on the retinue of AWS products and services selected.

Two of the major factors that will affect the size of your cloud bill are:

Type of cloud services consumed: Commodity services, such as storage or raw compute power, are relatively less expensive compared to more specialized services, such as machine learning. Amazon, for instance, offers Rekognition which does image and video analysis, and Polly, which is a text-to-speech service. These services have higher workload costs than storage. The total cost will depend on the types of services your business needs.

Cloud consumption model: The on-demand model, where resources are deployed as needed, is the most popular cloud usage model. However, it is also the most expensive cloud consumption model. The other way to consume cloud services is to use a savings plan or prepaid option (reserved instances). You could also opt for a hybrid model. Your cloud costs will differ depending on the consumption model your business adopts.

3. Consultation and training costs

If your team lacks the expertise required for the migration process, factor in the cost of hiring consultants for training.

Step 3: Consider the intangible benefits of the cloud.

Beyond comparing the monetary implications of on-premise versus cloud solutions, there are opportunity costs associated with not switching to the cloud. You need to quantify what this means for your business.

Innovation—The cloud offers hundreds of services you can access on demand. By continuing with an on-premises system, you sacrifice the ability of developers to move fast and respond quickly to market changes.

Elasticity—Handling demand in an on-premise environment is always a challenge. The solution is usually to maintain redundant infrastructure in anticipation of peak loads. In the cloud, however, you could easily deploy instances to take care of the additional peak without any downtime. When the peak is over, you go back to operating at your normal capacity at no additional cost. While you may incur a larger monthly cloud bill at peak, you will experience no downtime nor would you need to maintain redundant infrastructure when the surge is over.

Comparing On-Premise TCO To Cloud TCO: Key Things To Keep In Mind

At the end of your cloud TCO analysis, you should have specific numbers that can help with your decision-making. A few things should guide understanding of the results:

1. Cloud computing is not inherently cheaper than an on-premise model.
2. Cloud adoption is rarely about pure cost savings. Often the end result is a larger ROI and better business outcomes, not lower TCO, even though it could be both.
3. Comparing the business value and opportunity cost of switching to the cloud versus using an on-premises model is just as important as comparing head-to-head costs.
4. Identifying cost savings and efficiencies is critical when performing a cloud TCO analysis.

Summary

At the end of your cloud TCO analysis, you should have specific numbers that can help with your decision-making. A few things should guide understanding of the results:

- Cloud computing is not inherently cheaper than an on-premise model.
- Cloud adoption is rarely about pure cost savings. Often the end result is a larger ROI and better business outcomes, not lower TCO, even though it could be both.
- Comparing the business value and opportunity cost of switching to the cloud versus using an on-premises model is just as important as comparing head-to-head costs.
- Identifying cost savings and efficiencies is critical when performing a cloud TCO analysis.
- definition of cloud computing as a baseline for our understanding of the technology. The five factors of CMMI-SVC practices provide a framework by which thousands of organizations adopt and support reliable private cloud services.

Keywords

Cloud storage: A form of networked data storage where data files are stored on multiple virtual servers.

Local database: A local database is one in which all the data is stored on an individual computer.

Networked database: A networked database is one in which the data is stored on a computer or server connected to a network, and accessible by all computers connected to that network.

QuickBase: A Website that lets you quickly and easily select, customize and share online workgroup applications that actually work the way your teams do.

TeamDesk: The TeamDesk Application Library includes applications for project management, marketing, sales, customer support, human resources, billing and other business functions.

Self-Assessment Questions

- 1) Cloud computing architecture is a combination of?

Notes

- a. service-oriented architecture and grid computing
 - b. Utility computing and event-driven architecture.
 - c. Service-oriented architecture and event-driven architecture.
 - d. Virtualization and event-driven architecture.
- 2) In how many parts we can broadly divide the architecture of the Cloud?
- a. 4
 - b. 3
 - c. 2
 - d. 5
- 3) Which one of the following refers to the user's part of the Cloud Computing system?
- a. back End
 - b. Management
 - c. Infrastructure
 - d. Front End
- 4) Which one of the following can be considered as the example of the Front-end?
- a. Web Browser
 - b. Google Compute Engine
 - c. Cisco Metapod
 - d. Amazon Web Services
- 5) By whom is the backend commonly used?
- a. Client
 - b. User
 - c. Stockholders
 - d. service provider

Review Questions

1. What is the Best Practice Cloud IT Model
2. Explain Analyze of Case Studies when deciding to adopt cloud computing architecture
3. How to decide if the cloud is right for your requirements
4. Discuss the Cloud based service, applications and development platform deployment so as to improve the total cost of ownership (TCO)